AOS-CX 10.06 Fundamentals Guide

8320, 8325, 8360 Switch Series

Hewlett Packard Enterprise

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Notices

Acknowledgments

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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 8320 Switch Series (JL479A, JL579A, JL581A)
- Aruba 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- Aruba 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in **Support and other resources**.

Command syntax notation conventions

Convention	Usage
`example-text`	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets (`[ ]`).
`example-text`	In code and screen examples, indicates text entered by a user.
Any of the following: • *`<example-text>`* • `<example-text>` • *`example-text`* • *example-text*	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (`< >`). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
`|`	Vertical bar. A logical `OR` that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.

Table Continued

Convention	Usage
`{ }`	Braces. Indicates that at least one of the enclosed items is required.
`[ ]`	Brackets. Indicates that the enclosed item or items are optional.
… or `. . .`	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

`switch>`

The CLI prompt indicates the current command context. For example:

`switch>`

Indicates the operator command context.

`switch#`

Indicates the manager command context.

`switch(`*`CONTEXT-NAME`*`)#`

Indicates the configuration context for a feature. For example:

`switch(config-if)#`

Identifies the `interface` context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

`switch(config-vlan-100)#`

When referring to this context, this document uses the syntax:

`switch(config-vlan-`*`<VLAN-ID>`*`)#`

Where *`<VLAN-ID>`* is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

member/slot/port

On the 83xx Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Line module number. Always 1.
- *port*: Physical number of a port on a line module

For example, the logical interface `1/1/4` in software is associated with physical port 4 in slot 1 on member 1.

NOTE: If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split to 4 x 10G or 4 x 25G. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

AOS-CX introduction

AOS-CX is a new, modern, fully programmable operating system built using a database-centric design that ensures higher availability and dynamic software process changes for reduced downtime. In addition to robust hardware reliability, the AOS-CX operating system includes additional software elements not available with traditional systems, including:

- Automated visibility to help IT organizations scale: The Aruba Network Analytics Engine allows IT to monitor and troubleshoot network, system, application, and security-related issues easily through simple scripts. This engine comes with a built-in time series database that enables customers and developers to create software modules that allow historical troubleshooting, as well as analysis of historical trends to predict and avoid future problems due to scale, security, and performance bottlenecks.
- Programmability simplified: A switch that is running the AOS-CX operating system is fully programmable with a built-in Python interpreter as well as REST-based APIs, allowing easy integration with other devices both on premise and in the cloud. This programmability accelerates IT organization understanding of and response to network issues. The database holds all aspects of the configuration, statistics, and status information in a highly structured and fully defined form.
- Faster resolution with network insights: With legacy switches, IT organizations must troubleshoot problems after the fact, using traditional tools like CLI and SNMP, augmented by separate, expensive monitoring, analytics, and troubleshooting solutions. These capabilities are built in to the AOS-CX operating system and are extensible.
- High availability: For switches that support active and standby management modules, the AOS-CX database can synchronize data between active and standby modules and maintain current configuration and state information during a failover to the standby management module.
- Ease of roll-back to previous configurations: The built-in database acts as a network record, enabling support for multiple configuration checkpoints and the ability to roll back to a previous configuration checkpoint.

AOS-CX system databases

The AOS-CX operating system is a modular, database-centric operating system. Every aspect of the switch configuration and state information is modeled in the AOS-CX switch configuration and state database, including the following:

- Configuration information
- Status of all features
- Statistics

The AOS-CX operating system also includes a time series database, which acts as a built-in network record. The time series database makes the data seamlessly available to Aruba Network Analytics Engine agents that use rules that evaluate network conditions over time. Time-series data about the resources monitored by agents are automatically collected and presented in graphs in the switch Web UI.

Aruba Network Analytics Engine introduction

The Aruba Network Analytics Engine is a first-of-its-kind built-in framework for network assurance and remediation. Combining the full automation and deep visibility capabilities of the AOS-CX operating system, this unique framework enables monitoring, collecting network data, evaluating conditions, and taking corrective actions through simple scripting agents.

This engine is integrated with the AOS-CX system configuration and time series databases, enabling you to examine historical trends and predict future problems due to scale, security, and performance bottlenecks. With that information, you can create software modules that automatically detect such issues and take appropriate actions.

With the faster network insights and automation provided by the Aruba Network Analytics Engine, you can reduce the time spent on manual tasks and address current and future demands driven by Mobility and IoT.

AOS-CX CLI

The AOS-CX CLI is an industry standard text-based command-line interface with hierarchical structure designed to reduce training time and increase productivity in multivendor installations.

The CLI gives you access to the full set of commands for the switch while providing the same password protection that is used in the Web UI. You can use the CLI to configure, manage, and monitor devices running the AOS-CX operating system.

Aruba CX mobile app

The Aruba CX mobile app enables you to use a mobile device to configure or access a supported ArubaOS-CX switch. You can connect to the switch through Bluetooth or Wi-Fi.

You can use this application to do the following:

- Connect to the switch for the first time and configure basic operational settings—all without requiring you to connect a terminal emulator to the console port.
- View and change the configuration of individual switch features or settings.
- Manage the running configuration and startup configuration of the switch, including the following:
    - Transferring files between the switch and your mobile device
    - Sharing configuration files from your mobile device
    - Copying the running configuration to the startup configuration
- Access the switch CLI.

More information about the Aruba CX mobile app

For more information about the Aruba CX mobile app, see:

www.arubanetworks.com/products/networking/switches/cx-mobileapp

Aruba NetEdit

Aruba NetEdit enables the automation of multidevice configuration change workflows without the overhead of programming.

The key capabilities of NetEdit include the following:

- Intelligent configuration with validation for consistency and compliance
- Time savings by simultaneously viewing and editing multiple configurations
- Customized validation tests for corporate compliance and network design
- Automated large-scale configuration deployment without programming
- Ability to track changes to hardware, software, and configurations (whether made through NetEdit or directly on the switch) with automated versioning

More information about Aruba NetEdit

For more information about Aruba NetEdit, search for NetEdit at the following website:

www.hpe.com/support/hpesc

Ansible modules

Ansible is an open-source IT automation platform.

Aruba publishes a set of Ansible configuration management modules designed for switches running AOS-CX software. The modules are available from the following places:

- The `arubanetworks.aoscx_role` role in the Ansible Galaxy at:

  https://galaxy.ansible.com/arubanetworks/aoscx_role
- The `aoscx-ansible-role` at the following GitHub repository:

  https://github.com/aruba/aoscx-ansible-role

AOS-CX Web UI

The Web UI gives you quick and easy visibility into what is happening on your switch, providing faster problem detection, diagnosis, and resolution. The Web UI provides dashboards and views to monitor the status of the switch, including easy to read indicators for: power supply, temperature, fans, CPU use, memory use, log entries, system information, firmware, interfaces, VLANs, and LAGs. In addition, you use the Web UI to access the Network Analytics Engine, run certain diagnostics, and modify some aspects of the switch configuration.

AOS-CX REST API

Switches running the AOS-CX software are fully programmable with a REST (REpresentational State Transfer) API, allowing easy integration with other devices both on premises and in the cloud. This programmability—combined with the Aruba Network Analytics Engine—accelerates network administrator understanding of and response to network issues.

The AOS-CX REST API enables programmatic access to the AOS-CX configuration and state database at the heart of the switch. By using a structured model, changes to the content and formatting of the CLI output do not affect the programs you write. And because the configuration is stored in a structured database instead of a text file, rolling back changes is easier than ever, thus dramatically reducing a risk of downtime and performance issues.

The AOS-CX REST API is a web service that performs operations on switch resources using HTTPS `POST`, `GET`, `PUT`, and `DELETE` methods.

A switch resource is indicated by its Uniform Resource Identifier (URI). A URI can be made up of several components, including the host name or IP address, port number, the path, and an optional query string. The AOS-CX operating system includes the AOS-CX REST API Reference, which is a web interface based on the Swagger UI. The AOS-CX REST API Reference provides the reference documentation for the REST API, including resources URIs, models, methods, and errors. The AOS-CX REST API Reference shows most of the supported read and write methods for all switch resources.

In-band and out-of-band management

Management communications with a managed switch can be either of the following:

In band

In-band management communications occur through ports on the line modules of the switch, using common communications protocols such as SSH and SNMP.

When you use an in-band management connection, management traffic from that connection uses the same network infrastructure as user data. User data uses the `data plane`, which is responsible for moving data from source to destination. Management traffic that uses the data plane is more likely to be affected by traffic congestion and other issues affecting the user network.

Out of band

OOBM (out-of-band management) communications occur through a dedicated serial or USB console port or though a dedicated networked management port.

OOBM operates on a `management plane` that is separate from the `data plane` used by data traffic on the switch and by in-band management traffic. That separation means that OOBM can continue to function even during periods of traffic congestion, equipment malfunction, or attacks on the network. In addition, it can provide improved switch security: a properly configured switch can limit management access to the management port only, preventing malicious attempts to gain access through the data ports.

Networked OOBM typically occurs on a management network that connects multiple switches. It has the added advantage that it can be done from a central location and does not require an individual physical cable from the management station to the console port of each switch.

SNMP-based management support

The ArubaOS-CX operating system provides SNMP read access to the switch. SNMP support includes support of industry-standard MIB (Management Information Base) plus private extensions, including SNMP events, alarms, history, statistics groups, and a private alarm extension group. SNMP access is disabled by default.

User accounts

To view or change configuration settings on the switch, users must log in with a valid account. Authentication of user accounts can be performed locally on the switch, or by using the services of an external TACACS+ or RADIUS server.

Two types of user accounts are supported:

- Operators: Operators can view configuration settings, but cannot change them. No operator accounts are created by default.
- Administrators: Administrators can view and change configuration settings. A default locally stored administrator account is created with username set to **admin** and no password. You set the administrator account password as part of the initial configuration procedure for the switch.

Perform the initial configuration of a factory default switch using one of the following methods:

- Load a switch configuration using zero-touch provisioning (ZTP). When ZTP is used, the configuration is loaded from a server automatically when the switch booted from the factory default configuration.
- Connect to the switch wirelessly with a mobile device through Bluetooth, and use the Aruba CX Mobile App to deploy an initial configuration from a provided template. The template you choose during the deployment process determines how the management interface is configured. Optionally, as the final deployment step, you can select to import the switch into NetEdit through a WiFI connection to the NetEdit server.

  Alternatively, you can use the Aruba CX Mobile App to manually configure switch settings and features for a subset of the features you can configure using the CLI. You can also access the CLI through the mobile application.
- Connect the management port on the switch to your network, and then use SSH client software to reach the switch from a computer connected to the same network. This requires that a DHCP server is installed on the network. Configure switch settings and features by executing CLI commands.
- Connect a computer running terminal emulation software to the console port on the switch. Configure switch settings and features by executing CLI commands.

Initial configuration using ZTP

Zero Touch Provisioning (ZTP) configures a switch automatically from a remote server.

Prerequisites

- The switch must be in the factory default configuration.

  Do not change the configuration of the switch from its factory default configuration in any way, including by setting the administrator password.
- Your network administrator or installation site coordinator must provide a Category 6 (Cat6) cable connected to the network that provides access to the servers used for Zero Touch Provisioning (ZTP) operations.

Procedure

1. Connect the network cable to the out-of-band management port on the switch.

   See the *Installation Guide* for switch to determine the location of the switch ports.
2. If the switch is powered on, power off the switch.
3. Power on the switch. A switch running the factory default configuration attempts the ZTP operation for the first 10 minutes after the switch is powered on. During the ZTP operation, the switch might reboot if a new firmware image is being installed.

Initial configuration using the Aruba CX mobile app

This procedure describes how to use your mobile device to connect to the Bluetooth interface of the switch to connect to the switch for the first time so that you can configure basic operational settings using the Aruba CX mobile app.

Prerequisites

- You have obtained the USB Bluetooth adapter that was shipped with the switch. Information about the make and model of the supported adapter is included in the information about the Aruba CX mobile app in the Apple Store or Google Play.
- The Aruba CX mobile app must be installed on your mobile device.
- Bluetooth must be enabled on your mobile device.
- Your mobile device must be within the communication range of the Bluetooth adapter.
- If you are planning to import the switch into NetEdit, your mobile device must be able to use a Wi-Fi connection—not Bluetooth—to access the NetEdit server.

  If your mobile device does not support simultaneous Bluetooth and Wi-Fi connections, you must use the NetEdit interface to import the switch at a later time. You can use the **Devices** tab to display the IP address of the switches you configured using your mobile device.
- The switch must be installed and powered on, with the network operating system boot sequence complete.

  For information about installing and powering on the switch, see the *Installation Guide* for the switch.

  Because you are using this mobile application to configure the switch through the Bluetooth interface, it is not necessary to connect a console to the switch.
- Bluetooth and USB must be enabled on the switch. On switches shipped from the factory, Bluetooth and USB are enabled by default.

Procedure

1. Install the USB Bluetooth adapter in the USB port of the switch.

   For switches that have multiple management modules, you must install the USB Bluetooth adapter in the USB port of the active management module. Typically, the active management module is the module in slot 5.

   Switches shipped from the factory have both USB and Bluetooth enabled by default.

   For information about the location of the USB port on the switch, see the *Installation Guide* for the switch.
2. Use the Bluetooth settings on your mobile device to pair and connect the switch to your mobile device.

   If you are in range of multiple Bluetooth devices, more than one device is displayed on the list of available devices. Switches running the AOS-CX operating system are displayed in the following format:

   `Switch_model``-`*`Serial_number`*

   For example: `8325-987654X1234567` or `8320-AB12CDE123`

   A switch supports one active Bluetooth connection at a time.

   On some Android devices, you might need to change the settings of the paired device to specify that it be used for Internet access.
3. Open the Aruba CX mobile app on your mobile device.

The application attempts to connect to the switch using the switch Bluetooth IP address and the default switch login credentials. The **Home** screen of the application shows the status of the connection to the switch:

- If the login attempt was successful, the Bluetooth icon is displayed and the status message shows the Bluetooth IP address of the switch. In addition, the connection graphic is green. You can continue to the next step.
- If the login attempt was not successful, but a response was received, the Bluetooth icon is displayed, but the status message is: `Login Required`.

  You can continue to the next step. When you tap one of the tiles, you will be prompted for login credentials.
- If the login attempt did not receive a response, the Bluetooth icon is not displayed, and the status message is: `No Connection`.

4. Create the initial switch configuration:

- You can deploy an initial configuration to the switch. Through this process, you supply the information required by a configuration template that you choose from a list of templates provided by the application. Then you deploy the configuration to the switch and, optionally, import the switch into NetEdit.

  CAUTION: When you deploy a switch configuration, it becomes the running configuration, replacing the entire existing configuration of the switch. All changes previously made to the factory default configuration are overwritten.

  If you plan to both deploy a switch configuration and customize the configuration of switch features, deploy the initial configuration first.

  To deploy an initial switch configuration, tap: **Initial Config** and follow the instructions in the application.
- Alternatively, you can complete the initial configuration of the switch by tapping **Modify Config** and then selecting the features and settings to configure.
- You can also use the Modify Config feature to configure some switch features after the initial configuration is complete. For more information about what you can configure using the Aruba CX mobile app, see the online help for the application.

Initial configuration using the CLI

This procedure describes how to connect to the switch for the first time and configure basic operational settings using the CLI. In this procedure, you use a computer to connect to the switch using the either the console port or management port.

Procedure

1. Connect to the **console port** or the **management port**.
2. **Log into the switch for the first time**.
3. **Configure switch time using the NTP client**.

Connecting to the console port

Prerequisites

- A switch installed as described in its hardware installation guide.
- A computer with terminal emulation software.
- A JL448A Aruba X2 C2 RJ45 to DB9 console cable.

Procedure

1. Connect the console port on the switch to the serial port on the computer using a console cable.
2. Start the terminal emulation software on the computer and configure a new serial session with the following settings:
    - Speed: 115200 bps
    - Data bits: 8
    - Stop bits: 1
    - Parity: None
    - Flow control: None
3. Start the terminal emulation session.
4. Press **`Enter`** once. If the connection is successful, you are prompted to login.

Connecting to the management port

Prerequisites

- Two Ethernet cables
- SSH client software

Procedure

1. By default, the management interface is set to automatically obtain an IP address from a DHCP server, and SSH support is enabled. If there is no DHCP server on your network, you must configure a static address on the management interface:
    a. Connect to the **console port**.
    b. Configure the **management interface**.
2. Use an Ethernet cable to connect the management port to your network.

3. Use an Ethernet cable to connect your computer to the same network.
4. Start your SSH client software and configure a new session using the address assigned to the management interface. (If the management interface is set to operate as a DHCP client, retrieve the IP address assigned to the management interface from your DHCP server.)
5. Start the session. If the connection is successful, you are prompted to log in.

Logging into the switch for the first time

The first time you log in to the switch you must use the default administrator account. This account has no password, so you will be prompted on login to define one to safeguard the switch.

Procedure

1. When prompted to log in, specify **`admin`**. When prompted for the password, press **ENTER**. (By default, no password is defined.)

   For example:

   ```
   switch login: admin
   password:
   ```

2. Define a password for the **`admin`** account. The password can contain up to 32 alphanumeric characters in the range ASCII 32 to 127, which includes special characters such as asterisk (*), ampersand (&), exclamation point (!), dash (-), underscore (_), and question mark (?).

   For example:

   ```
   Please configure the 'admin' user account password.
   Enter new password: *******
   Confirm new password: *******
   switch#
   ```

3. You are placed into the manager command context, which is identified by the prompt: `switch#`, where `switch` is the model number of the switch. Enter the command `config` to change to the global configuration context `config`.

   For example:

   ```
   switch# config
   switch(config)#
   ```

Setting switch time using the NTP client

Prerequisites

- The IP address or domain name of an NTP server.
- If the NTP server uses authentication, obtain the password required to communicate with the NTP server.

Procedure

1. If the NTP server requires authentication, define the authentication key for the NTP client with the command `ntp authentication`.
2. Configure an NTP server with the command `ntp server`.

3. By default, NTP traffic is sent on the default VRF. If you want to send NTP traffic on the management VRF, use the command `ntp vrf`.
4. Review your NTP configuration settings with the commands `show ntp servers` and `show ntp status`.
5. See the current switch time, date, and time zone with the command `show clock`.

Example

This example creates the following configuration:

- Defines the authentication key **1** with the password **myPassword**.
- Defines the NTP server **my-ntp.mydomain.com** and makes it the preferred server.
- Sets the switch to use the management VRF (**mgmt**) for all NTP traffic.

```
switch(config)# ntp authentication-key 1 md5 myPassword
switch(config)# ntp server my-ntp.mydomain.com key 10 prefer
switch(config)# ntp vrf mgmt
```

Configuring banners

Procedure

1. Configure the banner that is displayed when a user connects to a management interface. Use the command `banner motd`. For example:

```
switch(config)# banner motd ^
Enter a new banner. Terminate the banner with the delimiter you have chosen.
>> This is an example of a banner text which a connecting user
>> will see before they are prompted for their password.
>>
>> As you can see it may span multiple lines and the input
>> will be terminated when the delimiter character is
>> encountered.^
Banner updated successfully!
```

2. Configure the banner that is displayed after a user is authenticated. Use the command `banner exec`. For example:

```
switch(config)# banner exec &
Enter a new banner. Terminate the banner with the delimiter you have chosen.
>> This is an example of a different banner text. This time
>> the banner entered will be displayed after a user has
>> authenticated.
>>
>> & This text will not be included because it comes after the '&'
Banner updated successfully!
```

Configuring in-band management on a data port

Prerequisites

- A connection to the CLI via either the console port or the management port
- Ethernet cable

Procedure

1. Use an Ethernet cable to connect a data port to your network.
2. **Configure a layer 3 interface** on the data port.
3. Enable SSH support on the interface (on the default VRF) with the command `ssh server vrf default`.

   For example:

   ```
   switch# config
   switch(config)# ssh server vrf default
   ```
4. Enable the Web UI on the interface (on the default VRF) with the command `https-server vrf default`.

   For example:

   ```
   switch(config)# https-server vrf default
   ```

Using the Web UI

The Web UI is disabled by default. Follow these steps to enable it on the management port and log in.

Prerequisites

A connection to the switch CLI.

Procedure

1. Log in to the CLI.
2. Switch to `config` context and enable the Web UI on the management port VRF with the command `https-server vrf mgmt`.

   For example:

   ```
   switch# config
   switch(config)# https-server vrf mgmt
   ```
3. Start your web browser and enter the IP address of the management port in the address bar,

   For example: **`https://192.168.1.1`**
4. The Web UI starts and you are prompted to log in.

Configuring the management interface

Prerequisites

A connection to the console port.

Procedure

1. Switch to the management interface context with the command `interface mgmt`.
2. By default, the management interface on the management port is enabled. If it was disabled, reenable it with the command `no shutdown`.
3. Use the command `ip dhcp` to configure the management interface to automatically obtain an address from a DHCP server on the network (factory default setting). Or, assign a static IPv4 or IPv6 address, default gateway, and DNS server with the commands `ip static`, `default-gateway`, and `nameserver`.
4. SSH is enabled by default on the management VRF. If disabled, enable SSH with the command `ssh server vrf mgmt`.

Examples

This example enables the management interface with dynamic addressing using DHCP:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip dhcp
```

This example enables the management interface with static addressing creating the following configuration:

- Sets a static IPv4 address of **198.168.100.10** with a mask of **24** bits.
- Sets the default gateway to **198.168.100.200**.
- Sets the DNS server to **198.168.100.201**.

```
switch(config)# interface mgmt
switch(config-if-mgmt)# no shutdown
switch(config-if-mgmt)# ip static 198.168.100.10/24
switch(config-if-mgmt)# default-gateway 198.168.100.200
switch(config-if-mgmt)# nameserver 198.168.100.201
```

Configuring the hardware forwarding table

Procedure

1. Set the hardware forwarding table mode with the command `profile`.

   NOTE: The hardware forwarding table profile setting is saved separately and cannot be recovered with a checkpoint restore or configuration download.

2. Reboot the switch for the mode change to take effect with the command `boot system`.

Examples

Optimizing the hardware forwarding table mode for layer 2 forwarding (aggregation layer) on an 8320 switch:

```
switch# config
switch(config)# profile l3-agg
switch(config)# exit
switch# boot system
```

Optimizing the hardware forwarding table mode for layer 3 forwarding (core layer) on an 8320 switch:

```
switch# config
switch(config)# profile l3-core
switch(config)# exit
switch# boot system
```

Restoring the switch to factory default settings

Prerequisites

You are connected to the switch through its Console port.

Procedure

NOTE: This procedure erases all user information and configuration settings Consider backing up your running configuration first.

1. Optionally, back up the running configuration with either `copy running-config <REMOTE-URL>` or `copy running-config <STORAGE-URL>`. The `json` storage format is required for later configuration restoration.
2. Switch to the configuration context with the command `config`.
3. Erase all user information and configuration, restoring the switch to its factory default state with the command `erase all zeroize`. Enter `Y` when prompted to continue. The switch automatically restarts.
4. Optionally restore your saved configuration (it must be in `json` format) with either `copy <REMOTE-URL> running-config` or `copy <STORAGE-URL> running-config` followed by `copy running-config startup-config`.

Example

Backing up the running configuration to a file on a remote server (using TFTP), resetting the switch to its factory default state, and then restoring the saved configuration.

```
switch# copy running-config tftp://192.168.1.10/backup_cfg json vrf mgmt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 10340    0      0  100 10340      0  1329k --:--:-- --:--:-- --:--:-- 1329k
100 10340    0      0  100 10340      0  1313k --:--:-- --:--:-- --:--:-- 1313k
switch#
switch#
switch# erase all zeroize
This will securely erase all customer data and reset the switch
to factory defaults. This will initiate a reboot and render the
switch unavailable until the zeroization is complete.
```

```
This should take several minutes to one hour to complete.
Continue (y/n)? y

The system is going down for zeroization.
[  OK  ] Stopped PSPO Module Daemon.
[  OK  ] Stopped ArubaOS-CX Switch Daemon for BCM.
...
[  OK  ] Stopped Remount Root and Kernel File Systems.
[  OK  ] Reached target Shutdown.
reboot: Restarting system
Press Esc for boot options
ServiceOS Information:
    Version:          GT.01.03.0006
    Build Date:       2018-10-30 14:20:44 PDT
    Build ID:         ServiceOS:GT.01.03.0006:8ee0faaa52da:201810301420
    SHA:              xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
...

################ Preparing for zeroization #################

################ Storage zeroization ######################
################ WARNING: DO NOT POWER OFF UNTIL  ##########
################          ZEROIZATION IS COMPLETE ##########
################ This should take several minutes ##########
################ to one hour to complete          ##########

################ Restoring files ###########################

Boot Profiles:

0. Service OS Console
1. Primary Software Image [XL.10.02.0010]
2. Secondary Software Image [XL.10.02.0010]

Select profile(primary):

Booting primary software image...
Verifying Image...

Image Info:
        Name: ArubaOS-CX
     Version: XL.10.02.0010
    Build Id: ArubaOS-CX:XL.10.02.0010:feaf5b9b7f09:201901292014
  Build Date: 2019-01-29 12:43:50 PST

Extracting Image...
Loading Image...
Done.
kexec_core: Starting new kernel
System is initializing
fips_post_check[5473]: FIPS_POST: Cryptographic selftest started...SUCCESS
[  OK  ] Started Login banner readiness check.
...
8400X login: admin
Password:

switch#
switch#
switch# copy tftp://192.168.1.10/backup_cfg running-config json vrf mgmt
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 10340  100 10340    0     0  2858k      0 --:--:-- --:--:-- --:--:-- 2858k
100 10340  100 10340    0     0  2804k      0 --:--:-- --:--:-- --:--:-- 2804k
Large configuration changes will take time to process, please be patient.
```

```
switch#
switch#
switch# copy running-config startup-config
Large configuration changes will take time to process, please be patient.
switch#
```

Management interface commands

default-gateway

Syntax

`default-gateway <IP-ADDR>`

`no default-gateway <IP-ADDR>`

Description

Assigns an IPv4 or IPv6 default gateway to the management interface. An IPv4 default gateway can only be configured if a static IPv4 address was assigned to the management interface. An IPv6 default gateway can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The `no` form of this command removes the default gateway from the management interface.

Command context

`config-if-mgmt`

Parameters

`<IP-ADDR>`

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting a default gateway with the IPv4 address of **`198.168.5.1`**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 198.168.5.1
```

Setting an IPv6 address of **`2001:DB8::1`**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# default-gateway 2001:DB8::1
```

ip static

Syntax

`ip static <IP-ADDR>/<MASK>`

`no ip static <IP-ADDR>/<MASK>`

Description

Assigns an IPv4 or IPv6 address to the management interface.

The `no` form of this command removes the IP address from the management interface and sets the interface to operate as a DHCP client.

Command context

`config-if-mgmt`

Parameters

<IP-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

<MASK>

Specifies the number of bits in an IPv4 or IPv6 address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 32 for IPv4, and 0 to 128 for IPv6.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting an IPv4 address of **`198.51.100.1`** with a mask of **`24`** bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 198.51.100.1/24
```

Setting an IPv6 address of **`2001:DB8::1`** with a mask of **`32`** bits:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# ip static 2001:DB8::1/32
```

nameserver

Syntax

```
nameserver <PRIMARY-IP-ADDR> [ <SECONDARY-IP-ADDR> ]

no nameserver <PRIMARY-IP-ADDR> [ <SECONDARY-IP-ADDR> ]
```

Description

Assigns a primary or secondary IPv4 or IPv6 DNS server to the management interface. IPv4 DNS servers can only be configured if a static IPv4 address was assigned to the management interface. IPv6 DNS servers can only be configured if a static IPv6 address was assigned to the management interface. The default gateway should be on the same network segment.

The `no` form of this command removes the DNS servers from the management interface.

Command context

`config-if-mgmt`

Parameters

<PRIMARY-IP-ADDR>

Specifies the IP address of the primary DNS server. Specify the address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

<SECONDARY-IP-ADDR>

Specifies the IP address of the secondary DNS server. Specify the address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting primary and secondary DNS servers with the IPv4 addresses of **198.168.5.1** and **198.168.5.2** :

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 198.168.5.1 198.168.5.2
```

Setting primary and secondary DNS servers with the IPv6 addresses of **2001:DB8::1** and **2001:DB8::2**:

```
switch(config)# interface mgmt
switch(config-if-mgmt)# nameserver 2001:DB8::1 2001:DB8::2
```

show interface mgmt

Syntax

```
show interface mgmt [vsx-peer]
```

Description

Shows status and configuration information for the management interface.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show interface mgmt

  Address Mode                    : static
  Admin State                     : up
  Mac Address                     : 02:42:ac:11:00:02
```

```
  IPv4 address/subnet-mask       : 192.168.1.10/16
  Default gateway IPv4           : 192.168.1.1
  IPv6 address/prefix            : 2001:db8:0:1::129/64
  IPv6 link local address/prefix: fe80::7272:cfff:fefd:e485/64
  Default gateway IPv6           : 2001:db8:0:1::1
  Primary Nameserver             : 2001::1
  Secondary Nameserver           : 2001::2
```

NTP commands

ntp authentication

Syntax

`ntp authentication`

`no ntp authentication`

Description

Enables support for authentication when communicating with an NTP server.

The `no` form of this command disables authentication support.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling authentication support:

```
switch(config)# ntp authentication
```

Disabling authentication support:

```
switch(config)# no ntp authentication
```

ntp authentication-key

Syntax

`ntp authentication-key <KEY-ID> {md5 | sha1} <PASSWORD> [trusted]`

`no ntp authentication-key <KEY-ID>`

Description

Defines an authentication key that is used to secure the exchange with an NTP time server. This command provides protection against accidentally synchronizing to a time source that is not trusted.

The `no` form of this command removes an authentication key.

Command context

`config`

Parameters

authentication-key *<KEY-ID>*

Specifies an identification number to uniquely identify the key. Range: 1 to 65534.

md5

Specifies md5 encryption to protect the key.

sh1

Specifies sh1 encryption to protect the key

<PASSWORD>

Specifies the md5 or sh1 password associated with the key. Range: 8 to 16 characters.

trusted

Specifies that this is a trusted key. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining key 10 with the **md5** password **myPassword** as a trusted key:

```
switch(config)# ntp authentication-key 10 md5 myPassword trusted
```

Removing key 10:

```
switch(config)# no ntp authentication-key 10
```

ntp disable

Syntax

`ntp disable`

Description

Disables the NTP client on the switch. The NTP client is disabled by default.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling the NTP client.

```
switch(config)# ntp disable
```

ntp enable

Syntax

```
ntp enable

no ntp enable
```

Description

Enables the NTP client on the switch to automatically adjust the local time and date on the switch. The NTP client is disabled by default.

The `no` form of this command disables the NTP client.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling the NTP client.

```
switch(config)# ntp enable
```

Disabling the NTP client.

```
switch(config)# no ntp enable
```

ntp master

Syntax

```
ntp master vrf <VRF-NAME> [stratum <NUMBER>]

no ntp master vrf <VRF-NAME>
```

Description

Sets the switch as the master time source for NTP clients on the specified VRF. By default, the switch operates at stratum level 8. The switch cannot function as both NTP master and client on the same VRF.

The `no` form of this command stops the switch from operating as the master time source on the specified VRF.

Command context

`config`

Parameters

`vrf <VRF-NAME>`

Specifies the VRF on which to act as master time source.

`stratum <NUMBER>`

Specifies the stratum level at which the switch operates. Range: 1 - 15. Default: 8.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the switch to act as master time source on VRF **primary-vrf** with a stratum level of **9**.

```
switch(config)# ntp master vrf primary-vry statum 9
```

Stops the switch from acting as master time source on VRF **primary-vrf**.

```
switch(config)# no ntp master vrf primary-vry
```

ntp server

Syntax

```
ntp server <IP-ADDR> [key <KEY-NUM>] [minpoll <MIN-NUM>] [maxpoll <MAX-NUM>][burst | iburst] [prefer] [version <VER-NUM>]

no ntp server <IP-ADDR>

no ntp server <IP-ADDR> [burst] [iburst] [prefer] [key-id <KEY-NUM>]
```

Description

Defines an NTP server to use for time synchronization, or updates the settings of an existing server with new values. Up to eight servers can be defined.

The `no` form of this command removes a configured NTP server.

Command context

`config`

Parameters

server *<IP-ADDR>*

Specifies the address of an NTP server as a DNS name, an IPv4 address (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or an IPv6 address (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

When specifying an IPv4 address, you can remove leading zeros. For example, the address `192.169.005.100` becomes `192.168.5.100`.

When specifying an IPv6 address, you can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address `2222:0000:3333:0000:0000:0000:4444:0055` becomes `2222:0:3333::4444:55` .

key *<KEY-NUM>*

Specifies the key to use when communicating with the server. A trusted key must be defined with the command `ntp authentication-key` and authentication must be enabled with the command `ntp authentication`. Range: 1 to 65534.

minpoll *<MIN-NUM>*

Specifies the minimum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 6 (64 seconds).

maxpoll *<MAX-NUM>*

Specifies the maximum polling interval in seconds, as a power of 2. Range: 4 to 17. Default: 10 (1024 seconds).

burst

Send a burst of packets instead of just one when connected to the server. Useful for reducing phase noise when the polling interval is long.

iburst

Send a burst of six packets when not connected to the server. Useful for reducing synchronization time at startup . Range: 1 to 4094.

prefer

Make this the preferred server.

version *<VER-NUM>*

Specifies the version number to use for all outgoing NTP packets. Range: 3 or 4.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining the ntp server pool.ntp.org, using iburst, and NTP version 4.

```
switch(config)# ntp server pool.ntp.org iburst version 4
```

Removing the ntp server pool.ntp.org.

```
switch(config)# no ntp server pool.ntp.org
```

Defining the ntp server my-ntp.mydomain.com and makes it the preferred server.

```
switch(config)# ntp server my-ntp.mydomain.com prefer
```

ntp trusted-key

Syntax

```
ntp trusted-key <KEY-ID>

no ntp trusted-key <KEY-ID>
```

Description

Sets a key as trusted. When NTP authentication is enabled, the switch only synchronizes with time servers that transmit packets containing a trusted key.

The `no` form of this command removes the trusted designation from a key.

Command context

`config`

Parameters

<KEY-ID>

Specifies the identification number of the key to set as trusted. Range: 1 to 65534.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining key 10 as a trusted key.

```
switch(config)# ntp trusted-key 10
```

Removing trusted designation from key 10:

```
switch(config)# no ntp trusted-key 10
```

ntp vrf

Syntax

```
ntp vrf <VRF-NAME>
```

Description

Specifies the VRF on which the NTP client communicates with an NTP server. The switch cannot function as both NTP master and client on the same VRF.

Command context

`config`

Parameters

<VRF-NAME>

Specifies the name of a VRF.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the switch to use the default VRF for NTP client traffic.

```
switch(config)# ntp vrf default
```

Setting the switch to use the default management VRF for NTP client traffic.

```
switch(config)# ntp vrf mgmt
```

show ntp associations

Syntax

```
show ntp associations [vsx-peer]
```

Description

Shows the status of the connection to each NTP server. The following information is displayed for each server:

- Tally code : The first character is the Tally code:
  - (blank): No state information available (e.g. non-responding server)
  - x : Out of tolerance (discarded by intersection algorithm)
  - . : Discarded by table overflow (not used)

    - - : Out of tolerance (discarded by the cluster algorithm)
    - + : Good and a preferred remote peer or server (included by the combine algorithm)
    - # : Good remote peer or server, but not utilized (ready as a backup source)
    - * : Remote peer or server presently used as a primary reference
    - o : PPS peer (when the prefer peer is valid)
- ID: Server number.
- NAME: NTP server FQDN/IP address (Only the first 24 characters of the name are displayed).
- REMOTE: Remote server IP address.
- REF_ID: Reference ID for the remote server (Can be an IP address).
- ST: (Stratum) Number of hops between the NTP client and the reference clock.
- LAST: Time since the last packet was received in seconds unless another unit is indicated.
- POLL: Interval (in seconds) between NTP poll packets. Maximum (1024) reached as server and client sync.
- REACH: 8-bit octal number that displays status of the last eight NTP messages (377 = all messages received).

Command context

Manager (#)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ntp associations
----------------------------------------------------------------------
 ID             NAME           REMOTE          REF-ID ST LAST POLL REACH
----------------------------------------------------------------------
  1       192.0.1.1       192.0.1.1         .INIT. 16    -   64     0
* 2  time.apple.com    17.253.2.253         .GPSs.  2   70  128   377
----------------------------------------------------------------------
```

show ntp authentication-keys

Syntax

```
show ntp authentication-keys [vsx-peer]
```

Description

Shows the currently defined authentication keys.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show ntp authentication-keys
--------------------------------
Auth key  Trusted   MD5 password
--------------------------------
 10          No     **********
 20         Yes     **********
```

show ntp servers

Syntax

```
show ntp servers [vsx-peer]
```

Description

Shows all configured NTP servers.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ntp servers
------------------------------------------------
     NTP SERVER KEYID MINPOLL MAXPOLL OPTION VER
------------------------------------------------
     192.0.1.18     -       5      10 iburst   3
     192.0.1.19     -       6      10   none   4
     192.0.1.20     -       6       8  burst   3 prefer
------------------------------------------------
```

show ntp statistics

Syntax

```
show ntp statistics [vsx-peer]
```

Description

Shows global NTP statistics. The following information is displayed:

- Rx-pkts: Total NTP packets received.
- Current Version Rx-pkts: Number of NTP packets that match the current NTP version.
- Old Version Rx-pkts: Number of NTP packets that match the previous NTP version.
- Error pkts: Packets dropped due to all other error reasons.
- Auth-failed pkts: Packets dropped due to authentication failure.
- Declined pkts: Packets denied access for any reason.
- Restricted pkts: Packets dropped due to NTP access control.
- Rate-limited pkts: Number of packets discarded due to rate limitation.
- KOD pkts: Number of Kiss of Death packets sent.

Command context

```
config
```

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch(config)# show ntp statistics
                Rx-pkts 100
Current Version Rx-pkts 80
    Old Version Rx-pkts 20
               Err-pkts 2
       Auth-failed-pkts 1
          Declined-pkts 0
        Restricted-pkts 0
      Rate-limited-pkts 0
               KoD-pkts 0
```

show ntp status

Syntax

```
show ntp status [vsx-peer]
```

Description

Shows the status of NTP on the switch.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying the status information when the switch is not synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Not synchronized with an NTP server.
```

Displaying the status information when the switch is synced to an NTP server:

```
switch# show ntp status
NTP is enabled.
NTP authentication is enabled.
NTP is using the default VRF for NTP server connections.

Wed Nov 23 23:29:10 PDT 2016
NTP uptime: 187 days, 1 hours, 37 minutes, 48 seconds

Synchronized to NTP Server 17.253.2.253 at stratum 2.
Poll interval = 1024 seconds.
Time accuracy is within 0.994 seconds
Reference time: Thu Jan 28 2016 0:57:06.647 (UTC)
```

Hardware forwarding table commands

profile

Syntax

8320 switch series

`profile {l3-agg | l3-core | leaf}`

8325 switch series

`profile {l3-agg | l3-core | leaf | spine}`

Description

Sets the hardware forwarding table profile on an 8320 and 8325 switch series.

NOTE: The switch must be rebooted for a mode change to take effect.

NOTE: For the 8320 switch series, prior to release 10.2, the forwarding table mode was configured with the command `platform forwarding-table-mode {3 | 4}`. When upgrading to release 10.02, any existing configuration is converted as follows: table mode `3` is converted to `l3-agg` and table mode `4` is converted to `l3-core`.

Command context

Manager (`#`)

Parameters

l3-agg

Optimizes the hardware forwarding mode for layer 2 forwarding with more table space allocated to host(ARP/ND) entries.

l3-core

Optimizes the hardware forwarding mode for layer 3 forwarding with more table space allocated to route entries. (Default on the 8320 switch series.)

leaf

Optimizes the hardware forwarding mode for layer 2 forwarding with more table space allocated to overlay host entries (VXLAN). (Default on the 8325 switch series.)

spine

Optimizes the hardware forwarding mode for layer 3 forwarding with more table space allocated to route entries. (8325 switch series only.)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Optimizing the hardware forwarding table mode for layer 2 forwarding (aggregation layer):

```
switch# config
switch(config)# profile l3-agg
switch(config)# exit
switch# boot system
```

Optimizing the hardware forwarding table mode for layer 3 forwarding (core layer):

```
switch# config
switch(config)# profile l3-core
switch(config)# exit
switch# boot system
```

show profiles available

Syntax

`show profiles available`

Description

Shows all available profiles for the 8320, 8325, and 8360.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all available profiles for an 8320 switch:

```
switch# show profiles available

Available profiles
-------------------
L3-agg   98304 L2 entries 120000 Host entries 16384  Route entries

L3-core  32768 L2 entries 14000  Host entries 131064 Route entries (Default)

Leaf     98304 L2 entries 120000 Host entries 16384  Route entries

Spine    32768 L2 entries 14000 Host entries 131064 Route entries
```

Showing all available profiles for an 8325 switch:

```
switch# show profiles available

Available profiles
-------------------
L3-agg   98304 L2 entries, 120000 Host entries (8190 unique overlay
          neighbors, 48638 unique underlay neighbors), 29696 Route entries

L3-core  32768 L2 entries, 28000 Host entries (12286 unique overlay
          neighbors, 32766 unique underlay neighbors), 163796 Route entries

Leaf     98304 L2 entries, 120000 Host entries (32766 unique overlay
          neighbors, 12286 unique underlay neighbors), 29696 Route entries
          (Default)

Spine    32768 L2 entries, 28000 Host entries (12286 unique overlay
          neighbors, 32766 unique underlay neighbors), 163796 Route entries
```

Showing all available profiles for an 8360 switch:

```
switch# show profiles available

Available profiles
-------------------
Aggregation-Leaf   114688 L2 entries, 163840 Host entries, 65536 Route entries

Core-Spine         32768 L2 entries, 65536 Host entries, 630784 Route entries
```

```
Leaf-Extended      212992 L2 entries, 16384 Host entries, 65536 Route entries
```

show profile current

Syntax

`show profile current`

Description

Shows current profile for 8320 and 8325 switch series.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing current profile for an 8320 switch:

```
switch# show profile current

Current profile
-------------------
L3-core
```

Configuring a layer 2 interface

Procedure

1. Change to the interface configuration context for the interface with the command `interface`.
2. By default, interfaces are layer 3. To create a layer 2 interface, disable routing with the command `no routing`.
3. Set the interface MTU (maximum transmission unit) with the command `mtu`.
4. Review interface configuration settings with the command `show interface`.

Example

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# mtu 1900
```

Configuring a layer 3 interface

Procedure

1. Change to the interface configuration context for the interface with the command `interface`.
2. Interfaces are layer 3 by default. If you previously set the interface to layer 2, then enable routing support with the command `routing`.
3. Assign an IPv4 address with the command `ip address`, or an IPv6 address with the command `ipv6 address`.
4. If required, enable support for layer 3 counters with the command `l3-counters`.
5. If required, set the IP MTU with the command `ip mtu`.
6. Review interface configuration settings with the command `show interface`.

Examples

This example creates the following configuration:

- Configures interface **1/1/1** as a layer 3 interface.
- Defines an IPv4 address of **10.10.20.209** with a 24-bit mask.

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# ip address 10.10.20.209/24
```

This example creates the following configuration:

- Configures interface **1/1/2** as a layer 3 interface.
- Defines an IPv6 address of **2001:0db8:85a3::8a2e:0370:7334** with a 24-bit mask.
- Enables layer 3 transmit and receive counters.

```
switch# config
switch(config)# interface 1/1/2
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
switch(config-if)# l3-counters tx
switch(config-if)# l3-counters rx
```

Single source IP address

Certain IP-based protocols used by the switch (such as RADIUS, sFlow, TACACS, and TFTP), use a client-server model in which the client's source IP address uniquely identifies the client in packets sent to the server. By default, the source IP address is defined as the IP address of the outgoing switch interface on which the client is communicating with the server. Since the switch can have multiple routing interfaces, outgoing packets can potentially be sent on different paths at different times. This can result in different source IP addresses being used for a client, which can create a client identification problem on the server. For example, it can be difficult to interpret system logs and accounting data on the server when the same client is associated with multiple IP addresses.

To resolve this issue, you can use the commands `ip source-interface` and `ipv6 source-interface` to define a single source IP address that applies to all supported protocols (RADIUS, sFlow, TACACS, and TFTP), or an individual address for each protocol. This ensures that all traffic sent by a client to a server uses the same IP address.

Unsupported transceiver support

Transceiver products (optical, DAC, AOCs) that are listed as supported by a switch model are detailed in the *Transceiver Guide*. Transceiver products that are not listed, are considered unsupported; this would include transceivers that are:

- Non-Aruba branded products
- HPE branded products that were designed for non-ArubaOS-CX switch models (e.g. Comware)
- HPE branded products designated for use in HPE Compute Servers or Storage
- Transceivers originally designated for use in Aruba WLAN controllers or former Mobility Access Switch (MAS) products
- End-of-life Aruba Transceivers

The unsupported transceiver mode (UT-mode) is designed to allow the possible use of these unsupported products. Not all unsupported products can be recognized and enabled; they may be unable to be identified (do not follow the proper MSA standards for identification). These unsupported transceiver products are enabled only on a best-effort basis and there are no guarantees implied for their continued operation.

The feature is disabled by default. A periodic system log will be generated by default at an interval of 24 hours listing the ports on which unsupported transceivers are present. The log interval is configurable and can be disabled by setting the log-interval to `none`.

Interface commands

allow-unsupported-transceiver

Syntax

`allow-unsupported-transceiver [confirm | log-interval {none | <INTERVAL>}]`

`no allow-unsupported-transceiver`

Description

Allows unsupported transceivers to be enabled or establish connections. Only 1G and 10G transceivers are enabled by this command and unsupported transceivers of other speeds will remain disabled.

The `no` form of this command disallows using unsupported transceivers. This is the default.

Command context

`config`

Parameters

`confirm`

Specifies that unsupported transceiver warnings are to be automatically confirmed.

`log-interval none`

Disables unsupported transceiver logging.

`log-interval <INTERVAL>`

Sets the unsupported transceiver logging interval in minutes. Default: 1440 minutes. Range: 1440 to 10080 minutes.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When none of the parameters are specified it will display a warning message to accept the warranty terms. With `confirm` option the warning message is displayed but the user is not prompted to `(y/n)` answering. Warranty terms must be agreed to as part of enablement and the support is on best effort basis.

Examples

Allowing unsupported transceivers with follow-up confirmation:

```
switch(config)# allow-unsupported-transceiver
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.

Do you agree and do you want to continue (y/n)? y
```

Allowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# allow-unsupported-transceiver confirm
Warning: The use of unsupported transceivers, DACs, and AOCs is at your
own risk and may void support and warranty. Please see HPE Warranty terms
and conditions.
```

Configuring unsupported transceiver logging with an interval of every 48 hours:

```
switch(config)# allow-unsupported-transceiver log-interval 2880
```

Disabling unsupported transceiver logging:

```
switch(config)# allow-unsupported-transceiver log-interval none
```

Disallowing unsupported transceivers with follow-up confirmation:

```
switch(config)# no allow-unsupported-transceiver
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow-unsupported-transceiver'
to identify unsupported transceivers, DACs, and AOCs.

Continue (y/n)? y
```

Disallowing unsupported transceivers with confirmation in command syntax:

```
switch(config)# no allow-unsupported-transceiver confirm
Warning: Unsupported transceivers, DACs, and AOCs will be disabled,
which could impact network connectivity. Use 'show allow unsupported-transceiver'
to identify unsupported transceivers, DACs, and AOCs.

switch(config)#
```

default interface

Syntax

`default interface <INTERFACE-ID>`

Description

Sets an interface (or a range of interfaces) to factory default values.

Command context

`config`

Parameters

<INTERFACE-ID>

Specifies the ID of a single interface or range of interfaces. Format: `member/slot/port` or `member/slot/port-member/slot/port` to specify a range.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting an interface:

```
switch(config)# default default interface 1/1/1
```

Resetting an range of interfaces:

```
switch(config)# default default interface 1/1/1-1/1/10
```

description

Syntax

```
description <DESCRIPTION>

no description
```

Description

Associates descriptive information with an interface to help administrators and operators identify the purpose or role of an interface.

The `no` form of this command removes a description from an interface.

Command context

`config-if`

Parameters

<DESCRIPTION>

Specify a description for the interface. Range: 1 to 64 ASCII characters (including space, excluding question mark).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the description for an interface to **`DataLink 01`**:

```
switch(config-if)# description DataLink 01
```

Removing the description for an interface.

```
switch(config-if)# no description
```

flow-control

Syntax

On the 8320:

```
flow-control rx

[no] flow-control rx
```

On the 8325 and 8360:

```
flow-control {rx | priority <PRIORITY>}

[no] flow-control [rx | priority <PRIORITY>]
```

Description

Enables negotiation of receive flow control on the current interface. The switch advertises rx support to the link partner. The final configuration is determined based on the capabilities of both partners.

Priority-based flow control (PFC), on the 8325, takes effect after the configuration is saved to startup-config and the switch is restarted.

Priority-based flow control (PFC), on the 8360, takes effect immediately after the configuration is saved to startup-config. On the JL720A, PFC is not supported on any link speed below 10 Gbps.

The no form disables flow control support on the current interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Parameters

`rx`

Honors received IEEE 802.3x link-level flow control requests.

`priority` ***`<PRIORITY>`***

On the 8325 and 8360, enables priority-based flow control on the current interface and sets the priority number. Range: 0 to 7.

Examples

Enable support for rx flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# flow-control rx
```

Disable support for rx flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# no flow-control rx
```

Enable support for priority flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# flow-control priority 2
```

Disable support for priority flow control:

```
switch(config)# interface 1/1/1
switch(config-if)# no flow-control priority 2
```

Enable support for priority flow control on the 8325:

```
switch(config)# interface 1/1/1
switch(config-if)# no flow-control priority 2

The setting will not be applied until configuration is saved to startup-config and the switch is rebooted.
```

interface

Syntax

`interface <PORT-NUM>`

Description

Switches to the `config-if` context for a physical port. This is where you define the configuration settings for the logical interface associated with the physical port.

Command context

`config`

Parameters

<PORT-NUM>

Specifies a physical port number. Format: `member/slot/port`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring an interface:

```
switch(config)# interface 1/1/1
switch(config-if)#
```

interface loopback

Syntax

`interface loopback <ID>`

`no interface loopback <ID>`

Description

Creates a loopback interface and changes to the `config-loopback-if` context. Loopback interfaces are layer 3.

The `no` form of this command deletes a loopback interface.

Command context

`config`

Parameters

<INSTANCE>

Specifies the loopback interface ID. Range: 1 to 256

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# config
switch(config)# interface loopback 1
switch(config-loopback-if)#
```

interface vlan

Syntax

`interface vlan <VLAN-ID>`

`no interface vlan <VLAN-ID>`

Description

Creates an interface VLAN also know as an SVI (switched virtual interface) and changes to the `config-if-vlan` context. The specified VLAN must already be defined on the switch.

The `no` form of this command deletes an interface VLAN.

Command context

`config`

Parameters

<VLAN-ID>

Specifies the loopback interface ID. Range: 1 to 4040

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# config
switch(config)# vlan 10
switch(config-vlan-10)# exit
switch(config)# interface vlan 10
switch(config-if-vlan)#
```

ip address

Syntax

`ip address <IPV4-ADDR>/<MASK> [secondary]`

`no ip address <IPV4-ADDR>/<MASK> [secondary]`

Description

Sets an IPv4 address for the current layer 3 interface.

The `no` form of this command removes the IPv4 address from the interface.

Command context

`config-if`

`config-loopback-if`

`config-if-vlan`

Parameters

<IPV4-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. You can remove leading zeros. For example, the address `192.169.005.100` becomes `192.168.5.100`.

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128.

secondary

Specifies a secondary IP address.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IP address on interface **1/1/1** to **192.168.100.1** with a mask of **24** bits:

```
switch(config)# interface 1/1/1
switch(config-if)# ip address 192.168.100.1/24
```

Removing the IP address **192.168.100.1** with a mask of **24** bits from interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no ip address 192.168.100.1/24
```

Assigning the IP address **192.168.20.1** with a mask of **24** bits to loopback interface **1**:

```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 192.168.20.1/24
```

Assigning the IP address **192.168.199.1** with a mask of **24** bits to interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-loopback-if)# ip address 192.168.199.1/24
```

Removing the IP address **192.168.199.1** with a mask of **24** bits from interface VLAN **10**:

```
switch(config)# interface vlan 10
switch(config-loopback-if)# no ip address 192.168.199.1/24
```

ip mtu

Syntax

`ip mtu <VALUE>`

`ip no mtu`

Description

Sets the IP MTU (maximum transmission unit) for an interface. This defines the largest IP packet that can be sent or received by the interface.

The `no` form of this command sets the IP MTU to the default value 1500.

Command context

`config-if`

`config-if-vlan`

Parameters

<VALUE>

Specifies the IP MTU in bytes. Range: 68 to 9198. Default: 1500.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IP MTU to 576 bytes:

```
switch(config-if)# ip mtu 576
```

Setting the IP MTU to the default value:

```
switch(config-if)# no ip mtu
```

ip source-interface

Syntax

```
ip source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog | ubt | dhcp-
relay | simplivity | dns | all} {interface <IFNAME> | <IPV4-ADDR>} [vrf <VRF-NAME>]

no ip source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog | ubt |
dhcp-relay | simplivity | dns | all} [interface <IFNAME> | <IPV4-ADDR>] [vrf <VRF-
NAME>]
```

Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The `no` form of this command deletes the single source IP address for all supported services, or a specific service.

Command contexts

`config`

Parameters

sflow | tftp | radius | tacacs | ntp | syslog | ubt | dhcp-relay | simplivity | dns | all

Sets a single source IP address for a specific service. The `all` option sets a global address that applies to all protocols that do not have an address set. For DHCP relay, the address is used as both the source IP and GIADDR.

interface *<IFNAME>*

Specifies the name of the interface from which the specified service obtains its source IP address. The interface must have a valid IP address assigned to it. If the interface has both a primary and secondary IP address, the primary IP address is used.

<IPV4-ADDR>

Specifies the source IP address to use for the specified service. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the `default` VRF, if the `vrf` option is not used). Specify the address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255.

vrf *<VRF-NAME>*

Specifies the name of a VRF.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IPv4 address `10.10.10.5` as the global single source address:

```
switch# config
switch(config)# ip source-interface all 10.10.10.5
```

Setting the secondary IPv4 address `10.10.10.5` on interface `1/1/1` as the global single source address:

```
switch# config
switch(config)# interface 1/1/1
switch(config-if)# ip address 10.10.10.1/24
switch(config-if)# ip address 10.10.10.5/24 secondary
switch(config)# exit
switch(config)# ip source-interface all 10.10.10.5
```

Setting the address `10.10.10.25` on VRF `sflow-vrf` on interface `1/1/2` as the single source address for sFlow:

```
switch(config)# vrf sflow-vrf
switch(config-vrf)# exit
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# vrf attach sflow-vrf
switch(config-if)# ip address 10.10.10.25/24
switch(config-if)# exit
switch(config)# ip source-interface sflow interface 1/1/2 vrf sflow-vrf
```

Clearing the global single source IP address **10.10.10.5**:

```
switch(config)# no ip source-interface all 10.10.10.5
```

ipv6 address

Syntax

`ipv6 address <IPV6-ADDR>/<MASK>{eui64 | [tag <ID>]}`

`no ipv6 address <IPV6-ADDR>/<MASK>`

Description

Sets an IPv6 address on the interface.

The `no` form of this command removes the IPv6 address on the interface.

NOTE: This command automatically creates an IPv6 link-local address on the interface. However, it does not add the `ipv6 address link-local` command to the running configuration. If you remove the IPv6 address, the link-local address is also removed. To maintain the link-local address, you must manually execute the `ipv6 address link-local` command.

Command context

`config-if`

Parameters

<IPV6-ADDR>

Specifies the IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F. You can use two colons (::) to represent consecutive zeros (but only once), remove leading zeros, and collapse a hextet of four zeros to a single 0. For example, this address `2222:0000:3333:0000:0000:0000:4444:0055` becomes `2222:0:3333::4444:55`.

<MASK>

Specifies the number of bits in the address mask in CIDR format (`x`), where `x` is a decimal number from 0 to 128.

`eui64`

Configure the IPv6 address in the EUI-64 bit format.

`tag` ***<ID>***

Configure route tag for connected routes. Range: 0 to 4294967295. Default: 0.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the IPv6 address **`2001:0db8:85a3::8a2e:0370:7334`** with a mask of 24 bits:

```
switch(config-if)# ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

Removing the IP address 2001:0db8:85a3::8a2e:0370:7334 with mask of 24 bits:

```
switch(config-if)# no ipv6 address 2001:0db8:85a3::8a2e:0370:7334/24
```

ipv6 source-interface

Syntax

```
ipv6 source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog | ubt |
dhcp-relay | simplivity | dns | all} {interface <IFNAME> | <IPV6-ADDR>} [vrf <VRF-
NAME>]

no ipv6 source-interface {sflow |  tftp |  radius | tacacs | ntp | syslog | ubt |
dhcp-relay | simplivity | dns | all} [interface <IFNAME> | <IPV6-ADDR>] [vrf <VRF-
NAME>]
```

Description

Sets a single source IP address for a feature on the switch. This ensures that all traffic sent the feature has the same source IP address regardless of how it egresses the switch. You can define a single global address that applies to all supported features, or an individual address for each feature.

This command provides two ways to set the source IP addresses: either by specifying a static IP address, or by using the address assigned to a switch interface. If you define both options, then the static IP address takes precedence.

The `no` form of this command deletes the single source IP address for all supported protocols, or a specific protocol.

Command context

`config`

Parameters

sflow | tftp | radius | tacacs | ntp | syslog | ubt | dhcp-relay | simplivity | dns | all

Sets a single source IP address for a specific protocol. The `all` option sets a global address that applies to all protocols that do not have an address set.

interface *<IFNAME>*

Specifies the name of the interface from which the specified protocol obtains its source IP address.

<IPV6-ADDR>

Specifies the source IP address to use for the specified protocol. The IP address must be defined on the switch, and it must exist on the specified VRF (which is the `default` VRF, if the `vrf` option is not used). Specify the IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

vrf *<VRF-NAME>*

Specifies the name of the VRF from which the specified protocol sets its source IP address.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the IPv6 address `2001:DB8::1` as the global single source address:

```
switch# config
switch(config)# ip source-interface all 2001:DB8::1/32
```

Configuring the IPv6 address `2001:DB8::1` on VRF `sflow-vrf` on interface `1/1/2` as the single source address for sFlow:

```
switch(config)# vrf sflow-vrf
switch(config-vrf)# exit
switch(config)# interface 1/1/2
switch(config-if)# no shutdown
switch(config-if)# vrf attach sflow-vrf
switch(config-if)# ipv6 address 2001:DB8::1/32
switch(config-if)# exit
switch(config)# ip source-interface sflow interface 1/1/2 vrf sflow-vrf
```

Stop the source IP address from using the IP address on interface **1/1/1** on VRF `one`.

```
switch(config)# no ip source-interface all interface 1/1/1 vrf one
```

Clear the source IP address `2001:DB8::1`.

```
switch(config)# no ip source-interface all 2001:DB8::1
```

l3-counters

Syntax

```
l3-counters [rx | tx]

no l3-counters [rx | tx]
```

Description

Enables counters on a layer 3 interface. By default, all interfaces are layer 3. To change a layer 2 interface to layer 3, use the `routing` command.

The `no` form of this command, with no specification, disables both transmit and receive counters on a layer 3 interface. To disable transmit (`tx`) or receive (`rx`) counters only, specify the counter type you want to disable.

Command context

`config-if`

Parameters

`rx`

Specifies receive counters.

`tx`

Specifies transmit counters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling layer 3 transmit counters on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# l3-counters tx
```

Disabling layer 3 transmit and receive counters on interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no l3-counters
```

mtu

Syntax

`mtu <VALUE>`

`no mtu`

Description

Sets the MTU (maximum transmission unit) for an interface. This defines the maximum size of a layer 2 (Ethernet) frame. Frames larger than the MTU (1500 bytes by default) are dropped and cause an ICMP fragmentation-needed message to be sent back to the originator.

To support jumbo frames (frames larger than 1522 bytes), increase the MTU as required by your network. A frame size of up to 9198 bytes is supported.

The largest possible layer 1 frame will be 18 bytes larger than the MTU value to allow for link layer headers and trailers.

The `no` form of this command sets the MTU to the default value 1500.

Command context

`config-if`

Parameters

<VALUE>

Specifies the MTU in bytes. Range: 46 to 9198. Default: 1500.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the MTU on interface **1/1/1** to 1000 bytes:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# mtu 1000
```

Setting the MTU on interface **1/1/1** to the default value:

```
switch(config)# interface 1/1/1
switch(config-if)# no routing
switch(config-if)# no mtu
```

routing

Syntax

`routing`

`no routing`

Description

Enables routing support on an interface, creating a L3 (layer 3) interface on which the switch can route IPv4/IPv6 traffic to other devices.

By default, routing is enabled on all interfaces.

The `no` form of this command disables routing support on an interface, creating a L2 (layer 2) interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling routing support on an interface:

```
switch(config-if)# routing
```

Disabling routing support on an interface:

```
switch(config-if)# no routing
```

show allow-unsupported-transceiver

Syntax

```
show allow-unsupported-transceiver
```

Description

Displays configuration and status of unsupported transceivers.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing unallowed unsupported transceivers:

```
switch(config)# show allow-unsupported-transceiver

Allow unsupported transceivers : no
Logging interval               : 1440 minutes

---------------------------------------------
Port       Type           Status
---------------------------------------------
1/1/31     SFP-SX         unsupported
1/1/32     SFP-1G-BXD     unsupported
1/1/2      SFP28DAC3      unsupported
```

Showing allowed unsupported transceivers:

```
switch# show allow-unsupported-transceiver

Allow unsupported transceivers : yes
Logging interval               : 1440 minutes

---------------------------------------------
Port       Type           Status
---------------------------------------------
1/1/31     SFP-SX         unsupported-allowed
1/1/32     SFP-1G-BXD     unsupported-allowed
1/1/2      SFP28DAC3     unsupported
```

show interface

Syntax

```
show interface [<IFNNAME>|<IFRANGE>] [brief | physical | extended [non-zero]]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [brief | physical]
show interface [lag | loopback | tunnel | vlan ] [<ID>] [extended | non-zero]
show interface vxlan <ID> [brief | physical]
show interface vxlan <ID> [brief | physical]
```

Description

Displays active configurations and operational status information for interfaces.

Command context

`config`

Parameters

<IFNAME>

Specifies a interface name.

<IFRANGE>

Specifies the port identifier range.

brief

Displays brief info in tabular format.

physical

Display the physical connection info in tabular format.

extended

Displays additional statistics.

non-zero

Displays only non zero statistics.

LAG

Displays LAG interface information.

LOOPBACK

Displays loopback interface information.

TUNNEL

Displays tunnel interface information.

VLAN

Displays VLAN interface information.

<LAG-ID>

Specifies the LAG number. Range: 1-256

<LOOPBACK-ID>

Specifies the LOOPBACK number. Range: 0-255

<TUNNEL-ID>

Specifies the tunnel ID. Range: 1-255

<VLAN-ID>

Specifies the VLAN ID. Range: 1-4094

VXLAN

Displays the VXLAN interface information.

<VXLAN-ID>

Specifies the VXLAN interface identifier. Default: 1

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

The following example shows when the interface is configured as a route-only port:

```
switch# show interface 1/1/1
Interface 1/1/1 is up
 Admin state is up
 Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
 Link transitions: 1
 Description: Backup data center link
 Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
 MTU 1500
 Type 1GbT
 Full-duplex
 qos trust none
 Speed 1000 Mb/s
 Auto-negotiation is on
 Flow-control: off
 Error-control: off

MDI mode: MDIX
    L3 Counters: Rx Enabled, Tx Enabled
 Rate collection interval: 300 seconds

  Rates              Rx            Tx              Total
 -------------------------------------------------------
 Mbits/sec         0.00          0.00            0.00
 Kbits/sec         0.00          0.00            0.00
  Unicast          0.00          0.00            0.00
  Multicast        0.00          0.00            0.00
  Broadcast        0.00          0.00            0.00
 Utilization%      0.00          0.00            0.00

 Statistics          Rx            Tx            Total
 -------------------------------------------------------
 Packets              0             0              0
  Unicast             0             0              0
  Multicast           0             0              0
  Broadcast           0             0              0
 Bytes                0             0              0
 Jumbos               0             0              0
 Dropped              0             0              0
 Filtered             0             0              0
 Pause Frames         0             0              0
 L3 Packets           0             0              0
 L3 Bytes             0             0              0
 Errors               0             0              0
  CRC/FCS             0            n/a             0
  Collision          n/a            0              0
  Runts               0            n/a             0
  Giants              0            n/a             0
```

When the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
```

```
...
Auto-negotiation is on with downshift active
```

show interface dom

Syntax

`show interface [<INTERFACE-ID>] dom [detail] [vsx-peer]`

Description

Shows diagnostics information and alarm/warning flags for the optical transceivers (SFP, SFP+, QSFP+). This information is known as DOM (Digital Optical Monitoring). DOM information also consists of vendor determined thresholds which trigger high/low alarms and warning flags.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: `member/slot/port`.

detail

Show detailed information.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show interface dom

-------------------------------------------------------------------------------------
Port     Type       Channel  Temperature Voltage  Tx Bias  Rx Power       Tx Power
                             (Celsius)   (Volts)  (mA)     (mW/dBm)       (mW/dBm)
-------------------------------------------------------------------------------------
1/1/1    SFP+SR              47.65       3.31     8.40     0.08, -10.96   0.63, -2.49
1/1/2    SFP+SR              n/a         n/a      n/a      n/a            n/a
1/1/3    SFP+DA3             42.10       3.24     n/a      n/a            n/a
1/1/4    QSFP+SR4   1        44.46       3.30     6.12     0.08, -10.96   0.63, -1.95
                    2        44.46       3.30     6.04     0.08, -10.96   0.63, -2.00
                    3        44.46       3.30     6.51     0.08, -10.96   0.60, -2.16
                    4        44.46       3.30     6.19     0.08, -10.96   0.63, -1.94
```

show interface transceiver

Syntax

`show interface [<INTERFACE-ID>] transceiver [detail | threshold-violations] [vsx-peer]`

Description

Displays information about transceivers present in the switch. The information shown varies for different transceiver types and manufacturers. Only basic information is shown for unsupported HPE and third-party transceivers installed in the switch and they are also identified with an asterisk in the output.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies the name or range of an interface on the switch. Use the format `member/slot/port` (for example, `1/3/1`).

`detail`

Show detailed information for the interfaces.

`threshold-violations`

Show threshold violations for transceivers.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing summary transceiver information with identification of unsupported transceivers:

```
switch(config)# show interface transceiver
-------------------------------------------------------------------
Port      Type           Product      Serial          Part
                         Number       Number          Number
-------------------------------------------------------------------
1/1/1     SFP+SR         J9150A       MYxxxxxxxx      1990-3657
1/1/2     SFP+ER*        --           --              --
1/2/1     QSFP+SR4       JH233A       MYxxxxxxxx      2005-1234
1/2/2     QSFP+ER4*      --           --              --
1/3/1     SFP28DAC3      844477-B21   MYxxxxxxxx      77fc-7ce7

* unsupported transceiver
```

Showing detailed transceiver information:

```
switch(config)# show interface transceiver detail
Transceiver in 1/1/1
   Interface Name      : 1/1/1
   Type                : SFP+SR
   Connector Type      : LC
   Wavelength          : 850nm
   Transfer Distance   : 0m (SMF), 30m (OM1), 80m (OM2), 300m (OM3)
   Diagnostic Support  : DOM
   Product Number      : J9150A
   Serial Number       : MYxxxxxxx
```

```
   Part Number        : 1990-3657

Status
   Temperature : 47.65C
   Voltage     : 3.31V
   Tx Bias     : 8.40mA
   Rx Power    : 0.08mW, -10.96dBm
   Tx Power    : 0.56mW, -2.49dBm

  Recent Alarms :
    Rx power low alarm
    Rx power low warning

  Recent Errors :
    Rx loss of signal

Transceiver in 1/1/2
   Interface Name     : 1/1/2
   Type               : unknown
   Connector Type     : ??
   Wavelength         : ??
   Transfer Distance  : ??
   Diagnostic Support : ??
   Product Number     : ??
   Serial Number      : ??
   Part Number        : ??

Transceiver in 1/2/1
   Interface Name     : 1/2/1
   Type               : QSFP+SR4
   Connector Type     : MPO
   Wavelength         : 850nm
   Transfer Distance  : 0m (SMF), 0m (OM1), 0m (OM2), 100m (OM3)
   Diagnostic Support : DOM
   Product Number     : JH233A
   Serial Number      : MYxxxxxxx
   Part Number        : 2005-1234

Status
   Temperature : 44.46C
   Voltage     : 3.30V

  ----------------------------------------------
            Tx Bias  Rx Power     Tx Power
  Channel#  (mA)     (mW/dBm)     (mW/dBm)
  ----------------------------------------------
  1         6.12     0.00, -inf   0.63, -1.95
  2         6.04     0.00, -inf   0.63, -2.00
  3         6.51     0.00, -inf   0.60, -2.16
  4         6.19     0.00, -inf   0.63, -1.94

  Recent Alarms :
    Channel 1 :
       Rx power low alarm
       Rx power low warning
    Channel 2 :
       Rx power low alarm
       Rx power low warning
    Channel 3 :
       Rx power low alarm
       Rx power low warning
    Channel 4 :
       Rx power low alarm
       Rx power low warning
```

```
  Recent Errors :
    Channel 1 :
       Rx Loss of Signal
    Channel 2 :
       Rx Loss of Signal
    Channel 3 :
       Rx Loss of Signal
    Channel 4 :
       Rx Loss of Signal

Transceiver in 1/2/2
   Interface Name     : 1/2/2
   Type               : unknown
   Connector Type     : ??
   Wavelength         : ??
   Transfer Distance  : ??
   Diagnostic Support : ??
   Product Number     : ??
   Serial Number      : ??
   Part Number        : ??

Transceiver in 1/3/1
   Interface Name     : 1/3/1
   Type               : SFP28DAC3
   Connector Type     : Copper Pigtail
   Transfer Distance  : 0.00km (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
   Diagnostic Support : None
   Product Number     : 844477-B21
   Serial Number      : MYxxxxxxx
   Part Number        : 77fc-7ce7
```

Showing detailed transceiver information with identification of unsupported transceivers:

```
switch# show interface transceiver detail
Transceiver in 1/1/2
  Interface Name      : 1/1/2
  Type                : SFP+ER (unsupported)
  Connector Type      : LC
  Wavelength          : 3590nm
  Transfer Distance   : 80m (SMF), 0m (OM1), 0m (OM2), 0m (OM3)
  Diagnostic Support  : DOM
  Vendor Name         : INNOLIGHT
  Vendor Part Number  : TR-PX15Z-NHP
  Vendor Part Revision: 1A
  Vendor Serial number: MYxxxxxxx

Status
  Temperature : 28.88C
  Voltage     : 3.30V
  Tx Bias     : 65.53mA
  Rx Power    : 0.00mW, -inf
  Tx Power    : 1.47mW, 1.67dBm

 Recent Alarms:
  Rx Power low alarm
  Rx Power low warning

 Recent Errors:
  Rx loss of signal
```

Showing transceiver threshold-violations:

```
switch(config)# show interface transceiver threshold-violations
-----------------------------------------------------
Port      Type       Channel  Type(s) of Recent
                              Threshold Violation(s)
-----------------------------------------------------
1/1/1     SFP+SR              Tx bias high warning
                              50.52 mA > 40.00 mA
1/1/2     SFP+ER*             ??
1/2/1     QSFP+SR4   1        Tx power low alarm
                              -17.00 dBm < -0.50 dBm
                     2        Tx bias low warning
                              3.12 mA < 4.00 mA
1/2/2     QSFP+ER4*           ??
1/3/1     SFP28DAC3           n/a

* unsupported transceiver
```

show ip interface

Syntax

`show ip interface <INTERFACE-ID> [vsx-peer]`

Description

Shows status and configuration information for an IPv4 interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies the name of an interface. Format: `member/slot/port`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show ip interface 1/1/1

Interface 1/1/1 is up
 Admin state is up
 Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
 IPv4 address 192.168.1.1/24
 MTU 1500
 RX
          0 packets, 0 bytes
 TX
          0 packets, 0 bytes
```

show ip source-interface

Syntax

```
show ip source-interface {sflow |  tftp |  radius | tacacs | all}
     [vrf <VRF-NAME>] [vsx-peer]
```

Description

Shows single source IP address configuration settings.

Command context

Manager (#)

Parameters

sflow | tftp | radius | tacacs | all

Shows single source IP address configuration settings for a specific protocol. The `all` option shows the global setting that applies to all protocols that do not have an address set.

vrf *<VRF-NAME>*

Specifies the name of a VRF.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing single source IP address configuration settings for sFlow:

```
switch# show ip source-interface sflow

Source-interface Configuration Information
----------------------------------------
Protocol        Source Interface
--------        ----------------
 sflow            10.10.10.1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ip source-interface all

Source-interface Configuration Information
----------------------------------------
Protocol        Source Interface
--------        ----------------
 all             1/1/1
```

show ipv6 interface

Syntax

```
show ipv6 interface <INTERFACE-ID> [vsx-peer]
```

Description

Shows status and configuration information for an IPv6 interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface ID. Format: `member/slot/port`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

```
switch# show ipv6 interface 1/1/1

Interface 1/1/1 is up
 Admin state is up
 IPv6 address:
    2001:0db8:85a3:0000:0000:8a2e:0370:7334/24 [VALID]
 IPv6 link-local address: fe80::1e98:ecff:fee3:e800/64 (default)[VALID]
 IPv6 virtual address configured: none
 IPv6 multicast routing: disable
 IPv6 Forwarding feature: enabled
 IPv6 multicast groups locally joined:
    ff02::ff70:7334  ff02::ffe3:e800  ff02::1  ff02::1:ff00:0
    ff02::2
 IPv6 multicast (S,G) entries joined: none
 IPv6 MTU: 1524 (using link MTU)
 IPv6 unicast reverse path forwarding: none
 IPv6 load sharing: none
 RX
          0 packets, 0 bytes
 TX
          0 packets, 0 bytes
```

show ipv6 source-interface

Syntax

```
show ipv6 source-interface {sflow |  tftp |  radius | tacacs | all}
     [vrf <VRF-NAME>] [vsx-peer]
```

Description

Shows single source IP address configuration settings.

Command context

Manager (#)

Parameters

`sflow | tftp | radius | tacacs | all`

Shows single source IP address configuration settings for a specific protocol. The `all` option shows the global setting that applies to all protocols that do not have an address set.

`vrf` ***`<VRF-NAME>`***

Specifies the name of a VRF.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing single source IP address configuration settings for sFlow:

```
switch# show ipv6 source-interface sflow

Source-interface Configuration Information
----------------------------------------
Protocol        Source Interface
--------        ----------------
 sflow            2001:DB8::1
```

Showing single source IP address configuration settings for all protocols:

```
switch# show ipv6 source-interface all

Source-interface Configuration Information
----------------------------------------
Protocol        Source Interface
--------        ----------------
 all             1/1/1
```

shutdown

Syntax

`shutdown`

`no shutdown`

Description

Disables an interface. Interfaces are disabled by default when created.

The `no` form of this command enables an interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Disabling an interface:

```
switch(config-if)# shutdown
```

Enabling an interface:

```
switch(config-if)# no shutdown
```

VLANs are primarily used to provide network segmentation at layer 2. VLANs enable the grouping of users by logical function instead of physical location. They make managing bandwidth usage within networks possible by:

- Allowing grouping of high-bandwidth users on low-traffic segments
- Organizing users from different LAN segments according to their need for common resources and individual protocols
- Improving traffic control at the edge of networks by separating traffic of different protocol types.
- Enhancing network security by creating subnets to control in-band access to specific network resources

VLANs are generally assigned on an organizational basis rather than on a physical basis. For example, a network administrator could assign all workstations and servers used by a particular workgroup to the same VLAN, regardless of their physical locations.

Hosts in the same VLAN can directly communicate with one another. A router or a Layer 3 switch is required for hosts in different VLANs to communicate with one another.

VLANs help reduce bandwidth waste, improve LAN security, and enable network administrators to address issues such as scalability and network management.

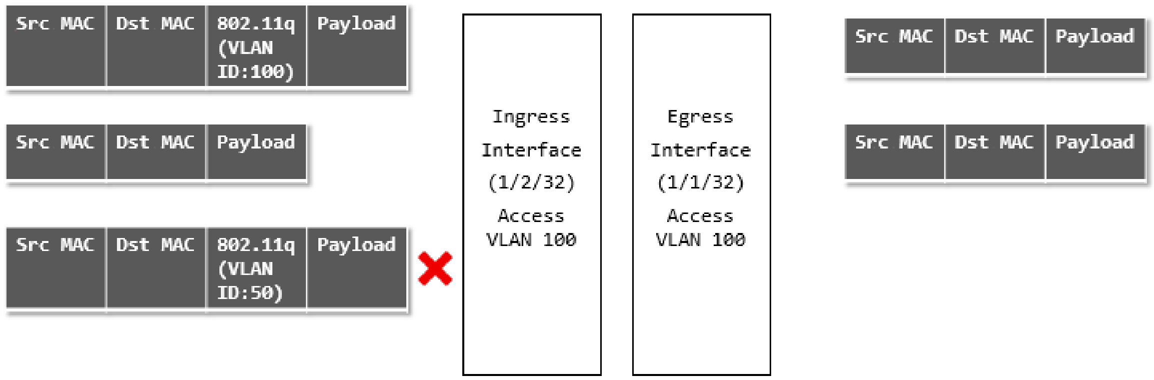
VLAN interfaces

Access interface

An access interface carries traffic for a single VLAN ID. Access interfaces are generally used to connect end devices that do not support VLANs to the network. The devices connected to an access interface are not aware of the VLAN. Access interface can carry traffic on only one VLAN, either tagged or untagged.

Example

This example shows ingress and egress traffic behavior for an access interface.



- An ingress tagged frame with VLAN ID of 100 arrives on interface 1/2/32. The switch accepts this frame and sends it to its target address on interface 1/1/32, where it egresses untagged.
- An ingress untagged frame arrives on interface 1/2/32. The switch accepts this frame and sends it to its target address on interface 1/1/32, where it egresses untagged.
- An ingress tagged frame with VLAN ID of 50 arrives on interface 1/2/32. The switch drops this frame as VLAN ID 50 is not configured on the interface.

Trunk interface

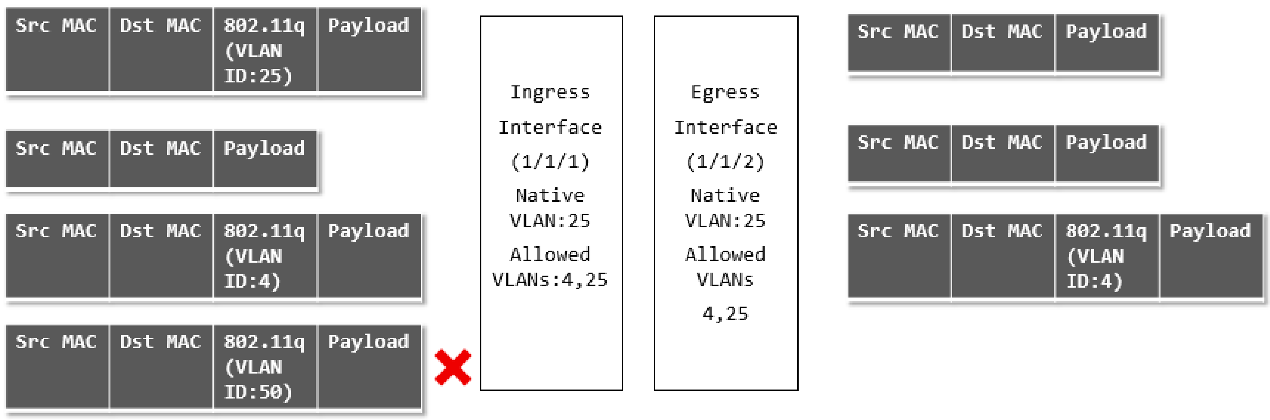
A trunk interface can carry traffic for one or more VLAN IDs. In most cases, a trunk interface is used to transport data to other switches or routers.

A trunk interface has two important settings:

- Native VLAN: This is the VLAN to which incoming untagged traffic is assigned. Only one VLAN can be assigned as the native VLAN. By default, VLAN 1 is assigned as the native VLAN for all trunk interfaces.
- Allowed VLANs: This is the list of VLANs that can be transported by the trunk. If the native VLAN is not included in the allowed list, all untagged frames that ingress on the trunk interface are dropped.

Example 1: Native untagged VLAN

This example shows ingress and egress traffic behavior when a trunk interface has a native untagged VLAN.



- An ingress tagged frame with VLAN ID of 25 arrives on interface 1/1/1. The switch accepts this frame and sends it to its target address on interface 1/1/2, where it egresses with a VLAN ID of 25 untagged since port 1/1/2 is configured with a native VLAN ID of 25.
- An ingress untagged frame arrives on interface 1/1/1. The switch accepts this frame and sends it to its target address on interface 1/1/2, where it egresses with a VLAN ID of 25 untagged since port 1/1/2 is configured with a native VLAN ID of 25.
- An ingress tagged frame with VLAN ID of 4 arrives on interface 1/1/1. The switch accepts this frame and sends it to its target address on interface 1/1/2, where it egresses with a VLAN ID of 4 tagged since port 1/1/2 is configured to allow traffic with a VLAN ID of 4.
- An ingress tagged frame with VLAN ID of 50 arrives on interface 1/1/1. The switch drops this frame as VLAN ID 50 is not in the allowed list for interface 1/1/1.

Example 2: Native tagged VLAN

This example shows ingress and egress traffic behavior when a trunk interface has a native tagged VLAN.

- An ingress tagged frame with VLAN ID of 6 arrives on interface 1/1/13. The switch accepts this frame and sends it to its target address on interface 1/1/21, where it egresses with a VLAN ID of 6 tagged since port 1/1/2 is configured with a native VLAN ID of 6.
- An ingress untagged frame arrives on interface 1/1/13. The switch drops this frame since the interface is configured as native tagged (all untagged frames a dropped in such a configuration).
- An ingress tagged frame with VLAN ID of 17 arrives on interface 1/1/13. The switch accepts this frame and sends it to its target address on interface 1/1/21, where it egresses with a VLAN ID of 17 tagged since port 1/1/2 is configured to allow traffic with a VLAN ID of 17.
- An ingress tagged frame with VLAN ID of 50 arrives on interface 1/1/13. The switch drops this frame as VLAN ID 50 is not in the allowed list for interface 1/1/13.

Traffic handling summary

VLAN configuration	Ingress traffic	Egress traffic
Access interface with: VLAN ID = X	**1.** Untagged **2.** Tagged with VLAN ID = X **3.** Tagged with any other VLAN ID	**1.** Untagged on VLAN X **2.** Untagged on VLAN X **3.** Dropped
Trunk interface with: • Untagged Native VLAN ID = X • Allowed VLAN IDs = X, Y, Z	**1.** Untagged **2.** Tagged with VLAN ID = X **3.** Tagged with VLAN ID = Y **4.** Tagged with VLAN ID = Z **5.** Tagged with any other VLAN ID	**1.** Untagged on VLAN X **2.** Untagged on VLAN X **3.** Tagged on VLAN Y **4.** Tagged on VLAN Z **5.** Dropped

Table Continued

VLAN configuration	Ingress traffic	Egress traffic
Trunk interface with: • Untagged Native VLAN ID = X • Allowed VLAN IDs = ALL	**1.** Untagged **2.** Tagged with VLAN ID = X **3.** Tagged with a VLAN ID defined on the switch **4.** Tagged with a VLAN ID not defined on the switch	**1.** Untagged on VLAN X **2.** Untagged on VLAN X **3.** Tagged on the matching VLAN **4.** Dropped
Trunk interface with: • Tagged Native VLAN ID = X • Allowed VLAN IDs = X, Y, Z	**1.** Untagged **2.** Tagged with VLAN ID = X **3.** Tagged with VLAN ID = Y **4.** Tagged with VLAN ID = Z **5.** Tagged with any other VLAN ID	**1.** Dropped **2.** Tagged on VLAN X **3.** Tagged on VLAN Y **4.** Tagged on VLAN Z **5.** Dropped
Trunk interface with: • Tagged Native VLAN ID = X • Allowed VLAN IDs = ALL	**1.** Untagged **2.** Tagged with VLAN ID = X **3.** Tagged with a VLAN ID defined on the switch **4.** Tagged with a VLAN ID not defined on the switch	**1.** Dropped **2.** Tagged on VLAN X **3.** Tagged on the matching VLAN **4.** Dropped
Trunk interface with: • Untagged Native VLAN ID = A • Allowed VLAN IDs = X, Y, Z	**1.** Untagged **2.** Tagged with VLAN ID = X **3.** Tagged with VLAN ID = Y **4.** Tagged with VLAN ID = Z **5.** Tagged with any other VLAN ID	**1.** Dropped **2.** Tagged on VLAN X **3.** Tagged on VLAN Y **4.** Tagged on VLAN Z **5.** Dropped

Comparing VLAN commands on PVOS, Comware, and AOS-CX

The following examples compare the commands needed to implement typical VLAN configurations on different HPE products.

AOS-CX	PVOS	Comware

interface 1/1/1
no routing
vlan trunk native 1
vlan trunk allowed 10,30,50
 	
A native VLAN must be defined on the switch. By default, this is VLAN 1. Since only VLANs 10, 30, and 50 are allowed on the trunk, all untagged traffic is dropped.
	interface A1
	  tagged vlan
	10,30,50
	  no untagged vlan 1
		Interface G1/0/1
		  port link type trunk
		  port trunk permit vlan
		10,30,50
		  port trunk pvid vlan 1
		
		PVID 1 is the default setting.

AOS-CX	PVOS	Comware

interface 1/1/1
no routing
vlan trunk native 10 tag
vlan trunk allowed 10,30,50
 	
Same as scenario 1, but allows untagged traffic on VLAN 10 as well.
	Not directly supported in PVOS. Scenario 1 is a workaround if there is no need to support untagged traffic.	Not directly supported in Comware. A possible workaround is:
		
		interface g1/0/1
		  port link-mode bridge
		  port link-type hybrid
		  port hybrid protocol-vlan
		vlan 10
		  port hybrid vlan 10 tagged
		  port hybrid vlan 30 tagged
		  port hybrid vlan 50 tagged

AOS-CX	PVOS	Comware

interface 1/1/1
no routing
vlan trunk native 5
vlan trunk allowed 5,
10,30,50
 	
VLAN 5 must be allowed on the trunk so that untagged traffic is not dropped.
	interface A1
	  untagged vlan 5
	  no tagged vlan
	10,30,50
		interface G1/0/1
		  Port link-mode bridge
		  port link-type trunk
		  port trunk pvid vlan 5
		  port trunk permit vlan
		5,10,30,50
		
		link-mode is only needed on later Comware 7 devices. 5930 is port link-mode route by default. 5900 is bridge by default.

ArubaOS-CX	PVOS	Comware

interface 1/1/1
no routing
vlan access 5
	interface A1
	  untagged vlan 5
		interface G1/0/0
		  port link-mode bridge
		  port access vlan 5

VLAN numbering

VLANs are numbered in the range 1 to 4040.

By default, VLAN 1 (the default VLAN) is associated with all interfaces on the switch. VLAN 1 cannot be removed from the switch.

Configuring VLANs

Creating and enabling a VLAN

Procedure

1. Switch to the configuration context with the command `config`.
2. Create a new VLAN with the command `vlan`.

Example

This example creates **`VLAN 10`**. The VLAN is enabled by default.

```
switch# config
switch(config)# vlan 10
switch(config-vlan-10)#
```

Disabling a VLAN

Procedure

1. Switch to configuration context with the command `config`.
2. Switch to configuration context for the VLAN you want to disable with the command `vlan`.
3. Disable the VLAN with the command `shutdown`.

Example

This example disables **`VLAN 10`**.

```
switch(config)# config
switch(config)# vlan 10
switch(config-vlan-10)# shutdown
```

Assigning a VLAN to an interface

To use a VLAN, it must be assigned to an interface on the switch. VLANs can only be assigned to non-routed (layer 2) interfaces. All interfaces are routed (layer 3) by default when created. Use the `no routing` command to disable routing on an interface.

Assigning a VLAN ID to an access interface

Prerequisites
At least one defined VLAN.

Procedure

1. Switch to configuration context with the command `config`.
2. Switch to the interface that you want to define as an access interface with the command `interface`.
3. Disable routing with the command `no routing`.
4. Configure the access interface and assign a VLAN ID with the command `vlan access`.

Examples

This example configures interface **1/1/2** as an access interface with VLAN ID set to **20**.

```
switch# config
switch(config)# vlan 20
switch(config-vlan-20)# exit
switch(config)# interface 1/1/2
switch(config-lag-if)# no routing
switch(config-if)# vlan access 20
```

This example configures LAG **1** as an access interface with VLAN ID set to **30**.

```
switch# config
switch(config)# vlan 30
switch(config-vlan-30)# exit
switch(config)# interface lag 1
switch(config-lag-if)# no shutdown
switch(config-lag-if)# no routing
switch(config-lag-if)# vlan access 30
```

Assigning a VLAN ID to a trunk interface

Prerequisites
At least one defined VLAN.

Procedure

1. Switch to configuration context with the command `config`.
2. Switch to the interface that you want to define as a trunk interface with the command `interface`.
3. Disable routing with the command `no routing`.
4. Configure the trunk interface and assign a VLAN ID with the command `vlan trunk allowed`.

Examples

This example configures interface **1/1/2** as a trunk interface allowing traffic with VLAN ID set to **20**.

```
switch# config
switch(config)# vlan 20
switch(config-vlan-20)# exit
switch(config)# interface 1/1/2
switch(config-lag-if)# no routing
switch(config-if)# vlan trunk allowed 20
```

This example configures interface **1/1/2** as a trunk interface allowing traffic with VLAN IDs **2**, **3**, and **4**.

```
switch# config
switch(config)# vlan 2,3,4
switch(config)# interface 1/1/2
switch(config-lag-if)# no routing
switch(config-if)# vlan trunk allowed 2,3,4
```

This example configures interface **1/1/2** as a trunk interface allowing traffic with VLAN IDs **2** to 8.

```
switch# config
switch(config)# vlan 2-8
switch(config)# interface 1/1/2
switch(config-lag-if)# no routing
switch(config-if)# vlan trunk allowed 2-8
```

This example configures interface **1/1/2** as a trunk interface allowing traffic with VLAN IDs **2** to 8 and **10**.

```
switch# config
switch(config)# vlan 2-8,10
switch(config)# interface 1/1/2
switch(config-lag-if)# no routing
switch(config-if)# vlan trunk allowed 2-8,10
```

This example configures interface **1/1/2** as a trunk interface allowing traffic on all configured VLAN IDs (20-100).

```
switch# config
switch(config)# vlan 20-100
switch(config)# interface 1/1/2
switch(config-lag-if)# no routing
switch(config-if)# vlan trunk allowed all
```

Assigning a native VLAN ID to a trunk interface

Prerequisites
At least one defined VLAN.

Procedure

1. Switch to configuration context with the command `config`.
2. Switch to the trunk interface to which you want to assign the native VLAN ID with the command `interface`.
3. Disable routing with the command `no routing`.
4. Assign the native VLAN ID with the command `vlan trunk native`. If tagging is required, use the command `vlan trunk native tag`.
5. Allow traffic tagged with the native VLAN ID to be transported by the trunk using the command `vlan trunk allowed`.

Example

This example assigns native VLAN ID **20** to trunk interface **1/1/2**.

```
switch# config
switch(config)# vlan 20
switch(config-vlan-20)# exit
switch(config)# interface 1/1/2
```

```
switch(config-if)# no routing
switch(config-if)# vlan trunk native 20
```

This example assigns native VLAN ID **40** to trunk interface **1/1/5**, enables tagging, and allows traffic with VLAN ID 40 to be transported by the trunk.

```
switch# config
switch(config)# vlan 40
switch(config-vlan-40)# exit
switch(config)# interface 1/1/5
switch(config-if)# no routing
switch(config-if)# vlan trunk native 40 tag
switch(config-if)# vlan trunk allow 40
```

Viewing VLAN configuration information

Prerequisites
At least one defined VLAN.

Procedure

1. View a summary of VLAN configuration information with the command `show vlan summary`.
2. View VLAN configuration settings with the command `show vlan`.
3. View VLANs configured for a specific layer 2 interface with the command `show vlan port`.
4. View the commands used to configure VLAN settings with the command `show running-config interface`.

Example

This example displays a summary of all VLANs.

```
switch# show vlan summary

Number of existing VLANs: 11
Number of static VLANs:   11
Number of dynamic VLANs:  0
```

This example displays configuration information for all defined VLANs.

```
switch# show vlan

-----------------------------------------------------------------------------------
VLAN  Name              Status  Reason          Type      Interfaces
-----------------------------------------------------------------------------------
1     DEFAULT_VLAN_1     up      ok               static    1/1/3-1/1/4
2     UserVLAN1          up      ok               static    1/1/1,1/1/3,1/1/5
3     UserVLAN2          up      ok               static    1/1/2-1/1/3,1/1/5-1/1/6
5     UserVLAN3          up      ok               static    1/1/3
10    TestNetwork        up      ok               static    1/1/3,1/1/5
11    VLAN11             up      ok               static    1/1/3
12    VLAN12             up      ok               static    1/1/3,1/1/6,lag1-lag2
13    VLAN13             up      ok               static    1/1/3,1/1/6
14    VLAN14             up      ok               static    1/1/3,1/1/6
20    ManagementVLAN     down    admin_down       static    1/1/3,1/1/10
```

This example displays configuration information for **VLAN 2**.

```
switch# show vlan 2
-----------------------------------------------------------------------------------
VLAN  Name                      Status  Reason          Type      Interfaces
-----------------------------------------------------------------------------------
2     UserVLAN1                 up      ok              static    1/1/1,1/1/3,1/1/5
```

This example displays the VLANs configured on interface **1/1/3**.

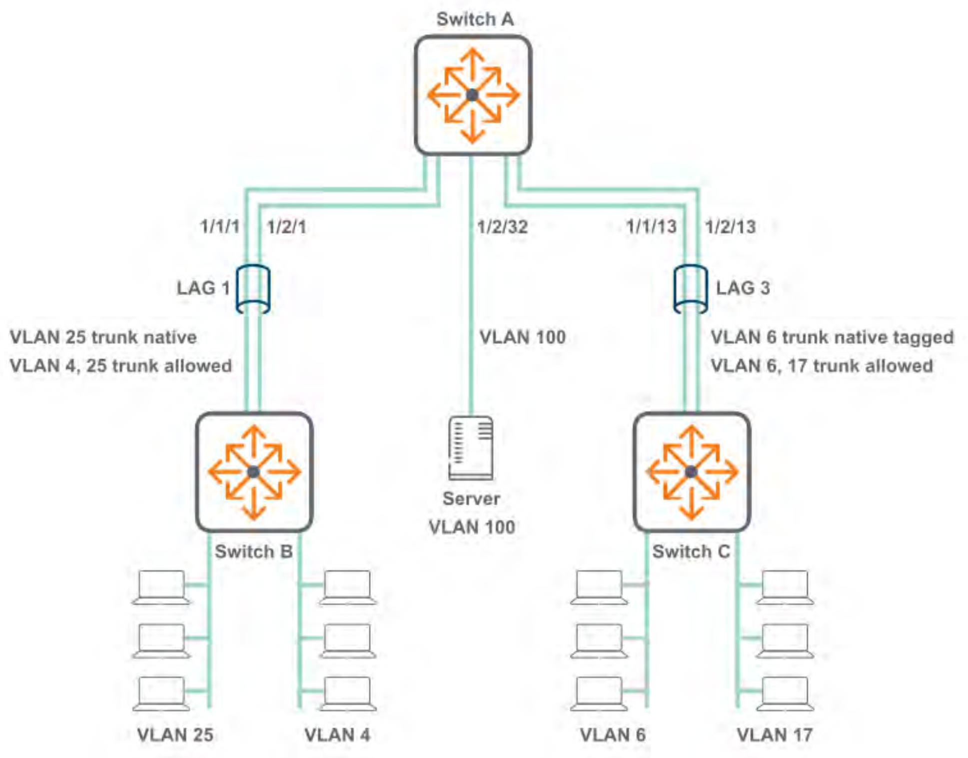
```
switch# show vlan port 1/1/3
------------------------------------------------------
VLAN  Name                            Mode
------------------------------------------------------
1     DEFAULT_VLAN_1                  native-untagged
2     UserVLAN1                       trunk
3     UserVLAN2                       trunk
5     UserVLAN3                       trunk
10    TestNetwork                     trunk
11    VLAN11                          trunk
12    VLAN12                          trunk
13    VLAN13                          trunk
14    VLAN14                          trunk
20    ManagementVLAN                  trunk
```

This example displays VLAN configuration commands for interface **1/1/16**.

```
switch#  show running-config interface 1/1/16
interface 1/1/16
   no routing
   vlan trunk native 108
   vlan trunk allowed all
   exit
```

VLAN scenario

This scenario shows how to assign VLAN IDs to access and trunk interfaces for the following deployment:



In this scenario, VLANs are used to isolate the traffic from different devices.

- VLAN 25 carries tagged and untagged traffic from computers connected to switch B.
- VLAN 4 carries tagged traffic from computers connected to switch B.
- VLAN 6 carries tagged and untagged traffic from computers connected to switch C.
- VLAN 17 carries tagged traffic from computers connected to switch C.
- VLAN 100 carries untagged traffic from the server.

Procedure

1. Execute the following commands on switch A and B.

   a. Create VLANs 4 and 25.

   ```
   switch# config
   switch(config)# vlan 4,25
   ```

   b. Define LAG `1` and assign the VLANs to it.

   ```
   switch(config)# interface lag 1
   switch(config-lag-if)# no shutdown
   switch(config-lag-if)# no routing
   switch(config-lag-if)# vlan trunk native 25
   switch(config-lag-if)# vlan trunk allowed 4,25
   ```

   c. Add ports `1/1/1` and `1/2/1` to LAG `1`.

```
switch(config-lag-if)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-lag-if)# no routing
switch(config-if)# lag 1
switch(config-if)# interface 1/2/1
switch(config-if)# no shutdown
switch(config-lag-if)# no routing
switch(config-if)# lag 1
```

2. Execute the following commands on switch A and C.

   a. Create VLANs 6 and 17.

   ```
   switch# config
   switch(config)# vlan 6,17
   ```

   b. Define LAG 3 and assign the VLANs to it.

   ```
   switch(config)# interface lag 3
   switch(config-lag-if)# no shutdown
   switch(config-lag-if)# no routing
   switch(config-lag-if)# vlan trunk native 6 tag
   switch(config-lag-if)# vlan trunk allowed 6,17
   ```

   c. Add ports **1/1/13** and **1/2/13** to LAG 3.

   ```
   switch(config-lag-if)# interface 1/1/13
   switch(config-if)# no shutdown
   switch(config-lag-if)# no routing
   switch(config-if)# lag 3
   switch(config-if)# interface 1/2/13
   switch(config-if)# no shutdown
   switch(config-if)# no routing
   switch(config-if)# lag 3
   ```

3. Execute the following commands on switch A to configure the connection to the server.

   a. Configure interface **1/2/13** as an access interface with VLAN ID set to 100.

   ```
   switch# config
   switch (config)# vlan 100
   switch(config-vlan-100)# interface 1/2/32
   switch(config-if)# no shutdown
   switch(config-lag-if)# no routing
   switch(config-if)# vlan access 100
   switch(config-if)# exit
   ```

4. Verify VLAN configuration by running the command `show vlan`. For example:

```
switch# show vlan

---------------------------------------------------------------------------------------------------------------------
VLAN  Name                              Status  Reason                Type      Interfaces
---------------------------------------------------------------------------------------------------------------------
1     DEFAULT_VLAN_1                    down    no_member_port        default
4     VLAN4                             up      ok                    static    lag1
6     VLAN6                             up      ok                    static    lag3
17    VLAN17                            up      ok                    static    lag3
25    VLAN25                            up      ok                    static    lag1
100   VLAN100                           up      ok                    static    1/2/32
```

5. Verify that the connection to the DHCP server is sending/receiving data with the command `show interface`. Check that the **Rx** and **Tx** fields are incrementing. For example:

```
switch# show interface 1/2/32
Interface 1/2/32 is up
 Admin state is up
 Description:
 Hardware: Ethernet, MAC Address: 70:72:cf:3a:8a:0b
 MTU 1500
 Type SFP+LR
 qos trust none
 Speed 10000 Mb/s
 Auto-Negotiation is off
 Input flow-control is off, output flow-control is off
 VLAN Mode: access
 Access VLAN: 100

 Rx
            20 input packets              1280 bytes
            0 input error                0 dropped
            0 CRC/FCS
 Tx
            9 output packets             1054 bytes
            0 input error                0 dropped
            0 collision
```

6. Verify LAG interface configuration with the command `show interface`. Check the fields admin state, MAC address, Aggregated-interfaces, VLAN Mode, Native VLAN, Allowed VLAN, Rx count, and Tx count. For example:

```
switch# show interface lag1
Aggregate-name lag1
 Description :
 Admin state            : up
 MAC Address            : 94:f1:28:21:63:00
 Aggregated-interfaces : 1/1/1 1/2/1
 Aggregation-key        : 1
 Speed 1000 Mb/s
 L3 Counters: Rx Disabled, Tx Disabled
 qos trust none
 VLAN Mode: native-untagged
 Native VLAN: 25
 Allowed VLAN List: 4,25
 Rx
            10 input packets              1280 bytes
            0 input error                0 dropped
            0 CRC/FCS
 Tx
            8 output packets             980 bytes
            0 input error                0 dropped
            0 collision
```

```
switch# show interface lag3
Aggregate-name lag3
 Description :
 Admin state            : up
 MAC Address            : 94:f1:28:21:63:00
 Aggregated-interfaces : 1/1/13 1/2/13
 Aggregation-key        : 3
 Speed 1000 Mb/s
 L3 Counters: Rx Disabled, Tx Disabled
 qos trust none
 VLAN Mode: native-tagged
 Native VLAN: 6
 Allowed VLAN List: 6,17
```

```
 Rx
            19 input packets              1280 bytes
            0 input error                0 dropped
            0 CRC/FCS
Tx
            15 output packets             1000 bytes
            0 input error                0 dropped
0    Collision
```

7. Verify the physical interfaces (1/1/1, 1/2/1, 1/1/13, 1/2/13) with the command `show interface`. Check that the **`Rx`** and **`Tx`** fields are incrementing. For example:

```
switch# show interface 1/1/1
Interface 1/1/1 is up
 Admin state is up
 Description:
 Hardware: Ethernet, MAC Address: 94:f1:28:21:73:ff
 MTU 1500
 Type SFP+LR
 qos trust none
 Speed 1000 Mb/s
 Auto-Negotiation is off
 Input flow-control is off, output flow-control is off
 Rx
            6 input packets              620 bytes
            0 input error                0 dropped
            0 CRC/FCS
 Tx
            4 output packets             422 bytes
            0 input error                0 dropped
0    collision
```

VLAN commands

description

Syntax

`description <DESCRIPTION>`

Description

Specifies a descriptive for a VLAN.

Command context

`config-vlan-<VLAN-ID>`

Parameters

<DESCRIPTION>

Specifies a description for the VLAN.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning a description to VLAN **`20`**:

```
switch(config)# vlan 20
switch(config-vlan-20)# description primary
```

name

Syntax

`name <VLAN-NAME>`

Description

Associates a name with a VLAN.

Command context

`config-vlan-<VLAN-ID>`

Parameters

<VLAN-NAME>

Specifies a name for a VLAN. Length: 1 to 32 alphanumeric characters, including underscore (_) and hyphen (-).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning the name **backup** to VLAN **20**:

```
switch(config)# vlan 20
switch(config-vlan-20)# name backup
```

show capacities svi-count

Syntax

`show capacities svi-count`

Description

Shows the maximum number of SVIs supported by the switch.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing switch SVI capacity:

```
switch# show capacities svi-count
System Capacities: Filter SVI count
Capacities Name                                                                                    Value
--------------------------------------------------------------------
Maximum number of SVIs supported in the system                    128
```

show system internal-vlan-range

Syntax

```
show system internal-vlan-range

no system vlan-client-presence-detect
```

Description

Shows the VLAN range reserved for internal use.

Command context

`manager`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing reserved VLANs:

```
switch(config)# show system internal-vlan-range
Internal VLAN range:          4041-4094
```

show vlan

Syntax

`show vlan [<VLAN-ID>] [vsx-peer]`

Description

Displays configuration information for all VLANs or a specific VLAN.

Command context

`Manager (#)`

Parameters

<VLAN-ID>

Specifies a VLAN ID.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying configuration information for VLAN **2**:

```
switch# show vlan 2
-------------------------------------------------------------------------------------
VLAN  Name                      Status  Reason          Type      Interfaces
-------------------------------------------------------------------------------------
2     UserVLAN1                 up      ok              static    1/1/1,1/1/3,1/1/5
```

Displaying configuration information for all defined VLANs:

```
switch# show vlan
-------------------------------------------------------------------------------------
VLAN  Name                  Status  Reason         Type      Interfaces
--------------------------------------------------------------------------------------
1     DEFAULT_VLAN_1        up      ok             static    1/1/3-1/1/4
2     UserVLAN1             up      ok             static    1/1/1,1/1/3,1/1/5
3     UserVLAN2             up      ok             static    1/1/2-1/1/3,1/1/5-1/1/6
5     UserVLAN3             up      ok             static    1/1/3
10    TestNetwork           up      ok             static    1/1/3,1/1/5
11    VLAN11                up      ok             static    1/1/3
12    VLAN12                up      ok             static    1/1/3,1/1/6,lag1-lag2
13    VLAN13                up      ok             static    1/1/3,1/1/6
14    VLAN14                up      ok             static    1/1/3,1/1/6
20    ManagementVLAN        down    admin_down     static    1/1/3,1/1/10
```

show vlan port

Syntax

`show vlan port <INTERFACE-ID> [vsx-peer]`

Description

Displays the VLANs configured for a specific layer 2 interface.

Command context

Manager (`#`)

Parameters

<INTERFACE-ID>

Specifies an interface ID. Format: `member/slot/port`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying the VLANs configured on interface **1/1/3**:

```
switch# show vlan port 1/1/3
------------------------------------------------------
VLAN  Name                            Mode
------------------------------------------------------
1     DEFAULT_VLAN_1                  native-untagged
```

```
2     UserVLAN1                         trunk
3     UserVLAN2                         trunk
5     UserVLAN3                         trunk
10    TestNetwork                       trunk
11    VLAN11                            trunk
12    VLAN12                            trunk
13    VLAN13                            trunk
14    VLAN14                            trunk
20    ManagementVLAN                    trunk
```

show vlan summary

Syntax

`show vlan summary [vsx-peer]`

Description

Displays a summary of the VLAN configuration on the switch.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying a summary of the VLAN configuration on the switch:

```
switch# show vlan summary
Number of existing VLANs: 11
Number of static VLANs:   11
Number of dynamic VLANs:  0
```

show vlan translation

Syntax

`show vlan translation [interface <INTERFACE-NAME>] [vsx-peer]`

Description

Shows a summary of all VLAN translations rules defined on the switch, or the rules defined for a specific interface.

Command context

Manager (#)

Parameters

interface *<INTERFACE-NAME>*

Specifies the name of a layer 2 interface. Format: `member/slot/port`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying a summary of all VLAN translations rules defined on the switch:

```
switch# show vlan translation
-------------------------------------------
Interface  VLAN-1       VLAN-2
-------------------------------------------
1/1/5      10           20
1/1/5      30           40
1/1/5      50           100
1/1/6      100          200

Total number of translation rules : 4
```

Displaying a summary of all VLAN translations rules defined on interface **1/1/5**:

```
switch# show vlan translation interface 1/1/5
-------------------------------------------
Interface  VLAN-1       VLAN-2
-------------------------------------------
1/1/5      10           20
1/1/5      30           40
1/1/5      50           100
```

shutdown

Syntax

```
shutdown

no shutdown
```

Description

Disables a VLAN. (By default, a VLAN is automatically enabled when it is created with the `vlan` command.)

The `no` form of this command enables a VLAN.

Command context

`config-vlan-<VLAN-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling VLAN 20:

```
switch(config)# vlan 20
switch(config-vlan-20)# no shutdown
```

Disabling VLAN 20:

```
switch(config)# vlan 20
switch(config-vlan-20)# shutdown
```

system internal-vlan-range

Syntax

`system internal-vlan-range {<VLAND-ID>-<VLAN-ID> | none } [confirm]`

`no system internal-vlan-range {<VLAND-ID>-<VLAN-ID> | none } [confirm]`

Description

Configures the VLAN range reserved for internal use for route-only ports and LAGs. The internal VLAN range cannot include any VLANs that are already in use.

If the number of internal VLANs is less than the number of route-only ports and LAGs, some ports will be blocked and unable to be used. When the internal VLAN range is modified, traffic on route-only ports and LAGs is briefly interrupted while they are moved to the new range.

The `no`' form of this command sets the range to the default of 4041 to 4094.

Command context

`config`

Parameters

<VLAND-ID>-<VLAN-ID>

Specifies the starting and ending VLAN number for the range. The reserved range must be between 2 and 4094 and cannot exceed 256 VLANs. Default: 4041-4094.

none

Do not reserve any internal VLANs.

confirm

Automatically acknowledge warning and skip confirmation prompt.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting a new internal VLAN range:

```
switch(config)# system internal-vlan-range 3041-3094
This will briefly interrupt traffic.

Continue (y/n)?
```

Setting a new internal VLAN range, skipping the prompt:

```
switch(config)# system internal-vlan-range 3041-3094 confirm
```

Removing all internal VLANs:

```
switch(config)# system internal-vlan-range none
All route-only ports and LAGs will be blocked.

Continue (y/n)?
```

system vlan-client-presence-detect

Syntax

`system vlan-client-presence-detect`

`no system vlan-client-presence-detect`

Description

Enables VNI mapped VLANs when detecting the presence of a client. When enabled, VNI mapped VLANs are *up* only if there are authenticated clients on the VLAN, or if the VLAN has statically configured ports and those ports are *up*. When not enabled, VNI mapped VLANs are always *up*.

The `no` form of this command disables detection of clients on VNI mapped VLANs.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling detection of clients:

```
switch(config)# system vlan-client-presence-detect
```

Disabling detection of clients:

```
switch(config)# no system vlan-client-presence-detect
```

vlan

Syntax

`vlan <VLAN-LIST>`

`no vlan <VLAN-LIST>`

Description

Creates a VLAN and changes to the `config-vlan-id` context for the VLAN. By default, the VLAN is enabled. To disable a VLAN, use the `no shutdown` command.

If the specified VLAN exists, this command changes to the `config-vlan-id` context for the VLAN. If a range of VLANs is specified, the context does not change.

The `no` form of this command removes a VLAN. VLAN 1 is the default VLAN and cannot be deleted.

Command context

`config`

Parameters

<VLAN-LIST>

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to 4040.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating VLAN **20**:

```
switch(config)# vlan 20
switch(config-vlan-20)#
```

Removing VLAN **20**:

```
switch(config)# no vlan 20
```

Creating VLANs **2** to **8** and **10**:

```
switch(config)# vlan 2-8,10
```

Removing VLANs **2** to **8** and **10**:

```
switch(config)# no vlan 2-8,10
```

vlan access

Syntax

`vlan access <VLAN-ID>`

`no vlan access [<VLAN-ID>]`

Description

Creates an access interface and assigns an VLAN ID to it. Only one VLAN ID can be assigned to each access interface.

VLANs can only be assigned to a non-routed (layer 2) interface or LAG interface. By default, all interfaces are routed (layer 3) when created. Use the `no routing` command to disable routing on an interface and change the interface to a layer 2 interface.

The `no` form of this command removes an access VLAN from the interface in the current context and sets it to the default VLAN ID of 1.

Command context

`config-if`

Parameters

<VLAN-ID>

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to 4040.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring interface **1/1/2** as an access interface with VLAN ID set to **20**:

```
switch(config)# interface 1/1/2
switch(config-if)# no routing
switch(config-if)# vlan access 20
```

Removing VLAN ID **20** from interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan access 20
```

or:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan access
```

vlan translate

Syntax

```
vlan translate <VLAN-1> <VLAN-2>

no vlan translate <VLAN-1> <VLAN-2>
```

Description

Defines a bidirectional VLAN translation rule that maps an external VLAN ID to an internal VLAN ID on a LAG or layer 2 interface. Applies to both incoming and outgoing traffic.

The `no` form of this command removes an existing VLAN translation rule on the current interface.

NOTE: VLAN translation and MVRP cannot be enabled on the same interface.

Command context

`config-if`

`config-lag-if`

Parameters

<VLAN-1>

Specifies the number of an external VLAN. Range: 1 - 4040.

<VLAN-2>

Specifies the number of an internal VLAN. Range: 1 - 4040.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Translates external VLAN **200** to internal VLAN **20** on interface **1/1/2**.

```
switch# config
switch(config)# vlan 20
switch(config-vlan-20)# exit
switch(config)# interface 1/1/2
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 20
switch(config-if)# vlan translate 200 20
```

Translates external VLANs **100** and **300** to internal VLANs **10** and **20** on interface **1/1/2**.

```
switch# config
switch(config)# vlan 10,30
switch(config-vlan-20)# exit
switch(config)# interface 1/1/2
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 10,30
switch(config-if)# vlan translate 100 10
switch(config-if)# vlan translate 300 30
```

vlan trunk allowed

Syntax

`vlan trunk allowed [<VLAN-LIST> | all]`

`no vlan trunk allowed [<VLAN-LIST>]`

Description

Assigns a VLAN ID to an trunk interface. Multiple VLAN IDs can be assigned to a trunk interface. These VLAN IDs define which VLAN traffic is allowed across the trunk interface.

VLANs can be assigned only to a non-routed (layer 2) interface or LAG interface. By default, all interfaces are routed (layer 3) when created. Use the `no routing` command to disable routing on an interface.

The `no` form of this command removes one or more VLAN IDs from a trunk interface. When the last VLAN is removed from a trunk interface, the interface continues to operate in trunk mode, and will trunk all the VLANs currently defined on the switch, and any new VLANs defined in the future. To disable the trunk interface, use the command **shutdown**.

Command context

`config-if`

Parameters

<VLAN-LIST>

Specifies a single ID, or a series of IDs separated by commas (2, 3, 4), dashes (2-4), or both (2-4,6). Range: 1 to 4040.

all

Configures the trunk interface to allow all the VLANs currently configured on the switch and any new VLANs that are configured in the future.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning VLANs **2**, **3**, and **4** to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 2,3,4
```

Assigning VLAN IDs **2** to **8** to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 2-8
```

Assigning VLAN IDs **2** to **8** and 10 to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no routing
switch(config-if)# vlan trunk allowed 2-8,10
```

Removing VLAN IDs **2**, **3**, and **4** from trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk allowed 2,3,4
```

Removing all VLANs assigned to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk allowed
```

vlan trunk native

Syntax

`vlan trunk native <VLAN-ID>`

`no vlan trunk native [<VLAN-ID>]`

Description

Assigns a native VLAN ID to a trunk interface. By default, VLAN ID 1 is assigned as the native VLAN ID for all trunk interfaces. VLANs can only be assigned to a non-routed (layer 2) interface or LAG interface. Only one VLAN ID can be assigned as the native VLAN.

NOTE: When a native VLAN is defined, the switch automatically executes the `vlan trunk allowed all` command to ensure that the default VLAN is allowed on the trunk. To only allow specific VLANs on the trunk, issue the `vlan trunk allowed` command specifying only specific VLANs.

The `no` form of this command removes a native VLAN from a trunk interface and assigns VLAN ID 1 as its native VLAN.

Command context

`config-if`

Parameters

<VLAN-ID>

Specifies a VLAN ID. Range: 1 to 4040.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Assigning native VLAN ID **20** to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no routing
switch(config-if)# vlan trunk native 20
```

Removing native VLAN **20** from trunk interface **1/1/2** and returning to the default VLAN 1 as the native VLAN.

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk native 20
```

or:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk native
```

Assigning native VLAN ID **20** to trunk interface **1/1/2** and then removing it from the list of allowed VLANs. (Only allow VLAN 10 on the trunk.)

```
switch(config)# interface 1/1/2
switch(config-if)# no routing
switch(config-if)# vlan trunk native 20
switch(config-if)# vlan trunk allowed 10
```

vlan trunk native tag

Syntax

`vlan trunk native <VLAN-ID> tag`

`no vlan trunk native <VLAN-ID> tag`

Description

Enables tagging on a native VLAN. Only incoming packets that are tagged with the matching VLAN ID are accepted. Incoming packets that are untagged are dropped except for BPDUs. Egress packets are tagged.

The `no` form of this command removes tagging on a native VLAN.

Command context

`config-if`

Parameters

<VLAN-ID>

Specifies the number of a VLAN. Range: 1 to 4040.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling tagging on native VLAN **20** on trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no routing
```

```
switch(config-if)# vlan trunk native 20
switch(config-if)# vlan trunk native 20 tag
```

Removing tagging on native VLAN **20** assigned to trunk interface **1/1/2**:

```
switch(config)# interface 1/1/2
switch(config-if)# no vlan trunk native 20 tag
```

Enabling tagging on native VLAN **20** assigned to LAG trunk interface **2**:

```
switch(config)# interface lag 2
switch(config-if)# no routing
switch(config-lag-if)# vlan trunk native 20
switch(config-lag-if)# vlan trunk native 20 tag
```

voice

Syntax

```
voice

no voice
```

Description

Configures a VLAN as a voice VLAN.

The `no` form of this command removes voice configuration from a VLAN.

Command context

`config-vlan-<VLAN-ID>`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring VLAN 10 as a voice VLAN:

```
switch(config)# vlan 10
switch(config-vlan-10)# voice
```

Removing voice from VLAN 10:

```
switch(config-vlan-10)# no voice
```

Checkpoints

A checkpoint is a snapshot of the running configuration of a switch and its relevant metadata during the time of creation. Checkpoints can be used to apply the switch configuration stored within a checkpoint whenever needed, such as to revert to a previous, clean configuration. Checkpoints can be applied to other switches of the same platform. A switch is able to store multiple checkpoints.

Checkpoint types

The switch supports two types of checkpoints:

- **System generated checkpoints**: The switch automatically generates a system checkpoint whenever a configuration change occurs.
- **User generated checkpoints**: The administrator can manually generate a checkpoint whenever required.

Maximum number of checkpoints

- Maximum checkpoints: 64 (including the startup configuration)
- Maximum user checkpoints: 32
- Maximum system checkpoints: 32

User generated checkpoints

User checkpoints can be created at any time, as long as one configuration difference exists since the last checkpoint was created. Checkpoints can be applied to either the running or startup configurations on the switch.

All user generated checkpoints include a time stamp to identify when a checkpoint was created.

A maximum of 32 user generated checkpoints can be created.

System generated checkpoints

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix `CPC` followed by a time stamp in the format `<YYYYMMDDHHMMSS>`. For example: `CPC20170630073127`.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Supported remote file formats

You can restore a switch configuration by copying a switch configuration stored on a USB drive or a remote network device through SFTP/TFTP. The remote file formats that the switch supports depends on where you plan to restore the checkpoint.

Restoring a checkpoint to a...	File type supported
Running configuration	• CLI • JSON • Checkpoint
Startup configuration	• JSON • Checkpoint
Specified checkpoint	Specified checkpoint

Rollback

The term rollback is used to refer to when a switch configuration is reverted to a pre-existing checkpoint.

For example, the following command applies the configuration from checkpoint `ckpt1`. All previous configurations are lost after the execution of this command: `checkpoint rollback` *`ckpt1`*

You can also specify the rollback of the running configuration or of the startup configuration with a specified checkpoint, as shown with the following command: `copy checkpoint` *`<checkpoint-name>`* `{running-config | startup-config}`

Checkpoint auto mode

Checkpoint auto mode configures the switch with failover support, causing it to automatically revert to a previous configuration if it becomes inoperable or inaccessible due to configuration changes that are being made.

After entering checkpoint auto mode, you have a set amount of time to add, remove, or modify the existing switch configuration. To save your changes, you must execute the `checkpoint auto confirm` command before the auto mode timer expires. If you do not execute the `checkpoint auto confirm` command within the specified time, all configuration changes you made are discarded and the running configuration reverts to the state it was before entering checkpoint auto mode.

Testing a switch configuration in checkpoint auto mode

Process overview:

Procedure

1. **Enable the checkpoint auto mode.**
2. To save the configuration, enter the **`checkpoint auto confirm`** command before the specified time set in step 1.

Checkpoint commands

checkpoint auto

Syntax

`checkpoint auto <TIME-LAPSE-INTERVAL>`

Description

Starts auto checkpoint mode. In auto checkpoint mode, the switch temporarily saves the runtime configuration as a checkpoint only for the specified time lapse interval. Configuration changes must be saved before the interval expires, otherwise the runtime configuration is restored from the temporary checkpoint.

Command context

Manager (#)

Parameters

<TIME-LAPSE-INTERVAL>

Specifies the time lapse interval in minutes. Range: 1 to 60.

Authority

Administrators or local user group members with execution rights for this command.

Usage

To save the runtime checkpoint permanently, run the `checkpoint auto confirm` command during the time lapse interval. The filename for the saved checkpoint is named `AUTO<YYYYMMDDHHMMSS>`. If the `checkpoint auto confirm` command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING  Please "checkpoint auto confirm" within 2 minutes
switch# checkpoint auto confirm
checkpoint AUTO20170801011154 created
```

In this example, the runtime checkpoint was saved because the `checkpoint auto confirm` command was entered within the value set by the `time-lapse-interval` parameter, which was 20 minutes.

Not confirming the auto checkpoint:

```
switch# checkpoint auto 20
Auto checkpoint mode expires in 20 minute(s)
switch# WARNING  Please "checkpoint auto confirm" within 2 minutes
WARNING: Restoring configuration. Do NOT add any new configuration.
Restoration successful
```

In this example, the runtime checkpoint was reverted because the `checkpoint auto confirm` command was not entered within the value set by the `time-lapse-interval` parameter, which was 20 minutes.

`checkpoint auto confirm`

Syntax

```
checkpoint auto confirm
```

Description

Signals to the switch to save the running configuration used during the auto checkpoint mode. This command also ends the auto checkpoint mode.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

To save the runtime checkpoint permanently, run the `checkpoint auto confirm` command during the time lapse value set by the `checkpoint auto <TIME-LAPSE-INTERVAL>` command. The generated checkpoint name will be in the format `AUTO<YYYYMMDDHHMMSS>`. If the `checkpoint auto confirm` command is not entered during the specified time lapse interval, the previous runtime configuration is restored.

Examples

Confirming the auto checkpoint:

```
switch# checkpoint auto confirm
```

`checkpoint diff`

Syntax

```
checkpoint diff {<CHECKPOINT-NAME1> | running-config | startup-config}
     {<CHECKPOINT-NAME2> | running-config | startup-config}
```

Description

Shows the difference in configuration between two configurations. Compare checkpoints, the running configuration, or the startup configuration.

Command context

Manager (#)

Parameters

`{<CHECKPOINT-NAME1> | running-config | startup-config}`

Selects either a checkpoint, the running configuration, or the startup configuration as the baseline.

`{<CHECKPOINT-NAME2> | running-config | startup-config}`

Selects either a checkpoint, the running configuration, or the startup configuration to compare.

Authority

Administrators or local user group members with execution rights for this command.

Usability

The output of the `checkpoint diff` command has several symbols:

- The plus sign (+) at the beginning of a line indicates that the line exists in the comparison but not in the baseline.
- The minus sign (-) at the beginning of a line indicates that the line exists in the baseline but not in the comparison.

Examples

In the following example, the configurations of checkpoints `cp1` and `cp2` are displayed before the `checkpoint diff` command, so that you can see the context of the `checkpoint diff` command.

```
switch# show checkpoint cp1
Checkpoint configuration:
!
!Version ArubaOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number jl363a
!
!
!
!
!
!
!
vlan 1,200
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24

switch# show checkpoint cp2
Checkpoint configuration:
!
!Version ArubaOS-CX XL.10.00.0002
!Schema version 0.1.8
module 1/1 product-number jl363a
!
!
!
!
!
!
!
vlan 1,200,300
interface 1/1/1
    no shutdown
    ip address 1.0.0.1/24
interface 1/1/2
    no shutdown
    ip address 2.0.0.1/24

switch# checkpoint diff cp1 cp2
--- /tmp/chkpt11501550258421    2017-08-01 01:17:38.420514016 +0000
+++ /tmp/chkpt21501550258421    2017-08-01 01:17:38.420514016 +0000
@@ -9,7 +9,7 @@
 !
```

```
 !
 !
-vlan 1,200
+vlan 1,200,300
 interface 1/1/1
     no shutdown
     ip address 1.0.0.1/24
```

checkpoint post-configuration

Syntax

```
checkpoint post-configuration

no checkpoint post-configuration
```

Description

Enables creation of system generated checkpoints when configuration changes occur. This feature is enabled by default.

The `no` form of this command disables system generated checkpoints.

Command context

Manager (`#`)

Authority

Administrators or local user group members with execution rights for this command.

Usage

System generated checkpoints are automatically created by default. Whenever a configuration change occurs, the switch starts a timeout counter (300 seconds by default). For each additional configuration change, the timeout counter is restarted. If the timeout expires with no additional configuration changes being made, the switch generates a new checkpoint.

System generated checkpoints are named with the prefix `CPC` followed by a time stamp in the format `<YYYYMMDDHHMMSS>`. For example: `CPC20170630073127`.

System checkpoints can be applied using the checkpoint rollback feature or copy command.

A maximum of 32 system checkpoints can be created. Beyond this limit, the newest system checkpoint replaces the oldest system checkpoint.

Examples

Enabling system checkpoints:

```
switch(config)# checkpoint post-configuration
```

Disabling system checkpoints:

```
switch(config)# no checkpoint post-configuration
```

checkpoint post-configuration timeout

Syntax

```
checkpoint post-configuration timeout <TIMEOUT>

no checkpoint post-configuration timeout <TIMEOUT>
```

Description

Sets the timeout for the creation of system checkpoints. The timeout specifies the amount of time since the latest configuration for the switch to create a system checkpoint.

The `no` form of this command resets the timeout to 300 seconds, regardless of the value of the *<TIMEOUT>* parameter.

Command context

Manager (#)

Parameters

timeout *<TIMEOUT>*

Specifies the timeout in seconds. Range: 5 to 600. Default: 300.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the timeout for system checkpoints to 60 seconds:

```
switch(config)# checkpoint post-configuration timeout 60
```

Resetting the timeout for system checkpoints to 300 seconds:

```
switch(config)# no checkpoint post-configuration timeout 1
```

checkpoint rename

Syntax

`checkpoint rename <OLD-CHECKPOINT-NAME> <NEW-CHECKPOINT-NAME>`

Description

Renames an existing checkpoint.

Command context

Manager (#)

Parameters

<OLD-CHECKPOINT-NAME>

Specifies the name of an existing checkpoint to be renamed.

<NEW-CHECKPOINT-NAME>

Specifies the new name for the checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).

NOTE: Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Renaming checkpoint **ckpt1** to **cfg001**:

```
switch# checkpoint rename ckpt1 cfg001
```

checkpoint rollback

Syntax

checkpoint rollback {*<CHECKPOINT-NAME>* | startup-config}

Description

Applies the configuration from a pre-existing checkpoint or the startup configuration to the running configuration.

Command context

Manager (#)

Parameters

<CHECKPOINT-NAME>

Specifies a checkpoint name.

startup-config

Specifies the startup configuration.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Applying a checkpoint named ckpt1 to the running configuration:

```
switch# checkpoint rollback ckpt1
Success
```

Applying a startup checkpoint to the running configuration:

```
switch# checkpoint rollback startup-config
Success
```

copy checkpoint *<CHECKPOINT-NAME>* *<REMOTE-URL>*

Syntax

copy checkpoint *<CHECKPOINT-NAME>* *<REMOTE-URL>* [vrf *<VRF-NAME>*]

Description

Copies a checkpoint configuration to a remote location as a file. The configuration is exported in checkpoint format, which includes switch configuration and relevant metadata.

Command context

Manager (#)

Parameters

<CHECKPOINT-NAME>

Specifies the name of a checkpoint.

<REMOTE-URL>

Specifies the remote destination and filename using the syntax: `{tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>][;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>`

vrf *<VRF-NAME>*

Specifies a VRF name.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying checkpoint configuration to remote file through TFTP:

```
switch# copy checkpoint ckpt1 tftp://192.168.1.10/ckptmeta vrf default
######################################################################### 100.0%
Success
```

Copying checkpoint configuration to remote file through SFTP:

```
switch# copy checkpoint ckpt1 sftp://root@192.168.1.10/ckptmeta vrf default
The authenticity of host '192.168.1.10 (192.168.1.10)' can't be established.
ECDSA key fingerprint is SHA256:FtOm6Uxuxumil7VCwLnhz92H9LkjY+eURbdddOETy50.
Are you sure you want to continue connecting (yes/no)? yes
root@192.168.1.10's password:
sftp> put /tmp/ckptmeta ckptmeta
Uploading /tmp/ckptmeta to /root/ckptmeta
Warning: Permanently added '192.168.1.10' (ECDSA) to the list of known hosts.
Connected to 192.168.1.10.
Success
```

copy checkpoint *<CHECKPOINT-NAME>* {running-config | startup-config}

Syntax

`copy checkpoint <CHECKPOINT-NAME> {running-config | startup-config}`

Description

Copies an existing checkpoint configuration to the running configuration or to the startup configuration.

Command context

Manager (#)

Parameters

<CHECKPOINT-NAME>

Specifies the name of an existing checkpoint.

{running-config | startup-config}

Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying **ckpt1** checkpoint to the running configuration:

```
switch# copy checkpoint ckpt1 running-config
Success
```

Copying **ckpt1** checkpoint to the startup configuration:

```
switch# copy checkpoint ckpt1 startup-config
Success
```

copy checkpoint *<CHECKPOINT-NAME>* *<STORAGE-URL>*

Syntax

```
copy checkpoint <CHECKPOINT-NAME> <STORAGE-URL>
```

Description

Copies an existing checkpoint configuration to a USB drive. The file format is defined when the checkpoint was created.

Command context

Manager (#)

Parameters

<CHECKPOINT-NAME>

Specifies the name of the checkpoint to copy. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).

<STORAGE-URL>>

Specifies the name of the target file on the USB drive using the following syntax: `usb:/<FILE>`

The USB drive must be formatted with the FAT file system.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying the `test` checkpoint to the `testCheck` file on the USB drive:

```
switch# copy checkpoint test usb:/testCheck
Success
```

copy *<REMOTE-URL>* checkpoint *<CHECKPOINT-NAME>*

Syntax

```
copy <REMOTE-URL> checkpoint <CHECKPOINT-NAME> [vrf <VRF-NAME>]
```

Description

Copies a remote configuration file to a checkpoint. The remote configuration file must be in checkpoint format.

Command context

Manager (`#`)

Parameters

<REMOTE-URL>

Specifies a remote file using the following syntax: `{tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>][;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>>`

<CHECKPOINT-NAME>

Specifies the name of the target checkpoint. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-). Required.

NOTE: Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

vrf *<VRF-NAME>*

Specifies a VRF name. Default: `default`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying a checkpoint format file to checkpoint **`ckpt5`** on the default VRF:

```
switch# copy tftp://192.168.1.10/ckptmeta checkpoint ckpt5
######################################################################## 100.0%
100.0%
Success
```

copy <REMOTE-URL> {running-config | startup-config}

Syntax

`copy <REMOTE-URL> {running-config | startup-config } [vrf <VRF-NAME>]`

Description

Copies a remote file containing a switch configuration to the running configuration or to the startup configuration.

Command context

Manager (`#`)

Parameters

<REMOTE-URL>

Specifies a remote file with the following syntax: `{tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>][;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>`

{running-config | startup-config}

Selects whether the running configuration or the startup configuration receives the copied checkpoint configuration. If the startup configuration is already present, the command overwrites the startup configuration.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The switch copies only certain file types. The format of the file is automatically detected from contents of the file. The `startup-config` option only supports the JSON file format and checkpoints, but the `running-config` option supports the JSON and CLI file formats and checkpoints.

When a file of the CLI format is copied, it overwrites the running configuration. The CLI command does not clear the running configuration before applying the CLI commands. All of the CLI commands in the file are applied line-by-line. If a particular CLI command fails, the switch logs the failure and it continues to the next line in the CLI configuration. The event log (`show events -d hpe-config`) provides information as to which command failed.

Examples

Copying a JSON format file to the running configuration:

```
switch# copy tftp://192.168.1.10/runjson running-config
######################################################################### 100.0%
Configuration may take several minutes to complete according to configuration file size
 --0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Success
```

Copying a CLI format file to the running configuration with an error in the file:

```
switch# copy tftp://192.168.1.10/runcli running-config
######################################################################### 100.0%
Configuration may take several minutes to complete according to configuration file size
 --0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Some of the configuration lines from the file were NOT applied. Use 'show
events -d hpe-config' for more info.
```

Copying a CLI format file to the startup configuration:

```
switch# copy tftp://192.168.1.10/startjson startup-config
########################################################################## 100.0%
100.0%
Success
```

Copying an unsupported file format to the startup configuration:

```
switch# copy tftp://192.168.1.10/startfile startup-config
########################################################################## 100.0%
100.0%
unsupported file format
```

copy running-config {startup-config | checkpoint *<CHECKPOINT-NAME>*}

Syntax

```
copy running-config {startup-config | checkpoint <CHECKPOINT-NAME>}
```

Description

Copies the running configuration to the startup configuration or to a new checkpoint. If the startup configuration is already present, the command overwrites the existing startup configuration.

Command context

Manager (#)

Parameters

`startup-config`

Specifies that the startup configuration receives a copy of the running configuration.

`checkpoint` ***<CHECKPOINT-NAME>***

Specifies the name of a new checkpoint to receive a copy of the running configuration. The checkpoint name can be alphanumeric. It can also contain underscores (_) and dashes (-).

> **NOTE:** Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying the running configuration to the startup configuration:

```
switch# copy running-config startup-config
Success
```

Copying the running configuration to a new checkpoint named **`ckpt1`**:

```
switch# copy running-config checkpoint ckpt1
Success
```

copy {running-config | startup-config} <REMOTE-URL>

Syntax

`copy {running-config | startup-config} <REMOTE-URL> {cli | json} [vrf <VRF-NAME>]`

Description

Copies the running configuration or the startup configuration to a remote file in either CLI or JSON format.

Command context

Manager (#)

Parameters

`{running-config | startup-config}`

Selects whether the running configuration or the startup configuration is copied to a remote file.

<REMOTE-URL>

Specifies the remote file using the syntax: `{tftp | sftp}://<IP-ADDRESS>[:<PORT-NUMBER>][;blocksize=<BLOCKSIZE-VALUE>]/<FILE-NAME>`

{cli | json}

Selects the remote file format: P: CLI or JSON.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Copying a running configuration to a remote file in CLI format:

```
switch# copy running-config tftp://192.168.1.10/runcli cli
######################################################################### 100.0%
Success
```

Copying a running configuration to a remote file in JSON format:

```
switch# copy running-config tftp://192.168.1.10/runjson json
######################################################################### 100.0%
Success
```

Copying a startup configuration to a remote file in CLI format:

```
switch# copy startup-config sftp://root@192.168.1.10/startcli cli
root@192.168.1.10's password:
sftp> put /tmp/startcli startcli
Uploading /tmp/startcli to /root/startcli
Connected to 192.168.1.10.
Success
```

Copying a startup configuration to a remote file in JSON format:

```
switch# copy startup-config sftp://root@192.168.1.10/startjson json
root@192.168.1.10's password:
sftp> put /tmp/startjson startjson
Uploading /tmp/startjson to /root/startjson
Connected to 192.168.1.10.
Success
```

copy {running-config | startup-config} *<STORAGE-URL>*

Syntax

`copy {running-config | startup-config} <STORAGE-URL> {cli | json}`

Description

Copies the running configuration or a startup configuration to a USB drive.

Command context

Manager (#)

Parameters

{running-config | startup-config}

Selects the running configuration or the startup configuration to be copied to the switch USB drive.

<STORAGE-URL>

Specifies a remote file with the following syntax: `usb:/<file>`

{cli | json}

Selects the format of the remote file: CLI or JSON.

Authority

Administrators or local user group members with execution rights for this command.

Usage

The switch supports JSON and CLI file formats when copying the running or starting configuration to the USB drive. The USB drive must be formatted with the FAT file system.

The USB drive must be enabled and mounted with the following commands:

```
switch(config)# usb
switch(config)# end
switch# usb mount
```

Examples

Copying a running configuration to a file named `runCLI` on the USB drive:

```
switch# copy running-config usb:/runCLI cli
Success
```

Copying a startup configuration to a file named `startCLI` on the USB drive:

```
switch# copy startup-config usb:/startCLI cli
Success
```

copy startup-config running-config

Syntax

`copy startup-config running-config`

Description

Copies the startup configuration to the running configuration.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy startup-config running-config
Success
```

copy <STORAGE-URL> running-config

Syntax

`copy <STORAGE-URL> {running-config | startup-config | checkpoint <CHECKPOINT-NAME>}`

Description

This command copies a specified configuration from the USB drive to the running configuration, to a startup configuration, or to a checkpoint.

Command context

Manager (#)

Parameters

<STORAGE-URL>

Specifies the name of a configuration file on the USB drive with the syntax: `usb:/<FILE>`

`running-config`

Specifies that the configuration file is copied to the running configuration. The file must be in CLI, JSON, or checkpoint format or the copy will fail. the copy will not work.

`startup-config`

Specifies that the configuration file is copied to the startup configuration. The switch stores this configuration between reboots. The startup configuration is used as the operating configuration following a reboot of the switch. The file must be in JSON or checkpoint format or the copy will fail.

`checkpoint` ***<CHECKPOINT-NAME>***

Specifies the name of a new checkpoint file to receive a copy of the configuration. The configuration file on the USB drive must be in checkpoint format.

NOTE: Do not start the checkpoint name with `CPC` because it is used for system-generated checkpoints.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command requires that the USB drive is formatted with the FAT file system and that the file be in the appropriate format as follows:

- `running-config`: This option requires the file on the USB drive be in CLI, JSON, or checkpoint format.
- `startup-config`: This option requires the file on the USB drive be in JSON or checkpoint format.
- `checkpoint <checkpoint-name>`: This option requires the file on the USB drive be in checkpoint format.

Examples

Copying the file **`runCli`** from the USB drive to the running configuration:

```
switch# copy usb:/runCli running-config
Configuration may take several minutes to complete according to configuration
file size
--0%----10%----20%----30%----40%----50%----60%----70%----80%----90%----100%--
Success
```

Copying the file **`startUp`** from the USB drive to the startup configuration:

```
switch# copy usb:/startUp startup-config
Success
```

Copying the file **testCheck** from the USB drive to the **abc** checkpoint:

```
switch# copy usb:/testCheck checkpoint abc
Success
```

erase {checkpoint *<CHECKPOINT-NAME>* | startup-config | all}

Syntax

`erase {checkpoint <CHECKPOINT-NAME> | startup-config | all}`

Description

Deletes an existing checkpoint, startup configuration, or all checkpoints.

Command context

Manager (#)

Parameters

checkpoint *<CHECKPOINT-NAME>*

Specifies the name of a checkpoint.

startup-config

Specifies the startup configuration.

all

Specifies all checkpoints.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Erasing checkpoint **ckpt1**:

```
switch# erase checkpoint ckpt1
```

Erasing the startup configuration:

```
switch# erase startup-config
```

Erasing all checkpoints:

```
switch# erase checkpoint all
```

show checkpoint *<CHECKPOINT-NAME>*

Syntax

`show checkpoint <CHECKPOINT-NAME> [json]`

Description

Shows the configuration of a checkpoint.

Command context

Manager (#)

Parameters

checkpoint ***<CHECKPOINT-NAME>***

Specifies the name of a checkpoint.

[json]

Specifies that the output is displayed in JSON format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the configuration of the `ckpt1` checkpoint in non-JSON format:

```
switch# show checkpoint ckpt1
!Description: 'switch'
!Version OpenSwitch 0.3.0-rc0 (Build: xxxxxxxxxxx)
!Schema version 0.1.8
hostname leaf07
logrotate period hourly
timezone set utc
ntp server 11.1.1.35
logging 11.0.1.11
!
!
!
!
vlan 1
    no shutdown
interface 30
    no shutdown
    ip address 12.20.27.2/30
interface 31
    no shutdown
    ip address 12.20.37.2/30
interface 32
    no shutdown
    ip address 12.20.47.2/30
```

Showing the configuration of the `ckpt1` checkpoint in JSON format:

```
switch# show checkpoint ckpt1 json
Checkpoint configuration:
{
    "AAA_Server_Group": {
        "local": {
            "group_name": "local"
        },
        "none": {
            "group_name": "none"
        }
    },
...
...
...
...
```

show checkpoint post-configuration

Syntax

`show checkpoint post-configuration`

Description

Shows the configuration settings for creating system checkpoints.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# show checkpoint post-configuration

Checkpoint Post-Configuration feature
------------------------------------

 Status         : enabled
 Timeout (sec)     : 300
```

show checkpoint list

Syntax

`show checkpoint list {all | <START-DATE> <END-DATE>}`

Description

Shows a list of saved checkpoints.

Command context

Manager (#)

Parameters

all

Shows a detailed list of all saved checkpoints.

<START-DATE>

Specifies the starting date for the range of saved checkpoints to show. Format: `YYYY-MM-DD`.

<END-DATE>

Specifies the ending date for the range of saved checkpoints to show. Format: `YYYY-MM-DD`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the list of saved checkpoints:

```
switch# show checkpoint list
ckpt1
ckpt2
ckpt3
startup-config
AUTO20170308214100
```

Showing a detailed list of all saved system and user checkpoints:

```
switch# show checkpoint list all
|NAME                 |TYPE          |WRITER   |DATE(UTC)              |HARDWARE |IMAGE VERSION        |
|CPC20171017195548    |checkpoint    |System   |2020-10-17T19:55:48Z  |00000    |xx.10.0x.xxxxX-1-g7691be0|
|ckpt23               |checkpoint    |User     |2020-10-17T20:09:00Z  |00000    |xx.10.0x.xxxxX-1-g7691be0|
|ckpt24               |checkpoint    |User     |2020-10-17T20:09:09Z  |00000    |xx.10.0x.xxxxX-1-g7691be0|
|ckpt30               |checkpoint    |User     |2020-10-17T20:10:36Z  |00000    |xx.10.0x.xxxxX-1-g7691be0|
|startup-config-backup|checkpoint    |System   |2020-10-17T20:17:56Z  |00000    |xx.10.0x.xxxxX-1-g7691be0|
|CPC20171017201712    |checkpoint    |System   |2020-10-17T20:19:12Z  |00000    |xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
|ckpt31               |checkpoint    |User     |2020-10-17T20:19:24Z  |00000    |xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
|startup-config       |startup       |User     |2020-10-17T20:47:11Z  |00000    |xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
|ckpt32               |checkpoint    |User     |2020-10-17T20:50:24Z  |00000    |xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
|CPC20171017205110    |checkpoint    |System   |2020-10-17T20:51:10Z  |00000    |xx.10.0x.xxxxAZ-10-g4c6b4446bd6|
```

Showing a detailed list of saved checkpoints for a specific date range:

```
switch# show checkpoint list date 2020-03-08 2020-03-12
|NAME  |TYPE       |WRITER |DATE                |HARDWARE |IMAGE VERSION |
|ckpt2 |checkpoint |User   |2020-03-08 18:10:01 |         |0.0.0         |
|ckpt3 |checkpoint |User   |2020-03-09 23:11:02 |         |0.0.0         |
|ckpt4 |checkpoint |User   |2020-03-11 00:00:03 |         |0.0.0         |
```

write memory

Syntax

```
write memory
```

Description

Saves the running configuration to the startup configuration. It is an alias of the command `copy running-config startup-config`. If the startup configuration is already present, this command overwrites the startup configuration.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# write memory
Success
```

Boot commands

boot set-default

Syntax

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Command context

Manager (#)

Parameters

primary

Selects the primary network operating system image.

secondary

Selects the secondary network operating system image.

Authority

Administrators or local user group members with execution rights for this command.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

boot system

Syntax

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Command context

Manager (#)

Parameters

primary

Selects the primary operating system image for this reboot and sets the configured default operating system image to `primary` for future reboots.

secondary

Selects the secondary operating system image for this reboot and sets the configured default operating system image to `secondary` for future reboots.

serviceos

Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the ArubaOS-CX operating system and is used in rare cases when troubleshooting a switch.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the `show images` command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the `primary` or `secondary` optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

  You can use the `boot set-default` command to change the configured default operating system image.
- If you select `serviceos` as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the `boot system` command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the `boot system` command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n
```

```
This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

show boot-history

Syntax

`show boot-history [all]`

Description

Shows boot information. When no parameters are specified, shows the most recent information about the boot operation, and the three previous boot operations for the active management module. When the `all` parameter is specified, shows the boot information for the active management module and all available line modules. To view boot-history on the standby, the command must be sent on the standby console.

Command context

Manager (#)

Parameters

`all`

Shows boot information for the active management module and all available line modules.

Authority

Administrators or local user group members with execution rights for this command.

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

`Index`

The position of the boot in the history file. Range: 0 to3.

`Boot ID`

A unique ID for the boot . A system-generated 128-bit string.

Current Boot, up for *<SECONDS>* seconds

For the current boot, the `show boot-history` command shows the number of seconds the module has been running on the current software.

Timestamp boot reason

For previous boot operations, the `show boot-history` command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values:

***<DAEMON-NAME>* `crash`**

The daemon identified by *<DAEMON-NAME>* caused the module to boot.

`Kernel crash`

The operating system software associated with the module caused the module to boot.

Reboot requested through database

The reboot occurred because of a request made through the CLI or other API.

Uncontrolled reboot

The reason for the reboot is not known.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=================

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
switch#
```

Showing the boot history of the active management module and all line modules:

```
switch# show boot-history all
Management module
=================

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=================
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...
```

Firmware management commands

copy {primary | secondary} <REMOTE-URL>

Syntax

`copy {primary | secondary} <REMOTE-URL> [vrf <VRF-NAME>]`

Description

Uploads a firmware image to a TFTP or SFTP server.

Command context

Manager (#)

Parameters

{primary | secondary}

Selects the primary or secondary image profile to upload. Required

<REMOTE-URL>

Specifies the URL to receive the uploaded firmware using SFTP or TFTP. For information on how to format the remote URL, see **URL formatting for copy commands**.

vrf *<VRF-NAME>*

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

TFTP upload:

```
switch# copy primary tftp://192.0.2.0/00_10_00_0002.swi
######################################################################## 100.0%
Verifying and writing system firmware...
```

SFTP upload:

```
switch# copy primary sftp://swuser@192.0.2.0/00_10_00_0002.swi
swuser@192.0.2.0's password:
Connected to 192.0.2.0.
sftp> put primary.swi XL_10_00_0002.swi
Uploading primary.swi to /users/swuser/00_10_00_0002.swi
primary.swi                               100%  179MB  35.8MB/s   00:05
```

copy {primary | secondary} <FIRMWARE-FILENAME>

Syntax

`copy {primary | secondary} <FIRMWARE-FILENAME>`

Description

Copies a firmware image to USB storage.

Command context

Manager (#)

Parameters

{primary | secondary}

Selects the primary or secondary image from which to copy the firmware. Required

<FIRMWARE-FILENAME>

Specifies the name of the firmware file to create on the USB storage device. Prefix the filename with `usb:/`. For example: `usb:/firmware_v1.2.3.swi`

For information on how to format the path to a firmware file on a USB drive, see **USB URL**.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy primary usb:/11.10.00.0002.swi
```

copy primary secondary

Syntax

`copy primary secondary`

Description

Copies the firmware image from the primary to the secondary location.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy primary secondary
The secondary image will be deleted.

Continue (y/n)? y
Verifying and writing system firmware...
```

copy *<REMOTE-URL>*

Syntax

`copy <REMOTE-URL> {primary | secondary} [vrf <VRF-NAME>]`

Description

Downloads and installs a firmware image from a TFTP or SFTP server.

Command context

Manager (#)

Parameters

<REMOTE-URL>

Specifies the URL from which to download the firmware using SFTP or TFTP. For information on how to format the remote URL, see **URL formatting for copy commands**.

{primary | secondary}

Selects the primary or secondary image profile for receiving the downloaded firmware. Required.

vrf *<VRF-NAME>*

Specifies the name of a VRF. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

TFTP download:

```
switch#  copy tftp://192.10.12.0/ss.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y
######################################################################## 100.0%
Verifying and writing system firmware...
```

SFTP download:

```
switch# copy sftp://swuser@192.10.12.0/ss.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.10.12.0 (192.10.12.0)' can't be established.
ECDSA key fingerprint is SHA256:L64khLwlyLgXlARKRMiwcAAK8oRaQ8C0oWP+PkGBXHY.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.10.12.0' (ECDSA) to the list of known hosts.
swuser@192.10.12.0's password:
Connected to 192.10.12.0.
Fetching /users/swuser/ss.10.00.0002.swi to ss.10.00.0002.swi.dnld
/users/swuser/ss.10.00.0002.swi                   100%  179MB  25.6MB/s   00:07
Verifying and writing system firmware...
```

copy secondary primary

Syntax

```
copy secondary primary
```

Description

Copies the firmware image from the secondary to the primary location.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy secondary primary
The primary image will be deleted.

Continue (y/n)? y
Verifying and writing system firmware...
```

```
switch# copy sftp://stor@192.22.1.0/im-switch.swi primary vrf mgmt
The primary image will be deleted.

Continue (y/n)? y
The authenticity of host '192.22.1.0 (192.22.1.0)' can't be established.
ECDSA key fingerprint is SHA256:MyI1xbdKnehYut0NLfL69gDpNzCmZqBVvBaRR46m7o8.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '192.22.1.0' (ECDSA) to the list of known hosts.
stor@192.22.1.0's password:
Connected to 192.22.1.0.
sftp> get c8d5b9f-topflite.swi c8d5b9f-topflite.swi.dnld
Fetching /home/dr/im-switch.swi to c8d5b9f-topflite.swi.dnld
/home/dr/im-switch.swi                100%  226MB  56.6MB/s   00:04

Verifying and writing system firmware...
```

copy *<STORAGE-URL>*

Syntax

`copy <STORAGE-URL> {primary | secondary}`

Description

Copies, verifies, and installs a firmware image from a USB storage device connected to the active management module.

Command context

Manager (#)

Parameters

<STORAGE-URL>

Specifies the name of the firmware file to copy from the USB storage device. Required. Prefix the filename with `usb:/`. For example, `usb:/firmware_v1.2.3.swi` For information on how to format the path to a firmware file on a USB drive, see **USB URL**.

{primary | secondary}

Selects the primary or secondary image profile for receiving the copied firmware.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch# copy usb:/11.10.00.0002.swi primary
The primary image will be deleted.

Continue (y/n)? y

Verifying and writing system firmware...
```

URL formatting for copy commands

TFTP URL

Syntax

tftp://<IP-ADDR>[:<PORT-NUM>][;blocksize=<Value>]/<FILENAME>

Examples

To specify a URL with:

- an IPv4 address: tftp://1.1.1.1/a.txt
- an IPv6 address: tftp://[2000::2]/a.txt
- a hostname: tftp://hpe.com/a.txt

To specify TFTP with:

- the port number of the server in the URL: tftp://1.1.1.1:12/a.txt
- the blocksize in the URL: tftp://1.1.1.1;blocksize=1462/a.txt

  The valid blocksize range is 8 to 65464.
- the port number of the server and blocksize in the URL: tftp://1.1.1.1:12;blocksize=1462/a.txt

To specify a file in a directory of URL: tftp://1.1.1.1/dir/a.txt

SFTP URL

Syntax

sftp://<USERNAME>@<IP-ADDR>[:<PORT-NUM>]/<FILENAME>

Examples

To specify:

- A URL with an IPv4 address: sftp://user@1.1.1.1/a.txt
- A URL with an IPv6 address: sftp://user@[2000::2]/a.txt
- A URL with a hostname: sftp://user@hpe.com/a.txt
- SFTP port number of a server in the URL: sftp://user@1.1.1.1:12/a.txt
- A file in a directory of URL: sftp://user@1.1.1.1/dir/a.txt
- To specify a file with absolute path in the URL: sftp://user@1.1.1.1//home/user/a.txt

USB URL

Syntax

usb:/<FILENAME>

Examples

To specify a file:

- In a USB storage device: `usb:/`*`a.txt`*
- In a directory of a USB storage device: `usb:/`*`dir`*`/`*`a.txt`*

Simple Network Management Protocol (SNMP) is an Internet-standard protocol used for managing and monitoring the devices connected to a network by collecting, organizing and modifying information about managed devices on IP networks.

Configuring SNMP

(The SNMP agent provides read-only access.)

Procedure

1. Enable SNMP on a VRF using the command `snmp-server vrf`.
2. Set the system contact, location, and description for the switch with the following commands:
   - `snmp-server system-contact`
   - `snmp-server system-location`
   - `snmp-server system-description`
3. If required, change the default SNMP port on which the agent listens for requests with the command `snmp-server agent-port`.
4. By default, the agent uses the community string **`public`** to protect access through SNMPv1/v2c. Set a new community string with the command `snmp-server community`.
5. Configure the trap receivers to which the SNMP agent will send trap notifications with the command `snmp-server host`.
6. Create an SNMPv3 context and associate it with any available SNMPv3 user to perform context specific v3 MIB polling using the command `snmpv3 user` .
7. Create an SNMPv3 context and associate it with an available SNMPv1/v2c community string to perform context specific v1/v2c MIB polling using the command `snmpv3 context`.
8. Review your SNMP configuration settings with the following commands:
   - `show snmp agent-port`
   - `show snmp community`
   - `show snmp system`
   - `show snmpv3 context`
   - `show snmp trap`
   - `show snmp vrf`
   - `show snmpv3 users`
   - `show tech snmp`

Example

Example 1

This example creates the following configuration:

- Enables SNMP on the out-of-band management interface (VRF **`mgmt`**).
- Sets the contact, location, and description for the switch to: **`JaniceM`**, **`Building2`**, **`LabSwitch`**.
- Sets the community string to **`Lab8899X`**.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server system-contact JaniceM
switch(config)# snmp-server system-location Building2
switch(config)# snmp-server system-description LabSwitch
switch(config)# snmp-server community Lab8899X
```

Example 2

This example creates the following configuration:

- Creates an SNMPv3 user named **`Admin`** using **`sha`** authentication with the plaintext password **`mypassword`** and using **`des`** security with the plaintext password **`myprivpass`**.
- Associates the SNMPv3 user `Admin` with a context named `newContext`.

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des
     priv-pass plaintext myprivpass
switch(config)# snmpv3 user Admin context newContext
```

SNMP commands

rmon alarm

Syntax

```
rmon alarm index <INDEX> snmp-oid <SNMP-OID> rising-threshold <RISING-THRESHOLD>
     falling-threshold <FALLING-THRESHOLD> [sample-interval <SAMPLE-INTERVAL>] [sample-type <ABSOLUTE|DELTA>]

no rmon alarm [index <INDEX>]
```

Description

Stores configuration entries in an alarm table that defines the sample interval, sample-type, and threshold parameters for an SNMP MIB object. Only the SNMP MIB objects that resolve to an ASN.1 primitive type of INTEGER (INTEGER, Integer32, Counter32, Counter64, Gauge32, or TimeTicks) will be monitored.

The `no` form of this command removes all RMON alarms and allows you to specify an index to remove a particular RMON alarm.

Command context

`config`

Parameters

`index <INDEX>`

Specifies the RMON alarm index. Range: 1 to 20.

snmp-oid ***<SNMP-OID>***

Specifies the SNMP MIB object to be monitored by RMON.

rising-threshold ***<RISING-THRESHOLD>***

Specifies the upper threshold value for the RMON alarm.

falling-threshold ***<FALLING-THRESHOLD>***

Specifies the falling threshold value for the RMON alarm. The falling threshold must be less than the rising threshold.

sample-interval ***<SAMPLE-INTERVAL>***

Sample interval in seconds. Default: 30.

sample-type ***<ABSOLUTE|DELTA>***

Specifies the method of sampling of the SNMP MIB object. Default: Absolute.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring RMON for the MIB object **ifOutErrors.15** with an index **1**, rising threshold of **2147483647** and falling threshold of **-2134** using **absolute** sampling for a sample interval of **100** seconds:

```
switch(config)# rmon alarm index 1 snmp-oid ifOutErrors.15 rising-threshold 2147483647
     falling-threshold -2134 sample-type absolute sample-interval 100
```

Removing RMON alarm with the index 5:

```
switch(config)# no rmon alarm index 5
```

show rmon alarm

Syntax

```
show rmon alarm [index <INDEX>]
```

Description

Displays the RMON alarm configurations.

Command context

```
config
```

Parameters

index ***<INDEX>***

Specifies the RMON alarm index. Range: 1 to 20.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all RMON alarm configurations:

```
switch# show rmon alarm
Index                : 1
Enabled              : true
Status               : valid
MIB object           : ifOutErrors.15
Sample type          : delta
Sampling interval    : 6535 seconds
Rising threshold     : 100
Falling threshold    : 10
Last sampled value   : 0
Last sample time     : 2020-09-21 05:58:11

Index                : 3
Enabled              : true
Status               : invalid
MIB object           : IF-MIB::ifDescr.19
Sample type          : absolute
Sampling interval    : 10000 seconds
Rising threshold     : 4000
Falling threshold    : 10
Last sampled value   : 0
```

Showing RMON alarm with alarm index 1:

```
switch# show rmon alarm index 1
Index                : 1
Enabled              : true
Status               : valid
MIB object           : ifOutErrors.15
Sample type          : delta
Sampling interval    : 6535 seconds
Rising threshold     : 100
Falling threshold    : 10
Last sampled value   : 0
Last sample time     : 2020-06-21 05:58:11
```

Showing disabled RMON alarm information:

```
switch# show  rmon   alarm
Index                : 1
Enabled              : false
Status               : valid
MIB object           : ifOutErrors.15
Sample type          : delta
Sampling interval    : 6535 seconds
Rising threshold     : 100
Falling threshold    : 10
Last sampled value   : 0
Last sample time     : 2020-09-21 05:58:11

Index                : 3
Enabled              : false
Status               : invalid
MIB object           : IF-MIB::ifDescr.19
Sample type          : absolute
Sampling interval    : 10000 seconds
Rising threshold     : 4000
Falling threshold    : 10
Last sampled value   : 0
```

show snmp agent-port

Syntax

```
show snmp agent-port
```

Description

Displays SNMP agent UDP port number.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP agent UDP port number:

```
switch# show snmp agent-port
SNMP agent port : 161
```

show snmp community

Syntax

```
show snmp community
```

Description

Displays a list of all configured SNMPv1/v2c communities.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

When a user creates a custom community before enabling an SNMP agent, AOS-CX automatically removes the default `public` community from the system.

Example

Displaying a list of all configured SNMPv1/v2c communities:

Before any community is created by user

```
switch# show snmp community
---------------------
SNMP communities
---------------------
public
```

After community is created by user

```
switch#show snmp community
---------------------
SNMP communities
---------------------
private
private2
```

show snmp system

Syntax

```
show snmp system
```

Description

Displays SNMP description, location, and contact information.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP description, location, and contact information:

```
switch# show snmp system
SNMP system information
----------------------------
System description : Aggregation router
System location : Main lab
System contact : John Smith, Lab Admin
```

show snmp trap

Syntax

```
show snmp trap
```

Description

Displays all configured SNMP traps/informs receivers.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying all configured SNMP trap and informs receivers:

```
switch# show snmp trap
HOST                                        PORT  TYPE   VER COMMUNITY/USER NAME VRF
```

```
--------------------------------------------------------------------------------------
10.10.10.10                             162   trap   v1  public             default
10.10.10.10                             162   inform v2c public             default
10.10.10.10                             162   inform v3  name               default
```

show snmp vrf

Syntax

`show snmp vrf`

Description

Displays the VRF on which the SNMP agent service is running.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying SNMP services enabled on VRF:

```
switch#show snmp vrf
SNMP enabled VRF
----------------------------
mgmt
default
```

show snmpv3 context

Syntax

`show snmpv3 context`

Description

Displays all configured SNMP contexts.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying all configured SNMP contexts:

```
switch# show snmpv3 context
--------------------------------------------------------------------------
name                            vrf                             community
--------------------------------------------------------------------------
```

```
contextA                        default                         private
contextB                        vrf_A                           public
```

show snmpv3 engine-id

Syntax

`show snmpv3 engine-id`

Description

Displays the configured SNMPv3 snmp engine-id.

If the SNMPv3 engine-id is not configured, by default a unique engine-id is created by the switch using a combination of the enterprise OID value and the switch's mac address.

Command context

Operator (`>`) or Manager (`#`)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying the configured SNMPv3 engine-id:

```
switch# show snmpv3 engine-id
SNMP engine-id : 80:00:B8:5C:08:00:09:1d:de:a5
```

show snmpv3 security-level

Syntax

`show snmpv3 security-level`

Description

Displays the configured SNMPv3 security level.

Command context

Operator (`>`) or Manager (`#`)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying the configured SNMPv3 security level:

```
switch# show snmpv3 security-level
SNMPv3 security-level : auth
```

show snmpv3 users

Syntax

```
show snmpv3 users
```

Description

Displays all configured SNMPv3 users.

For more details on the user enabled status, see **snmpv3 security-level**.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying all configured SNMPv3 users:

```
switch# show snmpv3 users
---------------------------------------------------------------------
User                            AuthMode  PrivMode  Context  Enabled
---------------------------------------------------------------------
name                            md5       none      none     False
name2                           sha       aes       none     True
```

snmp-server agent-port

Syntax

```
snmp-server agent-port <PORT>

no snmp-server agent-port [<PORT>]
```

Description

Sets the UDP port number that the SNMP master agent uses to communicate. UDP port 161 is the default port.

The `no` form of this command sets the SNMP master agent port to the default value.

Command context

`config`

Parameters

<PORT>

Specifies the UDP port number that the SNMP master agent will use. Range: 1 to 65535. Default: 161.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMP master agent port to **2000**:

```
switch(config)# snmp-server agent-port 2000
```

Resetting the SNMP master agent port to the default value:

```
switch(config-schedule)# no snmp-server agent-port 2000
```

snmp-server community

Syntax

```
snmp-server community <STRING>

no snmp-server community <STRING>
```

Description

Adds an SNMPv1/SNMPv2c community string. A community string is a password that controls read access to the SNMP agent. A network management program must supply this name when attempting to get SNMP information from the switch. A maximum of 10 community strings are supported. Once you create your own community string, the default community string (`public`) is deleted.

The `no` form of this command removes the specified SNMPv1/SNMPv2c community string. When no community string exists, a default community string with the value `public` is automatically defined.

Command context

`config`

Parameters

<STRING>

Specifies the SNMPv1/SNMPv2c community string. Range: 1 to 32 printable ASCII characters, excluding space and question mark.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMPv1/SNMPv2c community string to **private**:

```
switch(config)# snmp-server community private
```

Removing SNMPv1/SNMPv2c community string **private**:

```
switch(config)# no snmp-server community private
```

snmp-server host

Syntax

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]

no snmp-server host <IPv4-ADDR | IPv6-ADDR> trap version <VERSION> [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]
```

```
snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]

no snmp-server host <IPv4-ADDR | IPv6-ADDR> inform version v2c [community <STRING>]
[port <UDP-PORT>] [<VRF-NAME>]

snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [<VRF-NAME>]

no snmp-server host <IPv4-ADDR | IPv6-ADDR> [trap version v3 | inform version v3] user <NAME>
[port <UDP-PORT>] [<VRF-NAME>]
```

Description

Configures a trap/informs receiver to which the SNMP agent can send SNMP v1/v2c/v3 traps or v2c informs. A maximum of 30 SNMP traps/informs receivers can be configured.

The `no` form of this command removes the specified trap/inform receiver.

Command context

`config`

Parameters

<IPv4-ADDR>

Specifies the IP address of a trap receiver in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255. You can remove leading zeros. For example, the address `192.169.005.100` becomes `192.168.5.100`.

<IPv6-ADDR>

Specifies the IP address of a trap receiver in IPv6 format (`x:x::x:x`).

trap version *<VERSION>*

Specifies the trap notification type for SNMPv1, v2c or v3. Available options are: `v1`, `v2c` or `v3`.

inform version v2c

Specifies the inform notification type for SNMPv2c.

trap version v3

Specifies the trap notification type for SNMPv3.

user *<NAME>*

Specifies the SNMPv3 user name to be used in the SNMP trap notifications.

community *<STRING>*

Specifies the name of the community string to use when sending trap notifications. Range: 1 - 32 printable ASCII characters, excluding space and question mark. Default: `public`.

<UDP-PORT>

Specifies the UDP port on which notifications are sent. Range: 1 - 65535. Default: 162.

<VRF-NAME>

Specifies the VRF on which the SNMP agent listens for incoming requests.

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config)# snmp-server host 10.10.10.10 trap version v1
switch(config)# no snmp-server host 10.10.10.10 trap version v1

switch(config)# snmp-server host a:b::c:d trap version v1
switch(config)# no snmp-server host a:b::c:d trap version v1

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public

switch(config)# snmp-server host a:b::c:d trap version v2c community public
switch(config)# no snmp-server host a:b::c:d trap version v2c community public

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port 5000
switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v2c community public port 5000 vrf default

switch(config)# no snmp-server host 10.10.10.10 trap version v2c community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d trap version v2c community public port 5000
switch(config)# no snmp-server host a:b::c:d trap version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public

switch(config)# snmp-server host a:b::c:d inform version v2c community public
switch(config)# no snmp-server host a:b::c:d inform version v2c community public

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public port 5000
switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 inform version v2c community public port 5000 vrf default

switch(config)# no snmp-server host 10.10.10.10 inform version v2c community public port 5000 vrf default

switch(config)# snmp-server host a:b::c:d inform version v2c community public port 5000
switch(config)# no snmp-server host a:b::c:d inform version v2c community public port 5000

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin

switch(config)# snmp-server host 10.10.10.10 trap version v3 user Admin port 2000
switch(config)# no snmp-server host 10.10.10.10 trap version v3 user Admin port 2000

switch(config)# snmp-server host a:b::c:d trap version v3 user Admin port 2000
switch(config)# no snmp-server host a:b::c:d trap version v3 user Admin port 2000
```

snmp-server system-contact

Syntax

snmp-server system-contact *<INFO>*

no snmp-server system-contact [*<INFO>*]

Description

Sets SNMP contact information.

The no form of this command removes the SNMP contact information.

Command context

config

Parameters

<INFO>

Specifies SNMP contact information. Range: 1 to 128 printable ASCII characters, except for question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines SNMP contact information to be **`John Smith, Lab Admin`**:

```
switch(config)# snmp-server system-contact John Smith, Lab Admin
```

Removes SNMP contact information:

```
switch(config)# no snmp-server system-contact
```

snmp-server system-description

Syntax

```
snmp-server system-description <DESCRIPTION>
```

```
no snmp-server system-description
```

Description

Sets the SNMP system description.

The `no` form of this command removes the SNMP system description.

Command context

`config`

Parameters

<DESCRIPTION>

Specifies the SNMP system description. Typical content to include would be the full name and version of the following:

- Hardware type of the system
- Software operating system
- Networking software

Range: 1 to 64 printable ASCII characters, except for the question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines the SNMP system description to be **`mainSwitch`**:

```
switch(config)# snmp-server system-description mainSwitch
```

Removes the SNMP system description:

```
switch(config)# no snmp-server system-description mainSwitch
```

snmp-server system-location

Syntax

```
snmp-server system-location <INFO>

no snmp-server system-location
```

Description

Sets the SNMP location information.

The `no` form of this command removes the SNMP location information.

Command context

`config`

Parameters

<INFO>

Specifies the SNMP location information. Range: 1 to 128 printable ASCII characters, except for the question mark (?).

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defines the SNMP location information to be **`Main Lab`**:

```
switch(config)# snmp-server system-location Main Lab
```

Removes the SNMP location information:

```
switch(config)# no snmp-server system-location
```

snmp-server vrf

Syntax

```
snmp-server vrf <VRF-NAME>

no snmp-server vrf <VRF-NAME>
```

Description

Configures a VRF on which the SNMP agent listens for incoming requests. By default, the SNMP agent does not listen on any VRF. The SNMP agent can listen on multiple VRFs.

The `no` form of this command stops the SNMP agent from listening for incoming requests on the specified VRF.

Command context

`config`

Parameters

<VRF-NAME>

Specifies the name of a VRF.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the SNMP agent to listen on VRF `default`.

```
switch(config)# snmp-server vrf default
```

Configuring the SNMP agent to listen on VRF `mgmt`.

```
switch(config)# snmp-server vrf mgmt
```

Configuring the SNMP agent to listen on used-defined VRF `myvrf`.

```
switch(config)# snmp-server vrf myvrf
```

Stopping the SNMP agent from listening on VRF `default`.

```
switch(config)# no snmp-server vrf default
```

snmpv3 context

Syntax

```
snmpv3 context <NAME> vrf <VRF-NAME> [community <STRING>]

no snmpv3 context <NAME> [vrf <VRF-NAME>]
```

Description

Creates an SNMPv3 context on the specified VRF.

The `no` form of this command removes the specified SNMP context.

Command context

`config`

Parameters

<NAME>

Specifies the name of the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark (?).

vrf *<VRF-NAME>*

Specifies the VRF associated with the context. Default: default.

community *<STRING>*

Specifies the SNMP community string associated with the context. Range: 1 to 32 printable ASCII characters, excluding space and question mark. Default: public.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating an SNMPv3 context named **newContext**:

```
switch(config)# snmpv3 context newContext
```

Creating an SNMPv3 context named **newContext** on VRF **myVrf** and with community string **private**.

```
switch(config)# snmpv3 context newContext vrf myVrf community private
```

Removing the SNMPv3 context named **newContext** on VRF **myVrf**:

```
switch(config)# no snmpv3 context newContext vrf myVrf
```

snmpv3 engine-id

Syntax

```
snmpv3 engine-id <ENGINE-ID>

no snmpv3 engine-id <ENGINE-ID>
```

Description

Configures the SNMPv3 SNMP engine-id allowing an administrator to configure a unique SNMP engine-id for the switch. This engine-id is used by the NMS management tool to identify and distinguish multiple switches on the same network.

The `no` form of this command restores the default engine-id, created by the switch using a combination of the enterprise OID value and the switch's mac address.

Command context

`config`

Parameters

<ENGINE-ID>

SNMPv3 SNMP engine-id in colon separated hexadecimal notation.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Configuring the SNMPv3 engine-id:

```
switch(config)#
switch(config)# snmpv3 engine-id
  WORD  SNMPv3 snmp engine-id in colon seperated hexadecimal notation
switch(config)# snmpv3 engine-id 01:23:45:67:89:ab:cd:ef:01:23:45:67
```

Restoring the default SNMPv3 engine-id:

```
switch(config)# no snmpv3 engine-id
```

snmpv3 security-level

Syntax

```
snmpv3 security-level {auth | auth-privacy}

no snmpv3 security-level {auth | auth-privacy}
```

Description

Configures the SNMPv3 security level. The security level determines which SMNPv3 users defined by the command `snmpv3 user` are able to connect.

The `no` form of this command changes the security level as follows:

- `no snmpv3 security-level auth`: Sets the security level to `auth-privacy`.
- `no snmpv3 security-level auth-privacy`: Sets the security level to no authentication or privacy, allowing any SNMP user to connect.

Command context

`config`

Parameters

`auth`

SNMPv3 users that support authentication, or authentication and privacy are allowed.

`auth-privacy`

Only SNMPv3 users with both authentication and privacy are allowed. This is the highest level of SNMPv3 security. Default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the SNMPv3 security level to authentication and privacy:

```
switch(config)# snmpv3 security-level auth-privacy
```

Setting the SNMPv3 security level to authentication only:

```
switch(config)# snmpv3 security-level auth
```

Setting the SNMPv3 security level to no authentication and no privacy:

```
switch(config)# no snmpv3 security-level auth-privacy
```

Restoring the default SNMPv3 security level to authentication and privacy:

```
switch(config)# no snmpv3 security-level auth
```

snmpv3 user

Syntax

```
snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass {plaintext | ciphertext}
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass {plaintext | ciphertext} <PRIV-PWORD>] ]

no snmpv3 user <NAME> [auth <AUTH-PROTOCOL> auth-pass
<AUTH-PWORD> [priv <PRIV-PROTOCOL> priv-pass <PRIV-PWORD>] ]
```

Description

Creates an SNMPv3 user and adds it to an SNMPv3 context. The `no` form of this command removes the specified SNMPv3 user.

For more details on the user enabled status, see **snmpv3 security-level**.

Command context

`config`

Parameters

<NAME>

Specifies the SNMPv3 username. Range 1 to 32 printable ASCII characters, excluding space and question mark (?).

auth *<AUTH-PROTOCOL>*

Specifies the authentication protocol used to validate user logins. Available options are: `md5` or `sha`.

auth-pass {plaintext | ciphertext} *<AUTH-PWORD>*

Specifies the SNMPv3 user password. Range for `plaintext` is 8 to32 printable ASCII characters, excluding space and question mark (?).

Range for `ciphertext` is 1 to 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

priv *<PRIV-PROTOCOL>*

Specifies the SNMPv3 security protocol (encryption method). Available options are: `aes` or `des`.

priv-pass {plaintext | ciphertext} *<PRIV-PWORD>*

Specifies the SNMPv3 user privacy passphrase. Range for `plaintext` is 8 to 32 printable ASCII characters, excluding space and question mark (?).

Range for `ciphertext` is 1 to 120 printable ASCII characters. This option is only used when copying user configuration settings between switches. It enables you to duplicate a user's configuration on another switch without having to know their password.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des priv-pass plaintext myprivpass
```

Removing an SNMPv3 user named `Admin`:

```
switch(config)# no snmpv3 user Admin
```

Defining an SNMPv3 user named **Admin** using **sha** authentication with the plaintext password **mypassword** and using **des** security with the plaintext password **myprivpass**:

```
switch(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword priv des priv-pass plaintext myprivpass
```

Copying an SNMP user from switch 1 to switch 2.

On switch 1, configure a user called **Admin**, then issue the `show running-config` command to display switch configuration settings. The snmpv3 user command uses the `ciphertext` option to protect the users's passwords.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass plaintext mypassword
priv des priv-pass plaintext myprivpass
switch1(config)# exit
switch1# show running-config
Current configuration:
!
!Version ArubaOS-CX XL.10.04.0001AD
!
!
!
snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
ssh server vrf mgmt
!
!
!
!
interface mgmt
    no shutdown
    ip dhcp
vlan 1
```

On switch 2, execute the `snmpv3 user` command that was displayed by `show running-config` on switch 1. This creates the user on switch 2 with the same configuration settings.

```
switch1(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqbNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
```

CoPP provides a way for administrators to protect the management processor on the switch from high packet loads (generated by malicious or nonmalicious sources) that might interfere with its ability to keep data plane traffic flowing. For example, a denial of service attack can result in excessive traffic that would slow down the management processor and negatively affect switch throughput.

A CoPP policy is composed of one or more classes. Each class defines a target protocol and how its traffic is managed. Every policy also has a default class to regulate packets that do not match any other class. The following actions can be applied for all packets matching a class:

- Drop the packets. (Excluding the default class.)
- Set the processing priority in the range 0 to 7.
- Set the maximum data rate in packets per second (pps) at which each line module can send packets to the management processor.
- Set the maximum burst size in packets at which each line module can send packets to the management processor.

Up to 32 CoPP policies can be defined, but only one can be active on the switch at a time.

A CoPP policy must always be active on the switch. By default, the switch has a CoPP policy named **default** which is automatically applied at first boot.

When the switch is rebooted, the CoPP policy that was actively applied to the switch before the reboot occurred will be applied if it was saved to the startup configuration with the `copy running-config startup-config` command.

For GRE tunneled traffic, CoPP policies match on the payload.

CoPP policies do not regulate traffic received from the Out-of-Band-Management (OOBM) Ethernet port.

Configuring CoPP

Procedure

1. Configure the default CoPP policy, edit an existing policy, or create a policy with the command **`copp-policy`**.
2. Add, edit, or remove classes in the policy with the command **`class`**.
3. If the policy is not the active policy on the switch, apply it with the command **`apply copp-policy`**. (Changes made to an active policy take effect immediately and do not need to be applied.)
4. Review the CoPP policy configuration settings with the command **`show copp-policy`**.

Example

This example creates the following configuration:

- Defines a new policy named **My_CoppPolicy**.
- Adds two classes to the policy.
- Activates the policy.
- Displays policy configuration settings.

```
switch(config)# copp-policy My_CoppPolicy
switch(config-copp)# class bgp-ipv4 priority 6 rate 5000 burst 60
switch(config-copp)# class ospf-multicast priority 2 rate 2000
switch(config-copp)# exit
switch(config)# apply copp-policy My_CoppPolicy
switch(config)# exit
switch# show copp-policy My_CoppPolicy
class                 drop priority rate pps burst pkts
--------------------- ---- -------- --------- ---------
bgp-ipv4                   6        5000      5000
ospf-multicast-ipv4        2        2000      2000
default                    1        6000      6000
```

CoPP commands

Classes of traffic

The different classes of traffic that can be individually configured are:

- `acl-logging`: Access Control List logging packets.
- `arp-broadcast`: Address Resolution Protocol packets with a broadcast destination MAC address.
- `arp-unicast`: Address Resolution Protocol packets with a switch system destination MAC address.
- `bfd`: Bidirectional Forwarding Detection (BFD) packets with a destination IP address owned by the switch.
- `bgp-ipv4`: Border Gateway Protocol packets with a destination IPv4 address owned by the switch and the Layer 4 protocol is TCP.
- `bgp-ipv6`: Border Gateway Protocol packets with a destination IPv6 address owned by the switch and the Layer 4 protocol is TCP.
- `dhcp-ipv4`: Dynamic Host Configuration Protocol packets with a destination IPv4 address owned by the switch and the Layer 4 protocol is UDP.
- `dhcp-ipv6`: Dynamic Host Configuration Protocol packets with a destination IPv6 address owned by the switch and the Layer 4 protocol is UDP.
- `erps`: ERPS control packets with the destination MAC address 01:19:a7:00:00:XX snooped to the CPU, where XX can be any value.
- `hypertext`: Hypertext Transfer Protocol (HTTP) or Hypertext Transfer Protocol Secure (HTTPS) packets.
- `icmp-broadcast-ipv4`: Internet Control Message Protocol packets with the broadcast destination IPv4 address 255.255.255.255 or a destination IPv4 subnet broadcast address.
- `icmp-multicast-ipv6`: Internet Control Message Protocol packets with a well-known multicast destination IPv6 address.
- `icmp-unicast-ipv4`: Internet Control Message Protocol packets with a destination IPv4 address owned by the switch.

- `icmp-unicast-ipv6`: Internet Control Message Protocol packets with a destination IPv6 address owned by the switch.
- `igmp`: Internet Group Management Protocol packets.
- `ip-exceptions`: Internet Protocol exception packets.
- `ipsec`: Internet Protocol Security IPv4 or IPv6, unicast or configured multicast. All IPsec traffic received by the CPU will be regulated by the 'ipsec' class regardless of the encapsulated protocol.
- `ipv4-options`: Unicast IPv4 packets including option headers.
- `ipv6-options`: Unicast IPv6 packets including option headers.
- `lacp`: Link Aggregation Control Protocol packets with the destination MAC address 01:80:c2:00:00:02.
- `lldp`: Link Layer Discovery Protocol packets with the destination MAC address 01:80:c2:00:00:0e.
- `loop-protect`: Loop Protection packets with the destination MAC address 09:00:09:09:13:a6.
- `mirror-to-cpu`: Packets from mirroring session configured to deliver to the console.
- `mld`: Multicast Listener Discovery packets of type V1 or V2 with an IPv6 address of FF00::/8, FF02::16 or FF02::2.
- `mvrp`: Multiple VLAN Registration Protocol packets with the destination MAC address 01:80:c2:00:00:20 or 01:80:c2:00:00:21.
- `ntp`: Network Time Protocol packets with a destination address owned by the switch and the Layer 4 protocol is UDP.
- `ospf-multicast-ipv4`: Open Shortest Path First packets with the multicast destination IPv4 address 224.0.0.5 or 224.0.0.6.
- `ospf-multicast-ipv6`: Open Shortest Path First packets with the multicast destination IPv6 address FF02::5 or FF02::6.
- `ospf-unicast-ipv4`: Open Shortest Path First packets with a destination IPv4 address owned by the switch.
- `ospf-unicast-ipv6`: Open Shortest Path First packets with a destination IPv6 address owned by the switch.
- `pim`: Protocol Independent Multicast packets with the destination IPv4 address 224.0.0.13 or IPv6 address FF02::D, or with a destination IP address owned by the switch.
- `sflow`: Packet headers sampled by the switch that will be sent to the sFlow collector.
- `ssh`: Secure Shell (SSH) or Secure File Transfer Protocol (SFTP) packets. Dropping ssh packets will result in the connection to the CLI being lost.
- `stp`: Spanning Tree Protocol (STP) packets with the destination MAC address 01:80:c2:00:00:00 or Per-VLAN Spanning Tree (PVST) packets with the destination MAC address 01:00:0c:cc:cc:cd.
- `telnet`: Secure Telnet packets.
- `udld`: Unidirectional Link Detection packets with the destination MAC address 01:00:0c:cc:cc:cc or 00:e0:52:00:00:00.
- `unknown-multicast`: Packets with an unknown multicast destination IP address.
- `unresolved-ip-unicast`: Packets to be software forwarded by management processor.

- `vrrp-ipv4`: Virtual Router Redundancy Protocol packets with the destination IPv4 address 224.0.0.18 or VSX-Keepalive packets.
- `vrrp-ipv6`: Virtual Router Redundancy Protocol packets with the destination IPv6 address FF02:0:0:0:0:0:0:12.

To regulate any other traffic destined for the CPU, every CoPP policy has a class named 'default' that can also be configured to regulate other traffic to the CPU or prevent other traffic from being delivered.

NOTE: All IPsec traffic received by the CPU will be regulated by the 'ipsec' class regardless of the encapsulated protocol.

apply copp-policy

Syntax

```
apply copp-policy { <NAME> | default }
```

```
no apply copp-policy <NAME>
```

Description

Applies a CoPP policy to the switch, replacing the policy that is in effect. There may be a brief interruption in traffic flow to the management processor while the switch implements the change.

Enter the `no apply copp-policy <NAME>` command with the name of a CoPP policy to unapply a CoPP policy and apply the default CoPP policy. This will only take effect if the specified policy is actively applied. Since there must always be a CoPP policy applied, this command effectively attempts to replace the applied CoPP policy with the default CoPP policy. The default CoPP policy cannot be unapplied using this command.

Command context

`config`

Parameters

<NAME>

Specifies the name of the policy to apply. Length: 1 to 64 characters.

`default`

Applies the default policy.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If the new policy cannot be applied (for example, due to a lack of hardware resources), the previous policy remains in effect. Use the **`show copp-policy`** command to determine which policy is in effect.

Examples

Applying a policy named My_CoppPolicy:

```
switch(config)# apply copp-policy My_CoppPolicy
```

Applying the default policy:

```
switch(config)# apply copp-policy default
```

class

Syntax

```
class <CLASS> {drop | priority <PRIORITY> rate <RATE> [burst <BURST>]}

no class <CLASS> {drop | priority <PRIORITY> rate <RATE> [burst <BURST>]}
```

Description

Adds a class to a CoPP policy. If the class exists, the existing class is modified. Changes made to an active (applied) policy take effect immediately.

When adding or modifying a class in an active policy, CoPP immediately activates the change on the switch. In cases where insufficient hardware resources exist to support a class or its action, CoPP fails to activate the changed class on the switch. When this failure occurs, the active configuration on the switch will be out of sync with its definition. To diagnose and remedy this situation:

- Use the **`show copp-policy`** command to determine which classes are out of sync between the active policy and its definition.
- Use the **`reset copp-policy`** command to synchronize the active policy with its definition. This synchronization changes the classes in the definition to match the classes in the active policy.

The `no` form of this command removes a class from a CoPP policy. Traffic for the removed class that is destined for the processor will be included in the default class. To stop a class of traffic from reaching the processor, set the class action to drop.

Configuring the 'mirror-to-cpu' class after it was previously removed from the applied CoPP policy will also clear the statistics for the 'mirror-to-cpu' class, possibly resulting in a net decrease to the CoPP statistics represented in the default class.

Command context

`config-copp`

Parameters

`<CLASS>`

Specifies the **class** to add or edit.

`drop`

Drop packets matching the selected class.

`priority <PRIORITY>`

Specifies the priority for packets matching the selected class. Range: 0 to 7.

`rate <RATE>`

Specifies the maximum rate, in packets per second (pps), for packets matching the selected class. Range: 1 to 99999.

`burst <BURST>`

Specifies the maximum burst size, in packets, for packets matching the selected class. Range: 1 to 9999.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Adding a class to handle OSPF multicast traffic with priority of 2 and rate of 2000:

```
switch(config-copp)# class ospf-multicast priority 2 rate 2000
```

Adding a class to drop LLDP packets:

```
switch(config-copp)# class lldp drop
```

Removing the class that handles LLDP packets. LLDP traffic destined to the processor will be included in the default class.

```
switch(config-copp)# no class lldp
```

clear copp-policy statistics

Syntax

```
clear copp-policy statistics
```

Description

Resets statistics for all CoPP classes to zero.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying and then resetting statistics for all classes in the active policy:

```
switch# show copp-policy statistics
Statistics for CoPP policy 'default':
Totals:
    bytes passed      : 170000         bytes dropped     : 104000
    packets passed    : 32000          packets dropped   : 18000
Class: arp-broadcast
    bytes passed      : 40000          bytes dropped     : 40000
    packets passed    : 8000           packets dropped   : 8000
Class: arp-unicast
    bytes passed      : 8000           bytes dropped     : 64000
    packets passed    : 1000           packets dropped   : 10000
Class: bgp-ipv4
    bytes passed      : 12000          bytes dropped     : 64000
    packets passed    : 5000           packets dropped   : 10000
...
Class: default
    bytes passed      : 20000          bytes dropped     : 15000
    packets passed    : 1000           packets dropped   : 1100
...
switch# clear copp-policy statistics
switch# show copp-policy statistics
Statistics for CoPP policy 'default':
Totals:
    bytes passed      : 0              bytes dropped     : 0
    packets passed    : 0              packets dropped   : 0
```

```
Class: arp-broadcast
    bytes passed      : 0                bytes dropped    : 0
    packets passed    : 0                packets dropped  : 0
Class: arp-unicast
    bytes passed      : 0                bytes dropped    : 0
    packets passed    : 0                packets dropped  : 0
Class: bgp-ipv4
    bytes passed      : 0                bytes dropped    : 0
    packets passed    : 0                packets dropped  : 0
...
Class: default
    bytes passed      : 0                bytes dropped    : 0
    packets passed    : 0                packets dropped  : 0
...
```

copp-policy

Syntax

`copp-policy {<NAME> | default [revert]}`

`no copp-policy <NAME>`

Description

Creates a CoPP policy and switches to the `config-copp` context for the policy. Or, if the specified policy exists, switches to the `config-copp` context for the policy. A predefined policy, named `default`, contains factory default classes and is applied to the switch at first startup. This policy cannot be deleted, but its configuration can be changed.

The `no` form of this command removes a CoPP policy. If a policy is active (applied), it cannot be removed . It must be replaced with another policy before it can be removed.

Command context

`config`

Parameters

<NAME>

Specifies the name of the policy to add or edit. Length: 1 to 64 characters. The name must not be a substring of any of the following reserved words: default, factory-default, commands, configuration, or statistics.

`default`

Specifies the default CoPP policy. Use this default policy to configure the default policy.

`revert`

Sets the default CoPP policy to its factory settings.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Creating a policy named My_CoppPolicy:

```
switch(config)# copp-policy My_CoppPolicy
switch(config-copp)#
```

Removing a policy named My_CoppPolicy:

```
switch(config)# no copp-policy My_CoppPolicy
```

Setting the default policy to its factory settings:

```
switch(config)# copp-policy default revert
```

Unapplying the policy named My_CoppPolicy:

```
switch(config)# no apply copp-policy My_CoppPolicy
```

default-class

Syntax

`default-class priority <PRIORITY> rate <RATE> [burst <BURST>]`

Description

Configures the default class that is automatically defined for all CoPP policies. The default class cannot be removed, but its configuration can be changed. The default class is applied to traffic that does not match any other class defined for a policy.

Command context

`config-copp`

Parameters

priority <PRIORITY>

Specifies the priority for packets matching the selected class. Range: 0 to 7.

rate <RATE>

Specifies the maximum rate, in packets per second (pps), for packets matching the selected class. Range: 1 to 99999.

burst <BURST>

Specifies the maximum burst size, in packets, for packets matching the selected class. Range: 1 to 9999.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the default class to a priority of **2** and rate of **2000**:

```
switch(config-copp)# default-class priority 2 rate 2000
```

reset copp-policy

Syntax

`reset copp-policy { <NAME> | default }`

Description

Resets an active CoPP policy to match the settings that are currently in effect for the active policy on the switch. Changes made to the active policy that could not be activated are removed from the active policy.

When the switch fails to add or modify a class in an active CoPP policy, it is possible the active policy settings on the switch may be out of sync with those defined in the policy.

Command context

`config`

Parameters

<NAME>

Specifies the name of the policy to reset. Length: 1 to 64 characters.

`default`

Resets the default policy to match its active settings.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Resetting a policy named My_CoppPolicy (Output from an 8320 switch):

```
switch# show copp-policy My_CoppPolicy
class                 drop priority rate pps burst pkts
--------------------- ---- -------- --------- ---------
bgp-ipv4                   6        5000      5000
ospf-multicast-ipv4        2        2000      2000
default                    1        6000      6000
switch# config terminal
switch(config)# copp-policy My_CoppPolicy
switch(config-copp)# class stp drop
switch(config-copp)# do show copp-policy My_CoppPolicy
class                 drop priority rate pps  burst pkts
--------------------- ---- -------- --------- ----------
bgp-ipv4                   6        5000      5000
ospf-multicast-ipv4        2        2000      2000
default                    1        6000      6000
% Warning: user-specified classes in CoPP policy My_CoppPolicy do not match
 active configuration.
switch(config-copp)# do show copp-policy My_CoppPolicy configuration
class                 drop priority rate pps  burst pkts applied
--------------------- ---- -------- --------- ---------- -------
bgp-ipv4                   6        5000      5000       yes
ospf-multicast-ipv4        2        2000      2000       yes
stp                   drop                               no
default                    1        6000      6000       yes
% Warning: user-specified classes in CoPP policy My_CoppPolicy do not match
 active configuration.
switch(config-copp)# exit
switch(config)# reset copp-policy My_CoppPolicy
switch(config)# do show copp-policy My_CoppPolicy
class                 drop priority rate pps  burst pkts
--------------------- ---- -------- --------- ----------
bgp-ipv4                   6        5000      5000
ospf-multicast-ipv4        2        2000      2000
default                    1        6000      6000
switch(config)# exit
switch#
```

Resetting the default policy:

```
switch(config)# reset copp-policy default
```

show copp-policy

Syntax

`show copp-policy  [<NAME> | default] [commands] [configuration] [vsx-peer]`

Description

Shows CoPP policy settings for a specific CoPP policy. When entered without specifying either a name or the `default` parameter, shows all the CoPP policy settings that are active on the switch and have successfully been programmed into the hardware.

A warning is displayed if:

- The active and user-specified applications of a policy do not match.
- The active and user-specified configurations of a policy do not match.

Command context

Operator (>) or Manager (#)

Parameters

<NAME>

Specifies the name of the policy for which to display settings. Length: 1 to 64 characters.

default

Displays CoPP settings for the default policy.

commands

Displays output as CLI commands.

configuration

Displays user-specified CoPP settings and **not** the active settings.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying the CoPP policies defined in the configuration and the active application:

```
switch# show copp-policy
applied copp_policy_name
------- ----------------
        My_CoppPolicy
applied default
switch#
```

Displaying the active configuration of all CoPP policies as CLI commands:

Displaying the active configuration of all CoPP policies as CLI commands (Output from an 8320 switch):

```
switch# show copp-policy commands
copp-policy My_CoppPolicy
    class bgp-ipv4 priority 6 rate 5000 burst 5000
    class ospf-multicast-ipv4 priority 2 rate 2000
    default-class priority 1 rate 6000 burst 7000
copp-policy default
    class arp-broadcast priority 1 rate 1000 burst 1000
    class arp-unicast priority 1 rate 1000 burst 1000
    class bgp-ipv4 priority 1 rate 5000 burst 5000
        <--OUTPUT OMITTED FOR BREVITY-->
    default-class priority 1 rate 5000 burst 5000
apply copp-policy default
switch#
```

Entering the config-copp context and modify the **default** policy (Output from an 8320 switch):

```
switch# show copp-policy default
class                  drop priority rate pps  burst pkts  hardware rate pps
---------------------- ---- -------- --------- ----------  -----------------
acl-logging                 0        50        50          50
arp-broadcast          drop
bgp-ipv4                    9        5000      5000        5000
    <--OUTPUT OMITTED FOR BREVITY-->
default                     1        5000      5000        5000
```

show copp-policy factory-default

Syntax

`show copp-policy factory-default [commands] [vsx-peer]`

Description

Display the configuration for the factory-default CoPP policy.

Command context

Operator (>) or Manager (#)

Parameters

commands

Displays output as CLI commands.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Displaying the factory-default policy (Output from an 8320 switch):

```
switch# show copp-policy factory-default
class                  drop priority rate pps burst pkts hardware rate pps
---------------------- ---- -------- -------- ---------- ----------------
acl-logging                 0        50       50         50
arp-broadcast               3        7000     7000       7000
arp-unicast                 4        2500     2500       2500
bgp-ipv4                    6        1500     1500       1500
bgp-ipv6                    6        1500     1500       1500
dhcp-ipv4                   1        1000     1000       1000
dhcp-ipv6                   1        1000     1000       1000
erps                        7        1000     1000       1000
hypertext                   5        150      150        150
icmp-broadcast-ipv4         3        2000     2000       2000
icmp-multicast-ipv6         3        2000     2000       2000
icmp-unicast-ipv4           4        1000     1000       1000
icmp-unicast-ipv6           4        1000     1000       1000
igmp                        6        2500     2500       2500
ip-exceptions               0        150      150        150
ipv4-options                2        150      150        150
ipv6-options                2        150      150        150
lacp                        6        1000     1000       1000
lldp                        6        500      500        500
loop-protect                7        1000     1000       1000
mvrp                        6        1000     1000       1000
ntp                         5        150      150        150
ospf-multicast-ipv4         6        2500     2500       2500
ospf-multicast-ipv6         6        2500     2500       2500
ospf-unicast-ipv4           6        2500     2500       2500
ospf-unicast-ipv6           6        2500     2500       2500
pim                         6        1500     1500       1500
sflow                       1        2000     2000       2000
ssh                         5        500      500        500
stp                         7        2500     2500       2500
telnet                      5        500      500        500
udld                        7        500      500        500
unknown-multicast           2        1500     1500       1500
unresolved-ip-unicast       2        1000     1000       1000
vrrp-ipv4                   6        1000     1000       1000
vrrp-ipv6                   6        1000     1000       1000
default                     1        500      500        500
```

Displaying the active configuration of **My_CoppPolicy** as CLI commands:

```
switch# show copp-policy My_CoppPolicy commands
copp-policy My_CoppPolicy
    class bgp-ipv4 priority 6 rate 5000 burst 60
    class ospf-multicast priority 2 rate 2000
    default-class priority 1 rate 6000 burst 70
```

Displaying the user-specified configuration of **My_CoppPolicy** (Output from an 8320 switch):

```
switch# show copp-policy My_CoppPolicy configuration
class                  drop priority rate pps  burst pkts applied
---------------------- ---- -------- --------- ---------- -------
bgp-ipv4                    6        5000      5000       yes
ospf-multicast-ipv4         2        2000      2000       yes
default                     1        6000      6000       yes
```

Displaying the user-specified configuration of **My_CoppPolicy** as CLI commands:

```
switch# show copp-policy My_CoppPolicy commands configuration
copp-policy My_CoppPolicy
    class bgp-ipv4 priority 6 rate 5000 burst 60
```

```
    class ospf-multicast priority 2 rate 2000
    default-class priority 1 rate 6000 burst 70
```

show copp-policy statistics

Syntax

```
show copp-policy statistics [class <CLASS> | default-class] [vsx-peer]
```

Description

Displays statistics for all classes, or a single class, in the active CoPP policy.

Command context

Operator (>) or Manager (#)

Parameters

<CLASS>

Specifies the **class** for which to display statistics.

default-class

Displays statistics for the default class.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying statistics for all classes in the active policy:

NOTE: The rate displayed is the actual rate in hardware.

```
switch# show copp-policy statistics
Statistics for CoPP policy 'default':
Totals:
    bytes passed     : 170000         bytes dropped    : 104000
    packets passed   : 32000          packets dropped  : 18000
Class: arp-broadcast
    bytes passed     : 40000          bytes dropped    : 40000
    packets passed   : 8000           packets dropped  : 8000
Class: arp-unicast
    bytes passed     : 8000           bytes dropped    : 64000
    packets passed   : 1000           packets dropped  : 10000
Class: bgp-ipv4
    bytes passed     : 12000          bytes dropped    : 64000
    packets passed   : 5000           packets dropped  : 10000
...
Class: default
    bytes passed     : 20000          bytes dropped    : 15000
    packets passed   : 1000           packets dropped  : 1100
...
```

Displaying statistics for the default class in the active policy (Output from a 8320 switch):

```
switch(config)# show copp-policy statistics default-class
Class: default
    priority         : 1
    rate (pps)       : 5000
    burst size (pkts): 5000

    bytes passed     : 20000          bytes dropped    : 15000
    packets passed   : 1000           packets dropped  : 1100
    avg_packet_size_passed: 20        avg_packet_size_dropped : 14
```

Displ aying statistics for the class bgp-ipv4 in the actively applied policy (Output from an 8320 switch):

```
switch# show copp-policy statistics class bgp-ipv4
Statistics for CoPP policy 'default':
Class: bgp-ipv4
    priority         : 1
    rate (pps)        : 5000
    burst size (pkts) : 5000

    bytes passed     : 12000          bytes dropped    : 64000
    packets passed   : 5000           packets dropped  : 10000
    avg_packet_size_passed: 2         avg_packet_size_dropped : 6
```

The Aruba Central network management solution, a software-as-a-service subscription in the cloud, provides streamlined management of multiple network devices. ArubaOS-CX switches are able to talk to Aruba Central and utilize cloud-based management functionality. Cloud-based management functionality allows for the deployment of network devices at sites with no or few dedicated IT personnel (branch offices, retail stores, and so forth). ArubaOS-CX switches utilize secure communication protocols to connect to the Aruba Central cloud portal, and can coexist with corporate security standards, such as those mandating the use of firewalls.

This feature provides:

- Zero-touch provisioning
- Network Management/Remote monitoring
- Events/alerts notification
- Switch Configuration using templates
- Firmware management

Connecting to Aruba Central

ArubaOS-CX switch downloads the location of Aruba Central server using:

- Command-line interface (CLI).
- Aruba Activate server.
- DHCP options provided during ZTP.

DHCP servers are used to connect to Central on-premise management.

If switch is unable to connect to Activate server, it retries to establish connection in exponential back off of 1s, 2s, 8s, 16s, 32s, 64s, 128s, and 256s. After the maximum back off of 256s, switch retries happen for every 5 minutes.

NOTE: If the Network Time Protocol (NTP) is not enabled on the switch, it will synchronize the system time with the Activate server.

Custom CA certificate

To use custom CA certificate to connect to Aruba Central, AOS-CX switch downloads the certificate from Aruba Activate server.

NOTE:

- If there is no custom CA provided by Aruba Activate, the CA certificate present in the device is used.
- Duplicate CA certificates from Aruba Activate server will be ignored.
- If CA certificate is absent in consecutive responses from Aruba Activate server, the installed custom CA certificate in device will be removed.
- Switch will have only one custom CA certificate installed from Aruba Activate Server.
- The certificate installed from Aruba Activate server will not be displayed in the show commands.

Aruba Central Commands

aruba-central

Syntax

`aruba-central`

`no aruba-central`

Description

Creates or enters the Aruba Central configuration context (`config-aruba-central`).

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Example

Creating the Aruba Central configuration context:

```
switch(config)# aruba-central
switch(config-aruba-central)#
```

disable

Syntax

`disable`

`no disable`

Description

Disables connection to Aruba Central server.

When the connection is disabled, the switch does not attempt to connect to the Aruba Central server or fetch central location from any of the three sources (CLI/Aruba Activate/DHCP). It also disconnects any active connection to the Aruba Central server.

The `no` form of this command enables connection to the Aruba Central server. The `no disable` command is identical to the `enable` command.

Command context

`config-aruba-central`

Authority

Administrators or local user group members with execution rights for this command.

Example

```
switch(config-aruba-central)# disable
switch(config-aruba-central)#
```

enable

Syntax

`enable`

`no enable`

Description

Enables connection to Aruba Central server. When the connection is enabled, the switch attempts to download the location of the Aruba Central server in one of the following ways at startup and after the connection is lost:

- Using command-line interface (CLI).
- Connecting to Aruba Activate server.
- Using DHCP options provided during ZTP.

DHCP servers are used to connect to Central on-premise management.

The `no` form of this command disables connection to the Aruba Central server. The `no enable` command is identical to the `disable` command.

Command context

`config-aruba-central`

Authority

Administrators or local user group members with execution rights for this command.

Examples

```
switch(config-aruba-central)# enable
switch(config-aruba-central)#
```

location-override

Syntax

```
location-override <location> [vrf <VRF-NAME>
 no location-override
```

Description

When `location` and `vrf` are configured, the switch overrides existing connections to Aruba Central. The switch attempts to establish connection to Aruba Central with the specified location and VRF with highest priority.

The `no` form of this command removes location override values from the Aruba Central configuration context.

Command context

`config-aruba-central`

Authority

Administrators or local user group members with execution rights for this command.

Parameters

<location>

Specifies one of these values:

- *<FQDN>*: a fully qualified domain name.
- *<IPV4>*: an IPv4 address.
- *<IPV6>*: an IPv6 address.

vrf *<VRF-NAME>*

Specifies the VRF name to be used for communicating with the server. If no VRF name is provided, the default VRF named `default` is used.

Examples

Configuring location override with location and VRF:

```
switch(config-aruba-central)# location-override aruba-central.com vrf default
switch(config-aruba-central)#
```

Configuring location override with location only:

```
switch(config-aruba-central)# location-override aruba-central.com
switch(config-aruba-central)#
```

Removing location override values from the Aruba Central configuration context:

```
switch(config-aruba-central)# no location-override
switch(config-aruba-central)#
```

show aruba-central

Syntax

`show aruba-central`

Description

Shows information about Aruba Central connection and the status of the Activate server connection.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

This command shows the following information about the Aruba Network Analytics Engine agents that are available on the switch:

Examples

Example of a switch that has the Aruba Central connection:

```
switch# show aruba-central
Central admin state                       :enabled

Central location                          :N/A
VRF for connection                        :N/A
Central connection status                 :N/A

Central source                            : dhcp
Central source connection status          : connection_failure
Central source last connected on          : N/A
System time synchronized from Activate : True

Activate server URL                       : 172.17.0.1
CLI location                              : N/A
CLI VRF                                   : N/A
```

show running-config current-context

Syntax

`show running-config current-context`

Description

Shows the running configuration for the current-context. If user is in the context of Aruba-Central(`config-aruba-central`), then Aruba Central running configuration is displayed.

Command context

Operator (`>`) or Manager (`#`)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Shows the running configuration of Aruba Central:

```
switch(config-aruba-central)# show running-config current-context
aruba-central
    disable
```

portfilter

Syntax

```
portfilter <INTERFACE-LIST>

no portfilter [<INTERFACE-LIST>]
```

Description

Configures the specified ports so they do not egress any packets that were received on the source port specified in interface context.

The `no` form of this command removes the port filter setting from one or more ingress ports/LAGs.

Command context

`config-if`

`config-lag-if`

Parameters

<INTERFACE-LIST>

Specifies a list of ports/LAGs to be blocked for egressing. Specify a single interface or LAG, or a range as a comma-separated list, or both. For example: `1/1/1`, `1/1/3-1/1/6`,`lag2`, `lag1-lag4`.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When a port filter configuration is applied on the same ingress physical port/LAG, the configuration is updated with the new sets of egress ports/LAGs that are to be blocked for egressing and that are not a part of its previous configuration. Duplicate updates on an existing port filter configuration are ignored.

When egress ports/LAGs are removed from the existing port filter configuration of an ingress port/LAG, egressing is allowed again on those egress ports/LAGs for all packets originating from the ingress port/LAG.

The `no portfilter [<IF-NAME-LIST>]` command removes port filter configurations from the egress ports/LAGs listed in the *<IF-NAME-LIST>* parameter only. All other egress ports/LAGs in the port filter configuration of the ingress port/LAG remain intact.

If no physical ports or LAGs are provided for the `no portfilter` command, the command removes the entire port filter configuration for the ingress port/LAG.

Examples

Creating a filter that prevents packets received on port **1/1/1** from forwarding to ports **1/1/3-1/1/6** and to LAGs **1** through **4**:

```
switch(config)# interface 1/1/1
switch(config-if)# portfilter 1/1/3-1/1/6,lag1-lag4
```

Creating a filter that prevents packets received on LAG **1** from forwarding to ports **1/1/6** and LAGs **2** and **4**:

```
switch(config)# interface lag 1
switch(config-lag-if)# portfilter 1/1/6,lag2,lag4
```

Removing filters from an existing configuration that allows back packets received on port **1/1/1** to forward to ports **1/1/6** and LAGs **3** and **4**:

```
switch(config)# interface 1/1/1
switch(config-if)# no portfilter 1/1/6,lag3,lag4
```

Removing all filters from an existing configuration that allows back packets received on LAG **1** to forward to all the ports and LAGs:

```
switch(config)# interface lag 1
switch(config-lag-if)# no portfilter
```

show portfilter

Syntax

```
show portfilter [<IFNAME>] [vsx-peer]
```

Description

Displays filter settings for all interfaces or a specific interface.

Command context

Operator (>) or Manager (#)

Parameters

<IFNAME>

Specifies the ingress interface name.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Displaying all port filter settings on the switch:

```
switch# show portfilter
Incoming   Blocked
Interface  Outgoing Interfaces
--------------------------------------------------------------------------------
1/1/1      1/1/3-1/1/6,lag1-lag2
1/1/3      1/1/1,1/1/5,1/1/7,1/1/9,1/1/11,1/1/13,1/1/15,1/1/17,1/1/19,1/1/21,
           1/1/23,1/1/25,1/1/27,1/1/29,1/1/31,1/1/33,1/1/35
lag2       1/1/1,1/1/3-1/1/6
```

Displaying the port filter settings for port **1/1/1**:

```
switch# show portfilter 1/1/1
Incoming   Blocked
Interface  Outgoing Interfaces
------------------------------------------------------------------------------
1/1/1      1/1/3-1/1/6,lag1-lag2
```

Displaying the port filter settings for **LAG2**:

```
switch# show portfilter lag2
Incoming   Blocked
Interface  Outgoing Interfaces
------------------------------------------------------------------------------
lag2       1/1/1,1/1/3-1/1/6
```

The Domain Name System (DNS) is the Internet protocol for mapping a hostname to its IP address. DNS allows users to enter more readily memorable and intuitive hostnames, rather than IP addresses, to identify devices connected to a network. It also allows a host to keep the same hostname even if it changes its IP address.

Hostname resolution can be either static or dynamic.

- In static resolution, a local table is defined on the switch that associates hostnames with their IP addresses. Static tables can be used to speed up the resolution of frequently queried hosts.
- Dynamic resolution requires that the switch query a DNS server located elsewhere on the network. Dynamic name resolution takes more time than static name resolution, but requires far less configuration and management.

DNS client

The DNS client resolves hostnames to IP addresses for protocols that are running on the switch. When the DNS client receives a request to resolve a hostname, it can do so in one of two ways:

- Forward the request to a DNS name server for resolution.
- Reply to the request without using a DNS name server, by resolving the name using a statically defined table of hostnames and their associated IP addresses.

Configuring the DNS client

Procedure

1. Configure one or more DNS name servers with the command `ip dns server`.
2. To resolve DNS requests by appending a domain name to the requests, either configure a single domain name with the command `ip dns domain-name`, or configure a list of up to six domain names with the command `ip dns domain-list`.
3. To use static name resolution for certain hosts, associate an IP address to a host with the command `ip dns host`.
4. Review your DNS configuration settings with the command `show ip dns`.

Examples

This example creates the following configuration:

- Defines the domain **`switch.com`** to append to all requests.
- Defines a DNS server with IPv4 address of **`1.1.1.1`**.

- Defines a static DNS host named **`myhost1`** with an IPv4 address of **`3.3.3.3`**.
- DNS client traffic is sent on the default VRF (named **`default`**).

```
switch(config)# ip dns domain-name switch.com
switch(config)# ip dns server-address 1.1.1.1
switch(config)# ip dns host myhost1 3.3.3.3
switch(config)# exit
switch# show ip dns

VRF Name : vrf_mgmt

Host Name                                                         Address
--------------------------------------------------------------------------------


VRF Name : vrf_default
Domain Name : switch.com
DNS Domain list :
Name Server(s) : 1.1.1.1

Host Name                                                         Address
--------------------------------------------------------------------------------
myhost1
```

This example creates the following configuration:

- Defines three domains to append to DNS requests **`domain1.com`**, **`domain2.com`**, **`domain3.com`** with traffic forwarding on VRF **`mainvrf`**.
- Defines a DNS server with an IPv6 address of **`c::13`**.
- Defines a DNS host named **`myhost`** with an IPv4 address of **`3.3.3.3`**.

```
switch(config)# ip dns domain-list domain1.com vrf mainvrf
switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain3.com vrf mainvrf
switch(config)# ip dns server-address c::13
switch(config)# ip dns host myhost 3.3.3.3 vrf mainvrf
switch(config)# quit
switch# show ip dns mainvrf

VRF Name : mainvrf
Domain Name :
DNS Domain list : domain1.com, domain2.com, domain3.com
Name Server(s) : c::13

Host Name                                                         Address
--------------------------------------------------------------------------------
myhost                                                            3.3.3.3
```

DNS client commands

ip dns domain-list

Syntax

```
ip dns domain-list <DOMAIN-NAME> [vrf <VRF-NAME>]

no ip dns domain-list <DOMAIN-NAME> [vrf <VRF-NAME>]
```

Description

Configures one or more domain names that are appended to the DNS request. The DNS client appends each name in succession until the DNS server replies. Domains can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF.

The `no` form of this command removes a domain from the list.

Command context

`config`

Parameters

list *<DOMAIN-NAME>*

Specifies a domain name. Up to six domains can be added to the list. Length: 1 to 256 characters.

vrf *<VRF-NAME>*

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines a list with two entries: **domain1.com** and **domain2.com**.

```
switch(config)# ip dns domain-list domain1.com
switch(config)# ip dns domain-list domain2.com
```

This example defines a list with two entries, **domain2.com** and **domain5.com**, with requests being sent on **mainvrf**.

```
switch(config)# ip dns domain-list domain2.com vrf mainvrf
switch(config)# ip dns domain-list domain5.com vrf mainvrf
```

This example removes the entry **domain1.com**.

```
switch(config)# no ip dns domain-list domain1.com
```

ip dns domain-name

Syntax

```
ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]

no ip dns domain-name <DOMAIN-NAME> [ vrf <VRF-NAME> ]
```

Description

Configures a domain name that is appended to the DNS request. The domain can be either IPv4 or IPv6. By default, requests are forwarded on the default VRF. If a domain list is defined with the command **ip dns domain-list**, the domain name defined with this command is ignored.

The `no` form of this command removes the domain name.

Command context

`config`

Parameters

<DOMAIN-NAME>

Specifies the domain name to append to DNS requests. Length: 1 to 256 characters.

vrf *<VRF-NAME>*

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the default domain name to `domain.com`:

```
switch(config)# ip dns domain-name domain.com
```

Removing the default domain name `domain.com`:

```
switch(config)# no ip dns domain-name domain.com
```

ip dns host

Syntax

```
ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]

no ip dns host <HOST-NAME> <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Associates a static IP address with a hostname. The DNS client returns this IP address instead of querying a DNS server for an IP address for the hostname. Up to six hosts can be defined. If no VRF is defined, the default VRF is used.

The `no` form of this command removes a static IP address associated with a hostname.

Command context

`config`

Parameters

host *<HOST-NAME>*

Specifies the name of a host. Length: 1 to 256 characters.

<IP-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

vrf *<VRF-NAME>*

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines an IPv4 address of **`3.3.3.3`** for **`host1`**.

```
switch(config)# ip dns host host1 3.3.3.3
```

This example defines an IPv6 address of **`b::5`** for **`host 1`**.

```
switch(config)# ip dns host host1 b::5
```

This example defines removes the entry for **`host 1`** with address **`b::5`**.

```
switch(config)# no ip dns host host1 b::5
```

ip dns server address

Syntax

```
ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]

no ip dns server-address <IP-ADDR> [ vrf <VRF-NAME> ]
```

Description

Configures the DNS name servers that the DNS client queries to resolve DNS queries. Up to six name servers can be defined. The DNS client queries the servers in the order that they are defined. If no VRF is defined, the default VRF is used.

The `no` form of this command removes a name server from the list.

Command context

`config`

Parameters

<IP-ADDR>

Specifies an IP address in IPv4 format (`x.x.x.x`), where `x` is a decimal number from 0 to 255, or IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

vrf *<VRF-NAME>*

Specifies a VRF name. Default: default.

Authority

Administrators or local user group members with execution rights for this command.

Examples

This example defines a name server at **1.1.1.1**.

```
switch(config)# ip dns server-address 1.1.1.1
```

This example defines a name server at **a::1**.

```
switch(config)# ip dns server-address a::1
```

This example removes a name server at **a::1**.

```
switch(config)# no ip dns server-address a::1
```

show ip dns

Syntax

`show ip dns [vrf <VRF-NAME>] [vsx-peer]`

Description

Shows all DNS client configuration settings or the settings for a specific VRF.

Command context

Manager (#)

Parameters

vrf *<VRF-NAME>*

Specifies the VRF for which to show information. If no VRF is defined, the default VRF is used.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

These examples define DNS settings and then show how they are displayed with the `show ip dns` command.

```
switch(config)# ip dns domain-name domain.com
switch(config)# ip dns domain-list domain5.com
switch(config)# ip dns domain-list domain8.com
switch(config)# ip dns server-address 4.4.4.4
switch(config)# ip dns server-address 6.6.6.6
switch(config)# ip dns host host3 5.5.5.5
switch(config)# ip dns host host2 2.2.2.2
switch(config)# ip dns host host3 c::12
switch(config)# ip dns domain-name reddomain.com vrf red
switch(config)# ip dns domain-list reddomain5.com vrf red
switch(config)# ip dns domain-list reddomain8.com vrf red
switch(config)# ip dns server-address 4.4.4.5 vrf red
switch(config)# ip dns server-address 6.6.6.7 vrf red
switch(config)# ip dns host host3 5.5.5.6 vrf red
```

```
switch(config)# ip dns host host2 2.2.2.3 vrf red
switch(config)# ip dns host host3 c::13 vrf red
switch# show ip dns
VRF Name : default

Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6

Host Name    Address
-------------------------------
host2            2.2.2.2
host3            5.5.5.5
host3            c::12

VRF Name : red

Domain Name : reddomain.com
DNS Domain list : reddomain5.com, reddomain8.com
Name Server(s) : 4.4.4.5, 6.6.6.7

Host Name    Address
-------------------------------
host2            2.2.2.3
host3            5.5.5.6
host3            c::13
```

```
switch(config)# ip dns domain-name domain.com vrf red
switch(config)# ip dns domain-list domain5.com vrf red
switch(config)# ip dns domain-list domain8.com vrf red
switch(config)# ip dns server-address 4.4.4.4 vrf red
switch(config)# ip dns server-address 6.6.6.6 vrf red
switch(config)# ip dns host host3 5.5.5.5 vrf red
switch(config)# no ip dns host host2 2.2.2.2 vrf red
switch(config)# ip dns host host3 c::12 vrf red

switch# show ip dns vrf red
VRF Name : red

Domain Name : domain.com
DNS Domain list : domain5.com, domain8.com
Name Server(s) : 4.4.4.4, 6.6.6.6

Host Name    Address
-------------------------------
host3            5.5.5.5
host3            c::12
```

The switch provides support for LLDP and CDP to enable automatic discovery and configuration of other devices on the network.

LLDP

The IEEE 802.1AB Link Layer Discovery Protocol (LLDP) provides a standards-based method for network devices to discover each other and exchange information about their capabilities. An LLDP device advertises itself to adjacent (neighbor) devices by transmitting LLDP data packets on all interfaces on which outbound LLDP is enabled, and reading LLDP advertisements from neighbor devices on ports on which inbound LLDP is enabled. Inbound packets from neighbor devices are stored in a special LLDP MIB (management information base). This information can then be queried by other devices through SNMP.

LLDP information is used by network management tools to create accurate physical network topologies by determining which devices are neighbors and through which interfaces they connect. LLDP operates at layer 2 and requires an LLDP agent to be active on each interface that sends and receives LLDP advertisements. LLDP advertisements can contain a variable number of TLV (type, length, value) information elements. Each TLV describes a single attribute of a device such as: system capabilities, management IP address, device ID, port ID.

Packet boundaries

When multiple LLDP devices are directly connected, an outbound LLDP packet travels only to the next LLDP device. An LLDP-capable device does not forward LLDP packets to any other devices, regardless of whether they are LLDP-enabled.

An intervening hub or repeater forwards the LLDP packets it receives in the same manner as any other multicast packets it receives. Therefore, two LLDP switches joined by a hub or repeater handle LLDP traffic in the same way that they would if directly connected.

Any intervening 802.1D device or Layer-3 device that is either LLDP-unaware or has disabled LLDP operation drops the packet.

LLDP-MED

LLDP-MED (ANSI/TIA-1057/D6) extends the LLDP (IEEE 802.1AB) industry standard to support advanced features on the network edge for Voice Over IP (VoIP) endpoint devices with specialized capabilities and LLDP-MED standards-based functionality. LLDP-MED in the switches uses the standard LLDP commands described earlier in this section, with some extensions, and also introduces new commands unique to LLDP-MED operation. The show commands described elsewhere in this section are applicable to both LLDP and LLDP-MED operation. LLDP-MED enables:

- Configure Voice VLAN and advertise it to connected MED endpoint devices.
- Power over Ethernet (PoE) status and troubleshooting support via SNMP.

LLDP agent

When you enable LLDP on the switch, it is automatically enabled on all data plane interfaces. You can customize this behavior by manually enabling/disabling support on each interface.

Supported standards

The LLDP agent supports the following standards: IEEE 802.1AB-2005, Station, and Media Access Control Connectivity Discovery.

Supported interfaces

LLDP is supported on interfaces mapped to a physical port, and the Out-Of-Band Management (OOBM) port. It is not supported on logical interfaces, such as loopback, tunnels, and SVIs.

Operating modes

When LLDP is enabled, the switch periodically transmits an LLDP advertisement (packet) out each active port enabled for outbound LLDP transmissions and receives LLDP advertisements on each active port enabled to receive LLDP traffic.

The LLDP agent can operate in one of the following modes:

- Transmit and receive (TxRx): This is the default setting on all ports. It enables a given port to both transmit and receive LLDP packets and to store the data from received (inbound) LLDP packets in the switch's MIB.
- Transmit only (Tx): Enables a port to transmit LLDP packets that can be read by LLDP neighbors. However, the port drops inbound LLDP packets from LLDP neighbors without reading them. This prevents the switch from learning about LLDP neighbors on that port.
- Receive only (Rx): Enables a port to receive and read LLDP packets from LLDP neighbors and to store the packet data in the switch's MIB. However, the port does not transmit outbound LLDP packets. This prevents LLDP neighbors from learning about the switch through that port.
- Disabled: Disables LLDP packet transmissions and reception on a port. In this state, the switch does not use the port for either learning about LLDP neighbors or informing LLDP neighbors of its presence.

An LLDP agent operating in TxRx mode or Tx mode sends LLDP frames to its directly connected devices both periodically and when the local configuration changes.

Sending LLDP frames

Each time the LLDP operating mode of an LLDP agent changes, its LLDP protocol state machine reinitializes. A configurable reinitialization delay prevents frequent initializations caused by frequent changes to the operating mode. If you configure the reinitialization delay, an LLDP agent must wait the specified amount of time to initialize LLDP after the LLDP operating mode changes.

Receiving LLDP frames

An LLDP agent operating in TxRx mode or Rx mode confirms the validity of TLVs carried in every received LLDP frame. If the TLVs are valid, the LLDP agent saves the information and starts an aging timer. The initial value of the aging timer is equal to the TTL value in the Time To Live TLV carried in the LLDP frame. When the LLDP agent receives a new LLDP frame, the aging timer restarts. When the aging timer decreases to zero, all saved information ages out.

TLV support

By default, the agent sends and receives the following mandatory TLVs on each interface:

- Port ID
- Chassis ID
- TTL

By default, the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED) are enabled on an agent. Sending them depends on the configuration and reception of any MED TLVs:

- MAC/PHY status. Includes the bit rate and auto negotiation status of the link.
- Power Via MDI: Includes Power Over Ethernet related information for supported interfaces.
- Port description
- System name
- System description
- Management address
- System capabilities
- Port VLAN ID

By default, the agent sends and receives the following ANSI/TIA-1057 TLVs for LLDP Media Endpoint Discovery (MED):

- Capabilities: Indicates MED TLV capability.
- Power Via MDI: Includes Power Over Ethernet related information.
- Network Policy: Includes the VLAN configuration for voice application.
- Location: Location identification information.
- Extended Power Via MDI: Power Over Ethernet related information

TLV advertisements

The LLDP agent transmits the following:

- Chassis-ID: Base MAC address of the switch.
- Port-ID: Port number of the physical port.
- Time-to-Live (TTL): Length of time an LLDP neighbor retains advertised data before discarding it.
- System capabilities: Identifies the primary switch capabilities (bridge, router). Identifies the primary switch functions that are enabled, such as routing.
- System description: Includes switch model name and running software version, and ROM version.
- System name: Name assigned to the switch.
- Management address: Default address selection method unless an optional address is configured.
- Port description: Physical port identifier.
- Port VLAN ID: On an L2 port, contains access or native VLAN ID. On an L3 port, contains a value of 0. Trunk allowed VLANs information are not advertised as part of the Port VLAN ID TLV. (Not supported on the OOBM interface)

LLDP MED support

LLDP-MED interoperates with directly connected IP telephony (endpoint) clients and provides the following features:

- Advertisement of the voice VLAN configured on the interface which is used by connected IP telephony (endpoint) clients.
- Advertisement of the configured location on the switch that can be used by the connected endpoint.
- Support for the fast-start capability

NOTE: LLDP-MED is intended for use with VoIP endpoints and is not designed to support links between network infrastructure devices (such as switch-to-switch or switch-to-router links).

Configuring the LLDP agent

Procedure

1. By default, the LLDP agent is enabled on all active interfaces. If LLDP was disabled, enable it with the command **`lldp`**.
2. By default, the LLDP agent transmits and receive on all interfaces. To customize LLDP behavior on a specific interface, use the commands **`lldp transmit`** and **`lldp receive`**.
3. By default, the LLDP agent sets the management address in all TLVs in the following order:
   a. LLDP management IP address.
   b. Loopback interface IP.
   c. ROP (L3 ports) or SVI (L2 ports).
   d. OOBM (Management interface IP).
   e. Base MAC.

   On the OOBM port, the following order is used:

   a. LLDP management IP address,
   b. IP address of the management interface (OOBM port).
   c. IP address of the loopback interface.
   d. Base MAC address of the switch.

   To specify a different address, use the commands **`lldp management-ipv4-address`** and **`lldp management-ipv6-address`**
4. By default, all supported TLVs are sent and received. To customize the list, use the command **`lldp select-tlv`**.
5. By default, support for the LLDP-MED TLV is enabled. To customize settings, use the commands **`lldp med`** and **`lldp med-location`**.
6. If required, adjust LLDP timer, holdtime, reinitialization delay, and transmit delay from their default values with the commands **`lldp timer`**, **`lldp holdtime`**, **`lldp reinit`**, and **`lldp txdelay`**.

Example

This example creates the following configuration:

- Enables LLDP support.
- Disables LLDP transmission on interface **1/1/1**.

```
switch(config)# lldp
switch(config)# interface 1/1/1
switch(config-copp)# no lldp transmit
```

LLDP commands

clear lldp neighbors

Syntax

`clear lldp neighbors`

Description

Clears all LLDP neighbor details.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing all LLDP neighbor details:

```
switch# clear lldp neighbors
```

clear lldp statistics

Syntax

`clear lldp statistics`

Description

Clears all LLDP neighbor statistics.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Clearing all LLDP neighbor statistics:

```
switch# clear lldp statistics
```

lldp

Syntax

`lldp`

`no lldp`

Description

Enables LLDP support globally on all active interfaces. By default, LLDP is enabled.

The `no` form of this command disables LLDP support globally on all active interfaces. It does not remove any LLDP configuration settings.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling LLDP:

```
switch(config)# lldp
```

Disabling LLDP:

```
switch(config)# no lldp
```

lldp dot3

Syntax

`lldp dot3 {poe | macphy}`

`no lldp dot3 {poe | macphy}`

Description

Sets the 802.3 TLVs to be advertised. By default, advertisement of both POE and MAC/PHY TLVs is enabled. Not supported on the OOBM interface.

The `no` form of this command disables advertisement of 802.3 TLVs.

Command context

`config-if`

Parameters

`poe`

Specifies advertisement of power over Ethernet data link classification.

`macphy`

Specifies advertisement of media access control and physical layer information.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling advertisement of the POE TLV:

```
switch(config-if)# lldp dot3 poe
```

Disabling advertisement of the POE TLV:

```
switch(config-if)# no lldp dot3 poe
```

lldp holdtime

Syntax

```
lldp holdtime <TIME>
```

```
no lldp holdtime
```

Description

Sets the holdtime that is used to calculate the LLDP Time-to-Live value. Time-to-Live defines the length of time that neighbors consider LLDP information sent by this agent as valid. When Time-to-Live expires, the information is deleted by the neighbor. Time-to-live is calculated by multiplying holdtime by the value of **lldp timer**.

The `no` form of this command sets the holdtime to its default value of 4.

Command context

`config`

Parameters

<TIME>

Specifies the holdtime in seconds. Range: 2 to 10. Default: 4.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the holdtime to 8 seconds:

```
switch(config)# lldp holdtime 8
```

Setting the holdtime to the default value of 4 seconds:

```
switch(config)# no lldp holdtime
```

lldp management-ipv4-address

Syntax

```
lldp management-ipv4-address <IPV4-ADDR>
```

```
no lldp management-ipv4-address
```

Description

Defines the IPv4 management address of the switch which is sent in the management address TLV. One IPv4 and one IPv6 management address can be configured.

If you do not define an LLDP management address, then LLDP uses one of the following (in order):

- IP address of the port
- IP address of the management interface
- Base MAC address of the switch

The `no` form of this command removes the IPv4 management address of the switch.

Command context

`config`

Parameters

<IPV4-ADDR>

Specifies the management address of the switch as an IPv4 format (`x.x.x.x`), where `x` is a decimal value from 0 to 255.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the management address to **`10.10.10.2`**:

```
switch(config)# lldp management-ipv4-address 10.10.10.2
```

Removing the management address:

```
switch(config)# no lldp management-ipv4-address
```

lldp management-ipv6-address

Syntax

```
lldp management-ipv6-address <IPV6-ADDR>
```

```
no lldp management-ipv6-address
```

Description

Defines the IPv6 management address of the switch. The management address is encapsulated in the management address TLV.

If no management address is specified, LLDP uses the IP address of the management interface, and if this the management interface is not defined, the chassis ID.

The `no` form of this command removes the IPv6 management address of the switch.

Command context

`config`

Parameters

<IPV6-ADDR>

Specifies an IP address in IPv6 format (`xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`), where `x` is a hexadecimal number from 0 to F.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the management address to 2001:db8:85a3::8a2e:370:7334:

```
switch(config)# lldp management-ipv6-address 2001:0db8:85a3::8a2e:0370:7334
```

Removing the management address:

```
switch(config)# no lldp management-ipv6-address
```

lldp med

Syntax

`lldp med [poe [priority-override] | capability | network-policy]`

`no med [poe [priority-override] | capability | network-policy]`

Description

Configures support for the LLDP-MED TLV. LLDP-MED (media endpoint devices) is an extension to LLDP developed by TIA to support interoperability between VoIP endpoint devices and other networking end-devices. The switch only sends the LLDP MED TLV after receiving a MED TLV from and connected endpoint device.

Not supported on the OOBM interface.

The `no` form of this command disables support for the LLDP MED TLV.

Command context

`config-if`

Parameters

`poe [priority-override]`

Specifies advertisement of power over Ethernet data link classification. The `priority-override` option overrides user-configured port priority for Power over Ethernet. When both `lldp dot3 poe` and `lldp med poe` are enabled, the `lldp dot3 poe3` setting takes precedence. Default: enabled.

`capability`

Specifies advertisement of supported LLDP MED TLVs. The capability TLV is always sent with other MED TLVs, therefore it cannot be disabled when other MED TLVs are enabled. Default: enabled.

`network-policy`

Network policy discovery lets endpoints and network devices advertise their VLAN IDs, and IEEE 802.1p (PCP and DSCP) values for voice applications. This TLV is only sent when a voice VLAN policy is present. Default: enabled.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling advertisement of the network policy TLV:

```
switch(config-if)# lldp med network-policy
```

Disabling advertisement of the network policy TLV:

```
switch(config-if)# no lldp med network-policy
```

lldp med-location

Syntax

```
lldp med-location {civic-addr  | elin-addr }
```

```
no med-location {civic-addr | elin-addr }
```

Description

Configures support for the LLDP-MED TLV. Supports only civic address and emergency location information number (ELIN). Coordinate-based location is not supported.

The `no` form of this command disables support for the LLDP MED TLV.

Command context

`config-if`

Parameters

civic-addr

Configures the LLDP MED civic location TLV.

elin-addr

Configures support for the LLDP MED emergency location TLV.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# lldp med-location elin-addr gher
```

Disabling support for the LLDP MED emergency location TLV:

```
switch(config-if)# no lldp med-location elin-addr gher
```

Enabling support for the LLDP MED civic address TLV:

```
switch(config-if)# lldp med-location civic-addr US 1 4 ret 6 tyu 7 tiyuo
```

Disabling support for the LLDP MED civic address TLV:

```
switch(config-if)# no lldp med-location civic-addr US 1 4 ret 6 tyu 7 tiyuo
```

lldp receive

Syntax

```
lldp receive

no lldp receive
```

Description

Enables reception of LLDP information on an interface. By default, LLDP reception is enabled on all active interfaces, including the OOBM interface.

The `no` form of this command disables reception of LLDP information on an interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp receive
```

Disabling LLDP reception on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp receive
```

Enabling LLDP reception on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# lldp receive
```

Disabling LLDP reception on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# no lldp receive
```

lldp reinit

Syntax

```
lldp reinit <TIME>

no lldp reinit
```

Description

Sets the amount of time (in seconds) to wait before performing LLDP initialization on an interface.

The `no` form of this command sets the reinitialization time to its default value of 2 seconds.

Command context

`config`

Parameters

<TIME>

Specifies the reinitialization time in seconds. Range: 1 to 10. Default: 2 seconds.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the reinitialization time to 5 seconds:

```
switch(config)# lldp reinit 5
```

Setting the reinitialization time to the default value of 2 seconds:

```
switch(config)# no lldp reinit
```

lldp select-tlv

Syntax

`lldp select-tlv <TLV-NAME>`

`no lldp select-tlv <TLV-NAME>`

Description

Selects a TLV that the LLDP agent will send and receive. By default, all supported TLVs are sent and received.

The `no` form of this command stops the LLDP agent from sending and receiving a specific TLV.

Command context

`config`

Parameters

select-tlv *<TLV-NAME>*

Specifies the TLV name to send. The following TLV names are supported:

- `management-address`: Selected as follows:
  1. IPv4 or IPV6 management address.
  2. IP address of the lowest configured loopback interface.
  3. If layer 3, then the route-only port IP address. If layer 2, the IP address of the SVI.
  4. OOBM interface IP address.
  5. Base MAC address of the switch.
- `port-description`: A description of the port.
- `port-vlan-id`: VLAN ID assigned to the port.
- `system-capabilities`: Identifies the primary switch functions that are enabled, such as routing.

- `system-description:` Description of the system, comprised of the following information: hardware serial number, hardware revision number, and firmware version.
- `system-name:` Host name assigned to the switch.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Stopping the LLDP agent from sending the **`port-description`** TLV:

```
switch(config)# no lldp select-tlv port-description
```

Enabling the LLDP agent to send the **`port-description`** TLV:

```
switch(config)# lldp select-tlv port-description
```

lldp timer

Syntax

`lldp timer <TIME>`

`no lldp timer`

Description

Sets the interval (in seconds) at which local LLDP information is updated and TLVs are sent to neighboring network devices by the LLDP agent. The minimum setting for this timer must be four times the value of `lldp txdelay`.

For example, this is a valid configuration:

- `lldp timer` = 16
- `lldp txdelay` = 4

And, this is an invalid configuration:

- `lldp timer` = 5
- `lldp txdelay` = 2

NOTE: When copying a saved configuration to the running configuration, the value for `lldp timer` is applied before the value of `lldp txdelay`. This can result in a configuration error if the saved configuration has a value of `lldp timer` that is not four times the value of `lldp txdelay` in the running configuration.

For example, if the saved configuration has the settings:

- `lldp timer` = 16
- `lldp txdelay` = 4

And the running configuration has the settings:

- `lldp timer` = 30
- `lldp txdelay` = 7

Then you will see an error indicating that certain configuration settings could not be applied, and you will have to manually adjust the value of `lldp txdelay` in the running configuration.

The `no` form of this command sets the update interval to its default value of 30 seconds.

Command context

`config`

Parameters

<TIME>

Specifies the update interval (in seconds). Range: 5 to 32768. Default: 30.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the update interval to 7 seconds:

```
switch(config)# lldp timer 7
```

Setting the update interval to the default value of 30 seconds:

```
switch(config)# no lldp timer
```

lldp transmit

Syntax

`lldp transsmit`

`no lldp transmit`

Description

Enables transmission of LLDP information on specific interface. By default, LLDP transmission is enabled on all active interfaces, including the OOBM interface.

The `no` form of this command disables transmission of LLDP information on an interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp transmit
```

Disabling LLDP transmission on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp transmit
```

Enabling LLDP transmission on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# lldp transmit
```

Disabling LLDP transmission on the OOBM interface:

```
switch(config)# interface mgmt
switch(config-if)# no lldp transmit
```

lldp txdelay

Syntax

`lldp txdelay <TIME>`

`no lldp txdelay`

Description

Sets the amount of time (in seconds) to wait before sending LLDP information from any interface. The maximum value for `txdelay` is 25% of the value of `lldp tx timer`.

The `no` form of this command sets the delay time to its default value of 2 seconds.

Command context

`config`

Parameters

<TIME>

Specifies the delay time in seconds. Range: 0 to 10. Default: 2.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting the delay time to 8 seconds:

```
switch(config)# lldp txdelay 8
```

Setting the delay time to the default value of 2 seconds:

```
switch(config)# no lldp txdelay
```

show lldp configuration

Syntax

```
show lldp configuration [<INTERFACE-ID>] [vsx-peer]
```

Description

Shows LLDP configuration settings for all interfaces or a specific interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: `member/slot/port`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing configuration settings for all interfaces:

```
switch# show lldp configuration

LLDP Global Configuration
=========================

LLDP Enabled                 : Yes
LLDP Transmit Interval       : 8
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Time Interval    : 2

TLVs Advertised
===============

Management Address
Port Description
Port VLAN-ID
System Capabilities
System Description
System Name

LLDP Port Configuration
=======================
```

```
PORT            TX-ENABLED          RX-ENABLED
------------------------------------------------
1/1/1           Yes                 Yes
1/1/2           Yes                 Yes
1/1/3           Yes                 Yes
1/1/4           Yes                 Yes
1/1/5           Yes                 Yes
...
mgmt            Yes                 Yes
```

This example shows configuration settings for interface **1/1/1**.

```
switch# show lldp configuration 1/1/1

LLDP Global Configuration
=========================
LLDP Enabled                 : Yes
LLDP Transmit Interval       : 8
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Time Interval    : 2

LLDP Port Configuration
=======================

PORT            TX-ENABLED          RX-ENABLED
-----------------------------------------------
1/1/1           Yes                 Yes
```

show lldp configuration mgmt

Syntax

show lldp configuration mgmt

Description

Shows LLDP configuration settings for the OOBM interface.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing configuration settings for all interfaces:

```
switch# show lldp configuration

LLDP Global Configuration
=========================

LLDP Enabled                 : Yes
LLDP Transmit Interval       : 30
LLDP Hold Time Multiplier    : 4
LLDP Transmit Delay Interval : 2
LLDP Reinit Time Interval    : 2
```

```
LLDP Port Configuration
=======================

PORT            TX-ENABLED          RX-ENABLED
----------------------------------------------
mgmt           Yes                 Yes
```

show lldp tlv

Syntax

```
show lldp tlv [vsx-peer]
```

Description

Shows the LLDP TLVs that are configured for send and receive.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

```
switch# show lldp tlv

TLVs Advertised
===============

Management Address
Port Description
Port VLAN-ID
System Capabilities
System Description
System Name
```

show lldp statistics

Syntax

```
show lldp statistics [<INTERFACE-ID>] [vsx-peer]
```

Description

Shows global LLDP statistics or statistics for a specific interface.

Command context

Manager (#)

Parameters

<INTERFACE-ID>

Specifies an interface. Format: `member/slot/port`.

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing global statistics for all interfaces:

```
switch# show lldp statistics

LLDP Global Statistics
======================

Total Packets Transmitted            : 19
Total Packets Received               : 19
Total Packets Received And Discarded : 0
Total TLVs Unrecognized              : 0

LLDP Port Statistics
====================

PORT-ID        TX-PACKETS     RX-PACKETS     RX-DISCARDED   TLVS-UNKNOWN
--------------------------------------------------------------------------
1/1/1          7              7              0              0
1/1/2          7              7              0              0
1/1/3          0              0              0              0
1/1/4          0              0              0              0
1/1/5          0              0              0              0
...
mgmt           5              5              0              0

```

```
```

Showing statistics for interface **`1/1/1`**:

```
switch# show lldp statistics 1/1/1

LLDP Statistics
===============

Port Name                         : 1/1/1
Packets Transmitted               : 159
Packets Received                  : 163
Packets Received And Discarded    : 0
Packets Received And Unrecognized : 0
```

show lldp statistics mgmt

Syntax

```
show lldp statistics mgmt
```

Description

Shows LLDP statistics for the OOBM interface.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing LLDP statistics for the OOBM interface:

```
switch# show lldp statistics mgmt

LLDP Statistics
===============

Port Name                          : mgmt
Packets Transmitted                : 20
Packets Received                   : 23
Packets Received And Discarded     : 0
Packets Received And Unrecognized : 0
```

show lldp neighbor-info

Syntax

`show lldp neighbor-info [<INTERFACE-NAME>] [vsx-peer]`

Description

Displays information about neighboring devices for all interfaces or for a specific interface. The information displayed varies depending on the type of neighbor connected and the type of TLVs sent by the neighbor.

Command context

Manager (#)

Parameters

<INTERFACE-NAME>

Specifies the interface for which to show information for neighboring devices. Use the format `member/slot/port` (for example, `1/3/1`).

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing LLDP information for all interfaces:

```
switch# show lldp neighbor-info

LLDP Neighbor Information
=========================

Total Neighbor Entries          : 3
Total Neighbor Entries Deleted  : 0
Total Neighbor Entries Dropped  : 0
Total Neighbor Entries Aged-Out : 0

LOCAL-PORT  CHASSIS-ID         PORT-ID       PORT-DESC     TTL      SYS-NAME
--------------------------------------------------------------------------------
1/1/1       70:72:cf:a4:7d:50  1/1/1         1/1/1         32       switch
1/1/2       48:0f:cf:af:73:80  1/1/2         1/1/2         120      switch
1/1/46      48:0f:cf:af:73:80  1/1/46        1/1/46        120      switch
mgmt        48:0f:cf:af:73:80  mgmt          mgmt          120      switch
```

Showing information for interface `1/3/1` when it has only one switch connected as a neighbor:

```
switch# show lldp neighbor-info 1/1/1

Port                           : 1/1/1
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : HP-3800-24G-PoEP-2XG
Neighbor Chassis-Description   : HP J9587A 3800-24G-PoE+-2XG Switch, revision...
Neighbor Chassis-ID            : 10:60:4b:39:3e:80
Neighbor Management-Address    : 192.168.1.1
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge
Neighbor Port-ID               : 1/1/1
Neighbor Port-Desc             : 1/1/1
Neighbor Port VLAN ID          :
TTL                            : 120
```

Showing information for interface `1/3/10` when the neighbor sends a DOT3 power TLV:

```
switch# show lldp neighbor-info 1/3/10
Port                           : 1/3/10
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description   : ArubaOS (MODEL: 325), Version Aruba IAP
Neighbor Chassis-ID            : 84:d4:7e:ce:5d:68
Neighbor Management-Address    : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled   : WLAN
Neighbor Port-ID               : 84:d4:7e:ce:5d:68
Neighbor Port-Desc             : eth0
TTL                            : 120
Neighbor Port VLAN ID          :
Neighbor PoE information       : DOT3
Neighbor Power Type            : TYPE2 PD
Neighbor Power Priority        : Unkown
Neighbor Power Source          : Primary
PD Requested Power Value       : 25.0 W
PSE Allocated Power Value: 25.0 W
Neighbor Power Supported       : Yes
```

```
Neighbor Power Enabled          : Yes
Neighbor Power Class            : 5
Neighbor Power Paircontrol      : No
PSE Power Pairs                 : Signal
```

Showing information for interface `1/1/1` when it has multiple neighbors (displays a maximum of four):

```
switch# show lldp neighbor-info 1/1/1

Port                            : 1/1/1
Neighbor Entries                : 4
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:00
Neighbor Management-Address     : 10.1.1.2
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/1
Neighbor Port-Desc              : 1/1/1
Neighbor Port VLAN ID           :
TTL                             : 120
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:01
Neighbor Management-Address     : 10.1.1.3
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/1
Neighbor Port-Desc              : 1/1/1
Neighbor Port VLAN ID           :
TTL                             : 120
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:02
Neighbor Management-Address     : 10.1.1.4
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/1
Neighbor Port-Desc              : 1/1/1
Neighbor Port VLAN ID           : 50
TTL                             : 120
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:03
Neighbor Management-Address     : 10.1.1.5
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge, Router
Neighbor Port-ID                : 1/1/1
Neighbor Port-Desc              : 1/1/1
Neighbor Port VLAN ID           : 100
TTL                             : 120
```

show lldp neighbor-info detail

Syntax

```
show lldp neighbor-info detail [vsx-peer]
```

Description

Shows detailed LLDP neighbor information for all LLDP neighbor connected interfaces.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing detailed LLDP information for all interfaces:

```
switch# show lldp neighbor-info detail

LLDP Neighbor Information
=========================

Total Neighbor Entries          : 6
Total Neighbor Entries Deleted  : 2
Total Neighbor Entries Dropped  : 0
Total Neighbor Entries Aged-Out : 2

--------------------------------------------------------------------------------

Port                           : 1/1/1
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/4
Neighbor Port-Desc             : 1/1/4
Neighbor Port VLAN ID          : 1
TTL                            : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD

--------------------------------------------------------------------------------

Port                           : 1/1/2
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
```

```
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/5
Neighbor Port-Desc             : 1/1/5
Neighbor Port VLAN ID          : 1
TTL                            : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD

--------------------------------------------------------------------------------

Port                           : 1/1/3
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/6
Neighbor Port-Desc             : 1/1/6
Neighbor Port VLAN ID          : 1
TTL                            : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD

--------------------------------------------------------------------------------

Port                           : 1/1/46
Neighbor Entries               : 1
Neighbor Entries Deleted       : 0
Neighbor Entries Dropped       : 0
Neighbor Entries Aged-Out      : 0
Neighbor Chassis-Name          : 6300
Neighbor Chassis-Description   : Aruba ...
Neighbor Chassis-ID            : 38:11:17:1a:d5:00
Neighbor Management-Address    : 38:11:17:1a:d5:00
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/19
Neighbor Port-Desc             : 1/1/19
Neighbor Port VLAN ID          : 1
TTL                            : 120

Neighbor Mac-Phy details
Neighbor Auto-neg Supported    : true
Neighbor Auto-Neg Enabled      : true
Neighbor Auto-Neg Advertised   : 1000 BASE_TFD, 100 BASE_T4, 10 BASET_FD
Neighbor MAU type              : 1000 BASETFD

--------------------------------------------------------------------------------

Port                           : 1/1/47
Neighbor Entries               : 1
```

```
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : 6300
Neighbor Chassis-Description    : Aruba ...
Neighbor Chassis-ID             : 38:11:17:1a:d5:00
Neighbor Management-Address     : 38:11:17:1a:d5:00
Chassis Cap
```

show lldp neighbor-info mgmt

Syntax

```
show lldp neighbor-info mgmt
```

Description

Displays information about neighboring devices connected to the OOBM interface.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing LLDP information for the OOBM interface:

```
switch# show lldp neighbor-info mgmt

Port                            : mgmt
Neighbor Entries                : 1
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : HP-3800-24G-PoEP-2XG
Neighbor Chassis-Description    : HP J9587A 3800-24G-PoE+-2XG Switch, revision...
Neighbor Chassis-ID             : 10:60:4b:39:3e:80
Neighbor Management-Address     : 192.168.1.1
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled    : Bridge
Neighbor Port-ID                : mgmt
Neighbor Port-Desc              : mgmt
Neighbor Port VLAN ID           :
TTL                             : 120
```

Showing LLDP information for the OOBM interface when there are four neighbors:

```
switch# show lldp neighbor-info mgmt

Port                            : mgmt
Neighbor Entries                : 4
Neighbor Entries Deleted        : 0
Neighbor Entries Dropped        : 0
Neighbor Entries Aged-Out       : 0
Neighbor Chassis-Name           : switch
Neighbor Chassis-Description    : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID             : 1c:98:ec:fe:25:00
Neighbor Management-Address     : 10.1.1.2
```

```
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/1
Neighbor Port-Desc             : 1/1/1
Neighbor Port VLAN ID          :
TTL                            : 120

Neighbor Chassis-Name          : switch
Neighbor Chassis-Description   : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID            : 1c:98:ec:fe:25:01
Neighbor Management-Address    : 10.1.1.3
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/1
Neighbor Port-Desc             : 1/1/1
Neighbor Port VLAN ID          :
TTL                            : 120

Neighbor Chassis-Name          : switch
Neighbor Chassis-Description   : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID            : 1c:98:ec:fe:25:02
Neighbor Management-Address    : 10.1.1.4
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/1
Neighbor Port-Desc             : 1/1/1
Neighbor Port VLAN ID          :
TTL                            : 120

Neighbor Chassis-Name          : switch
Neighbor Chassis-Description   : Aruba JL375A 8400X XL.01.01.0001
Neighbor Chassis-ID            : 1c:98:ec:fe:25:03
Neighbor Management-Address    : 10.1.1.5
Chassis Capabilities Available : Bridge, Router
Chassis Capabilities Enabled   : Bridge, Router
Neighbor Port-ID               : 1/1/1
Neighbor Port-Desc             : 1/1/1
Neighbor Port VLAN ID          :
TTL                            : 120
```

show lldp local-device

Syntax

`show lldp local-device [vsx-peer]`

Description

Shows global LLDP information advertised by the switch, as well as port-based data. If VLANs are configured on any active interfaces, the VLAN ID is only shown for trunk native or untagged VLAN IDs on access interfaces.

Command context

Manager (`#`)

Parameters

`[vsx-peer]`

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing global LLDP information only (all ports including OOBM port are administratively down):

```
switch# show lldp local-device

Global Data
===========

Chassis-ID               : 1c:98:ec:e3:45:00
System Name              : switch
System Description       : Aruba JL375A 8400X XL.01.01.0001
Management Address       : 192.168.10.1
Capabilities Available   : Bridge, Router
Capabilities Enabled     : Bridge, Router
TTL                      : 120
```

Showing all ports except **1/1/11** and OOBM as administratively down:

```
switch# show lldp local-device

Global Data
===========

Chassis-ID               : 1c:98:ec:e3:45:00
System Name              : switch
System Description       : Aruba
Management Address       : 192.168.10.1
Capabilities Available   : Bridge, Router
Capabilities Enabled     : Bridge, Router
TTL                      : 120

Port Based Data
===============

Port-ID            : 1/1/11
Port-Desc          : "1/1/11"
Port Mgmt-Address  : 164.254.21.220
Port VLAN ID       : 0

Port-ID            : mgmt
Port-Desc          : "mgmt"
Port Mgmt-Address  : 164.254.21.220
```

In this example, all the ports except **1/1/11** are administratively down, and VLAN ID 100 is configured on this access interface.

```
switch# show lldp local-device

Global Data
===========

Chassis-ID               : 1c:98:ec:e3:45:00
System Name              : switch
System Description       : Aruba
Management Address       : 192.168.10.1
Capabilities Available   : Bridge, Router
Capabilities Enabled     : Bridge, Router
TTL                      : 120
```

```
Port Based Data
===============

Port-ID            : 1/1/11
Port-Desc          : "1/1/11"
Port VLAN ID       : 100
Parent Interface   : interface 1/1/11
```

Cisco Discovery Protocol (CDP)

Cisco Discovery Protocol (CDP) is a proprietary layer 2 protocol supported by most Cisco devices. It is used to exchange information, such as software version, device capabilities, and voice VLAN information, between directly connected devices, such as a VoIP phone and a switch.

CDP support

By default, CDP is enabled on each active switch port. This is a read-only capability, which means the switch can receive and store information about adjacent CDP devices, but does not generate CDP packets (except when communicating with Cisco IP phones.)

The switch supports CDPv2 only and does not support SNMP MIB traps.

When a CDP-enabled port receives a CDP packet from another CDP device, it enters data for that device into the CDP Neighbors table, along with the port number on which the data was received. It does not forward the packet. The switch also periodically purges the table of any entries that have expired. (The holdtime for any data entry in the switch CDP Neighbors table is configured in the device transmitting the CDP packet and cannot be controlled in the switch receiving the packet.) A switch reviews the list of CDP neighbor entries every three seconds and purges any expired entries.

Support for legacy Cisco IP phones

Autoconfiguration of legacy Cisco IP phones for tagged voice VLAN support requires CDPv2.

On initial boot-up, and sometimes periodically, a Cisco phone queries the switch and advertises information about itself using CDPv2. When the switch receives the VoIP VLAN Query TLV (type 0x0f) from the phone, the switch immediately responds with the voice VLAN ID in a reply packet using the VoIP VLAN Reply TLV (type 0x0e). This enables the Cisco phone to boot properly and send traffic on the advertised voice VLAN ID.

The switch CDP packet includes these TLVs:

- CDP Version: 2
- CDP TTL: 180 seconds
- Checksum
- Capabilities (type 0x04): 0x0008 (is a switch)
- Native VLAN: The PVID of the port
- VoIP VLAN Reply (type 0xe): voice VLAN ID (same as advertised by LLDP-MED)
- Trust Bitmap (type 0x12): 0x00
- Untrusted port CoS (type 0x13): 0x00

CDP commands

cdp

Syntax

```
cdp
```

Description

Configures CDP support globally on all active interfaces or on a specific interface. By default, CDP is enabled on all active interfaces.

When CDP is enabled, the switch adds entries to its CDP Neighbors table for any CDP packets it receives from neighboring CDP devices.

When CDP is disabled, the CDP Neighbors table is cleared and the switch drops all inbound CDP packets without entering the data in the CDP Neighbors table.

The `no` form of this command disables CDP support globally on all active interfaces or on a specific interface.

Command context

`config`

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling CDP globally:

```
switch(config)# cdp
```

Disabling CDP globally:

```
switch(config)# no cdp
```

Enabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# cdp
```

Disabling CDP on interface **1/1/1**:

```
switch(config)# interface 1/1/1
switch(config-if)# no cdp
```

clear cdp counters

Syntax

```
clear cdp counters
```

Description

Clears CDP counters.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing CDP counters:

```
switch(config) clear cdp counters
```

clear cdp neighbor-info

Syntax

`clear cdp neighbor-info`

Description

Clears CDP neighbor information.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing CDP neighbor information:

```
switch(config) clear neighbor-info
```

show cdp

Syntax

`show cdp`

Description

Shows CDP information for all interfaces.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing CDP information:

```
switch(config)# show cdp
CDP Global Information
======================
```

```
CDP            : Enabled
CDP Mode       : Rx only
CDP Hold Time  : 180 seconds

Port          CDP
--------   --------------
1/1/1         Enabled
1/1/2         Enabled
1/1/3         Enabled
1/1/4         Enabled
1/1/5         Enabled
1/1/6         Enabled
1/1/7         Enabled
1/1/8         Enabled
1/1/9         Enabled
1/1/10        Enabled
```

show cdp neighbor-info

Syntax

`show cdp neighbor-info <INTERFACE-ID>`

Description

Shows CDP information for all neighbors or for CDP information on a specific interface.

Command context

`config`

Parameters

<INTERFACE-ID>

Specifies an interface. Format: `member/slot/port`.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all CDP neighbor information:

```
switch(config)# show cdp neighbor-info
Port        Device ID                 Platform                  Capability
--------------------------------------------------------------------------------
1/1/1       myswitch                  cisco WS-C2950-12         SI
```

Showing CDP information for interface **1/1/1**:

```
switch(config)# show cdp neighbor-info 1/1/1
Local Port : 1/1/1
MAC                  : 3c:a8:2a:7b:6b:2b
Device ID            : SEPd4adbd2a30d6
Address              : 2.71.0.230
Platform             : Cisco IP Phone 3905
Duplex               : full
Capability           : host
Voice VLAN Support : Yes
Neighbor Port-ID     : Port 1
```

show cdp traffic

Syntax

```
show cdp neighbor-info
```

Description

Shows CDP statistics for each interface.

Command context

```
config
```

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing CDP traffic statistics:

```
switch(config)# show cdp traffic
CDP Statistics
====================
Port         Transmitted Frames       Received Frames          Discarded Frames
--------------------------------------------------------------------------------
1/1/1           0                        4                        0
1/1/2           0                        0                        0
1/1/3           0                        2                        0
1/1/4           0                        0                        0
1/1/5           0                        0                        0
```

DCBx is a discovery and capability exchange protocol to discover peers and negotiate Data Center Bridging configuration. DCBx is specified as part of IEEE 802.1Qaz-2011. DCBx uses LLDP as the underlying protocol for exchange of parameters with the peer. The DCBx parameters are exchanged as LLDP TLVs.

There are two main versions of DCBx: IEEE DCBx and CEE DCBx. AOS-CX switches support the IEEE DCBx version which uses an OUI of 0x0080c2.

DCBx supports VSX synchronization. For more information about enabling VSX synchronization, see the *Virtual Switching Extension (VSX) Guide* for your switch and software version.

DCBx LLDP TLVs supported by ArubaOS-CX:

- PFC (Priority Flow Control) TLV with subtype 0x0b.
  - Advertises priorities that are configured in the switch as lossy/lossless.
  - PFC TLVs are symmetrical which means they have to match between peers.
  - If a peer PFC priority configuration does not match switch configuration, a misconfiguration error is displayed by the command `show dcbx interface`.
- ETS (Enhanced Transmission Selection) configuration TLV with subtype 0x09.
  - Advertises the configured bandwidth reservation and the transmission algorithm used for each traffic class.
  - This is an asymmetric TLV which means the configuration does not have to match between peers.
- ETS (Enhanced Transmission Selection) recommendation TLV with subtype 0x0a.
  - If the peer device is willing to accept switch ETS configuration, then the contents of this TLV can be used by peer to configure itself.
  - The switch sends the current ETS configuration as the ETS recommended values.
- Application priority TLV with subtype 0x0c.
  - This is an informational TLV that tells the peer to map certain application traffic to a priority.
  - The user has to correctly configure this information using the application priority command.
  - This allows the peer to map applications to appropriate lossless priority configured on the switch.

DCBx guidelines

- DCBx is disabled by default.
- LLDP must be enabled on the interfaces supporting DCBx.
- DCBx is only supported on physical interfaces and not on management or logical interfaces, similar to how LLDP behaves.
- ArubaOS-CX supports only the IEEE 802.1Qaz 2011 version of DCBx with an OUI of 0x0080c2.

- ArubaOS-CX advertises DCBx with 'willing bit' set to 0 in all TLVs. This tells the peer that the switch is not willing to change its configuration to match the peer's configuration.
- When a peer switch does not support IEEE DCBx, a misconfiguration error will be displayed in the `show dcbx interface` output.

DCBx commands

`lldp dcbx`

Syntax

`lldp dcbx`

`no lldp dcbx`

Description

Globally enables DCBx.

The **no** form of this command disables DCBx advertisement.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling DCBx globally.

```
switch(config)# lldp dcbx
```

Disabling DCBx globally.

```
switch(config)# no lldp dcbx
```

`lldp dcbx (per interface)`

Syntax

`lldp dcbx`

`no lldp dcbx`

Description

Enables DCBx on an interface. Default is enabled. DCBx must be enabled globally for this configuration to take effect.

The **no** form of this command disables DCBx on an interface.

Command context

`config-if`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Enabling DCBx on interface 1/1/1.

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dcbx
```

Disabling DCBx on interface 1/1/1.

```
switch(config)# interface 1/1/1
switch(config-if)# no lldp dcbx
```

Enabling DCBx and configuring PFC for priority 4 on interface 1/1/1.

NOTE: Priority Flow Control (PFC) commands are only supported on the 8325 and 8360.

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dcbx
switch(config-if)# flow-control priority 4
```

dcbx application

Syntax

```
dcbx application {ISCSI | TCP-SCTP <PORT-NUM> | TCP-SCTP-UDP <PORT-NUM> | UDP <PORT-NUM> |

ether <ETHERTYPE>} priority <PRIORITY>

no dcbx application
```

Description

Configures application to priority map that gets advertised in DCBx application priority messages. This tells the DCBx peer to send the application traffic with the configured priority so that the traffic is treated as lossless. Multiple applications can be configured in this manner. PFC lossless priority configured on the switch should be the same as this priority.

NOTE: Priority Flow Control (PFC) commands are only supported on the 8325 and 8360.

The `no` form of this command removes the existing configuration.

Command context

`config`

Parameters

`ISCSI`

Specifies a physical port on the switch. TCP ports 860 and 3260.

`TCP-SCTP <PORT-NUM>`

Specifies the traffic for a specified TCP or SCTP port.

TCP-SCTP-UDP *<PORT-NUM>*

Specifies the traffic for a specified TCP or SCTP or UDP port.

UDP *<PORT-NUM>*

Specifies the traffic for a specified UDP port.

<ETHERTYPE>

Specifies the traffic for a specific Ethernet type. Range: 1536 to 65535.

<PRIORITY>

Specifies the application priority. Range: 0 to 7.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Mapping iSCSI traffic to priority 5.

```
switch(config)# dcbx application iscsi priority 5
```

Mapping TCP or SCTP traffic with port 860 to priority 3.

```
switch(config)# dcbx application tcp-sctp 860 priority 3
```

show dcbx interface

Syntax

`show dcbx interface <IFNAME>`

Description

Shows the current DCBx status and the configuration of PFC, ETS, and application priority applied on the interface and the status of the TLVs received from the peer.

Command context

Operator (>) or Manager (#)

Parameters

interface *<IFNAME>*

Specifies the interface name.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

Showing DCBx interface 1/1/1.

```
switch# show dcbx interface 1/1/1
DCBx admin state : enabled
DCBx operational state : active

Priority Flow Control (PFC)
```

```
---------------------------
Operational state : active

Local advertisement:
    Willing                  : No
    MacSec Bypass Capability : No
    Max traffic classes      : 4

    Priority    Enabled
    0           False
    1           False
    2           False
    3           False
    4           False
    5           False
    6           True
    7           False

Remote advertisement:
    Willing                  : Yes
    MacSec Bypass Capability : No
    Max traffic classes      : 4

    Priority    Enabled
    0           False
    1           False
    2           False
    3           False
    4           False
    5           False
    6           True
    7           False


Enhanced Transmission Selection (ETS)
-------------------------------------

Local advertisement:
    Willing             : No
    Credit Based Shaper : No
    Max traffic classes : 4

    Priority  Traffic Class
    0         0
    1         5
    2         0
    3         0
    4         0
    5         0
    6         0
    7         7

    Traffic  Bandwidth   Algorithm
    Class    Percentage
    0        0           Strict
    1        0           Strict
    2        5           ETS
    3        5           ETS
    4        5           ETS
    5        5           ETS
    6        5           ETS
    7        75          ETS

Remote advertisement:
```

```
    Willing            : No
    Credit Based Shaper : No
    Max traffic classes : 4

    Priority    Traffic Class
    0           0
    1           5
    2           0
    3           0
    4           0
    5           0
    6           0
    7           7

    Traffic  Bandwidth   Algorithm
    Class    Percentage
    0        0           Strict
    1        5           ETS
    2        5           ETS
    3        5           ETS
    4        5           ETS
    5        5           ETS
    6        5           ETS
    7        70          ETS

Application Priority Map
------------------------

Local advertisement:
Protocol        Port/Type        Priority
-----------------------------------------
tcp-sctp        320              5
ether           0x8906           4
iscsi                            7

Remote advertisement:
Protocol        Port/Type        Priority
-----------------------------------------
tcp-sctp        320              5
ether           0x8906           4
```

Zero Touch Provisioning (ZTP) enables the auto-configuration of factory default switches without a network administrator onsite.

When a switch is booted from its factory default configuration, ZTP autoprovisions the switch by automatically downloading and installing a firmware file, a configuration file, or both. With ZTP, even a nontechnical user (for example: a store manager in a retail chain or a teacher in a school) can deploy devices at a site.

ZTP support

The switch supports standards-based Zero Touch Provisioning (ZTP) operations as follows:

- The switch must be running the factory default configuration.
- The switch can connect to the DHCP server from the OOBM management port.
- ZTP operations are supported over IPv4 connections only. IPv6 connections are not supported for ZTP operations.
- You must configure the DHCP server to provide a standards-based ZTP server solution. Options and features that are specific to Network Management Solution (NMS) tools, such as AirWave, are not supported.
    - Aruba Central on-premise can manage AOS-CX switches on supported models through DHCP ZTP using two approaches:

      On the DHCP server, configure DHCP option-60 as "ArubaInstantAP" **90** and provide the value in option-43 in the format *<group-details>, <aruba-central-on-prem-ip-or-fqdn>, <shared-secret>.*

      On the DHCP server, configure DHCP option-60 as HPE vendor VCI and provide the value in option-43 in the tag-length-value (TLV) format with sub-option code of 146 as the Aruba Central on-premise FQDN or IPv4 address.

  Supported DHCP options are:

DHCP option	Description
43	Vendor Specific Information
43 suboption 144	Name of the configuration file
43 suboption 145	Name of the firmware image file
43 suboption 146	Aruba Central FQDN or IPv4 address
43 suboption 148	HTTP Proxy FQDN or IPv4 address

Table Continued

DHCP option	Description
60	Vendor Class Identifier (VCI)
66	IPv4 address of the TFTP server (Specifying a host name instead of an IP address is not supported.)
67	Name of the configuration file (Option 43 suboption 144 takes precedence over this option.)

- The configuration file is a text file or JSON file that becomes the startup and running configuration on the switch after the ZTP operation is complete. The configuration can be in CLI or in JSON format.
- When the switch is started using the factory default configuration, the ZTP operation is started automatically and is active until any running configuration of the switch is modified. There is no CLI command required to start the operation.

The switch supports the following standards:

- **RFC 2131**, *Dynamic Host Configuration Protocol*.
- **RFC 2132**, *DHCP Options and BOOTP Vendor Extensions*. Support is limited to the options listed in the table "Supported DHCP options for ZTP on AOS-CX."

Hewlett Packard Enterprise recommends that you implement ZTP in a secure and private environment. Any public access can compromise the security of the switch, as follows:

- ZTP is enabled only in the factory default configuration of the switch, DHCP snooping is not enabled. The Rogue DHCP server must be manually managed.
- The DHCP offer is in plain data without encryption.

Setting up ZTP on a trusted network

The following procedure is an overview of setting up a Zero Touch Provisioning (ZTP) environment to provision newly installed switches automatically. The procedure is intended for network administrators who are familiar with automatically provisioning switches in a network, and does not provide detailed information about configuring or managing switches.

Procedure

1. For each switch model to be provisioned using ZTP, do the following:
    a. Obtain the switch firmware image file.
    b. Prepare the switch configuration file. The configuration file becomes the running configuration and the startup configuration on the switch.
2. Set up a TFTP server and record its IP address. The address is required when you set up the DHCP server. The switch must be able to reach the TFTP server and DHCP server, either on the same subnet, or on a remote subnet via DHCP relay.

    For switches that do not support ZTP connections through a data port, use the management port and management network.

3. Publish the configuration files and image files to the TFTP server. You need to know the locations of the files and the IP address of the TFTP server when you set up the vendor class options on the DHCP server.
4. On the DHCP server, set up vendor classes for each switch model you plan to provision. To do this you need the following information:
   - The IP address of the TFTP server. Using a host name is not supported.
   - The path to the switch configuration and firmware image files on the TFTP server.
   - The vendor class identifier (VCI) for each switch model.

     You can obtain the VCI by entering the `show dhcp client vendor-class-identifier` command from a switch CLI command prompt in the manager context. The VCI is the text string in the response that starts with `Aruba`.

     For example:

     ```
     switch# show dhcp client vendor-class-identifier
     Vendor Class Identifier:  Aruba xxxxx xxxx
     ```

     Where x indicates the switch model number.
5. At the installation site, provide the switch installer with a Cat6 network cable connected to the network that includes the DHCP and TFTP servers, and information about the switch port to use. The switch installer plugs the cable into the data port you specify.

   The ZTP operation begins when power is applied to the switch after the network cable is installed.
6. Assuming the downloaded configuration includes a way to access the CLI of the switch, you can enter the following command to show the options offered by the DHCP server and the status of the ZTP operation:

   `show ztp information`

ZTP process during switch boot

1. The switch boots up with the factory default configuration.

   If the ZTP operation detects that the switch configuration is different from the factory default configuration, the ZTP operation ends. The switch must be configured at the installation site.
2. The switch sends out a DHCP discovery from the management port.

   The switch waits to receive DHCP options indefinitely or until the running configuration is modified. If a DHCP IP address is received but no DHCP options are received, the switch waits an additional minute before ending the ZTP operation.

   After the ZTP operation ends, there is no automatic retry. You can either attempt to boot the switch with the factory default configuration again, configure the switch at the installation site, or use the ZTP force-provision CLI to trigger the ZTP process, ignoring the present running configuration of the switch.

   - Once force-provision is enabled, new DHCP requests are sent from the switch. Disabling force-provision does not stop the DHCP already in progress, but only changes the switch configuration status of force-provision.
   - If ZTP fails while force-provision is enabled, there is no automatic retry. To retry, `ztp force-provision` should be disabled and re-enabled to clear the current ZTP state and send a new DHCP

request. When `ztp force-provision` is already enabled on the switch, re-enabling it results in no operation.

- If the DHCP server is configured to provide both ZTP image and configuration options and there is a non-default startup configuration present on the switch, clearing the non-default startup configuration before triggering `ztp force-provision` is recommended. If an image is downloaded via ZTP, the switch reboots once the image download is complete and ZTP force-provision configuration is lost, causing ZTP to enter into a failed state. ZTP force-provision will need to be enabled again to continue the process.

3. The DHCP server responds with an offer containing the following:

- The IPv4 address of the TFTP server
- One or both of the following:
  - The name of the firmware image file
  - The name of the configuration file
- Aruba Central Location (optional)
- HTTP Proxy Location (optional)

4. If a firmware image file is offered, the ZTP operation downloads the image file from the TFTP server to the switch. If the current switch image and downloaded firmware image version do not match, then the switch boots with the downloaded image:

- If the image upgrade fails, the switch retains its original firmware image and the ZTP operation ends with a failed status.
- If the image upgrade succeeds, the ZTP operation is started again after the switch reboots. Because the downloaded image file matches the image file installed on the switch, the ZTP operation continues, and checks if a configuration file is offered.

5. If a configuration file is offered, the ZTP operation downloads the configuration file copies the file to the startup configuration of the switch:

- If the startup configuration update fails, the switch retains its factory-default running configuration and the ZTP operation ends with a failed status.
- If the startup configuration update is successful, the startup configuration is copied to the running configuration:
  - If the copy operation fails, the ZTP operation ends with a failed status.
  - If the copy operation succeeds, the ZTP operation ends successfully.

ZTP VSF Switchover Support

ZTP status is not synced in the VSF stack. When the VSF stack is formed, configuration changes are applied on the master switch, which is then synced to standby switch. When the switchover is performed on the VSF stack, the standby becomes the new master switch.

As part of the switchover process, the ZTP daemon starts on the new master. The status of the ZTP is failed because there are configuration changes present.

ZTP commands

`show ztp information`

Syntax

```
show ztp information
```

Description

Shows information about Zero Touch Provisioning (ZTP) operations performed on the switch.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

When a switch configured to use ZTP is booted from a factory default configuration, the switch contacts a DHCP server, which offers options for obtaining files used to provision the switch:

- The IP address of the TFTP server
- The name of the image file
- The name of the configuration file

The `show ztp information` command shows the options offered by the DHCP server and the status of the ZTP operation.

The status of the ZTP operation is one of the following:

Success

The ZTP operation succeeded.

One of the following is true:

- Both the running configuration and the startup configuration were updated.
- The IP address of the TFTP server was received, but the offer did not include a configuration file or a firmware image file.
- Any combination of vendor encapsulated DHCP options are received as configured, along with the firmware image and switch configuration file.
- Only vendor encapsulated DHCP options are configured and are received accordingly.

Failed - Custom startup configuration detected

The switch was booted from a configuration that is not the factory default configuration. For example, the administrator password has been set.

Failed - Timed out while waiting to receive ZTP options

Either the switch received the DHCP IPv4 address but no ZTP options were received within 1 minute or ZTP force-provision is triggered and no ZTP options are received within 3 minutes.

Failed - Detected change in running configuration

The running configuration was modified by a user while the ZTP operation was in progress.

Failed - TFTP server unreachable

The TFTP server is not reachable at the specified IP address.

Failed - TFTP server information unavailable

The image file name or config file name is provided without the TFTP server location to fetch the files from and ZTP enters failed state.

Failed - Invalid configuration file received

Either the file transfer of the configuration file failed, or the configuration file is invalid (an error occurred while attempting to apply the configuration).

Failed - Invalid image file received

Either the file transfer of the firmware image file failed, or the firmware image file is invalid (an error occurred while verifying the image).

Examples

In the following example, the ZTP operation succeeded, and both an image file and a configuration file were provided.

```
VSF-10-Mbr# show ztp information
TFTP Server             : 10.1.84.160
Image File              : FL_10_06_0001CK.swi
Configuration File      : 102720-new-setup-config-updated.txt
Status                  : Success
Aruba Central Location : NA
Force-Provision         : Disabled
HTTP Proxy Location     : NA
VSF-10-Mbr#
```

In the following example, the ZTP option succeeded. A configuration file was not provided, but an image file was provided.

```
VSF-10-Mbr# show ztp information
TFTP Server             : 10.1.84.160
Image File              : TL_10_02_0001.swi
Configuration File      : NA
Status                  : Success
Aruba Central Location : NA
Force-Provision         : Disabled
HTTP Proxy Location     : NA
VSF-10-Mbr#
```

In the following example, the ZTP operation failed because the TFTP server was unreachable.

```
VSF-10-Mbr# show ztp information
TFTP Server             : 10.1.84.160
Image File              : TL_10_02_0001.swi
Configuration File      : 102720-new-setup-config-updated.txt
Status                  : Failed - TFTP server unreachable
Aruba Central Location : NA
Force-Provision         : Disabled
HTTP Proxy Location     : NA
VSF-10-Mbr#
```

In the following example, the ZTP operation was stopped because the switch did not receive any options from the DHCP server for ZTP within 1 minute of receiving the IP address from the server.

```
VSF-10-Mbr## show ztp information
TFTP Server              : NA
Image File               : NA
Configuration File       : NA
Status                   : Failed - Timed out while waiting to receive ZTP options
Aruba Central Location : NA
Force-Provision          : Disabled
HTTP Proxy Location      : NA
VSF-10-Mbr#
```

In the following example, the ZTP operation was stopped because the switch was booted from a configuration that was not the factory default configuration.

```
switch# show ztp information
TFTP Server              : 10.0.0.2
Image File               : TL_10_02_0001.swi
Configuration File       : ztp.cfg
Status                   : Failed - Custom startup configuration detected
Aruba Central Location : NA
Force-Provision          : Disabled
HTTP Proxy Location      : NA
```

ztp force provision

Syntax

```
ztp force-provision

no ztp force-provision
```

Description

Starts on-demand ZTP.

Command context

Operator (>) or Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Usage

DHCP options received are processed independent of he current state of configuration on the switch. Previous ZTP TFTP Server, Image File, Configuration File, Aruba Central Location, and HTTP Proxy location options are cleared and the switch sends a DHCP request.

Examples

In the following example, force-provision is enabled.

```
switch# configure terminal
switch(config)# ztp force-provision
```

In the following example, force-provision status is checked while enabled.

```
switch# show ztp information
TFTP Server              : 10.0.0.2
Image File               : TL_10_02_0001.swi
Configuration File       : ztp.cfg
Status                   : Success
Aruba Central Location : NA
```

```
Force-Provision         : Enabled
HTTP Proxy Location     : NA
```

In the following example, force-provision is disabled.

```
switch# configure terminal
switch(config)# no ztp force-provision
```

In the following example, force-provision status is checked while disabled.

```
switch# show ztp information
TFTP Server             : 10.0.0.2
Image File              : TL_10_02_0001.swi
Configuration File      : ztp.cfg
Status                  : Success
Aruba Central Location : NA
Force-Provision         : Disabled
HTTP Proxy Location     : NA
```

bluetooth disable

Syntax

```
bluetooth disable

no bluetooth disable
```

Description

Disables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE). Bluetooth is enabled by default.

The `no` form of this command enables the Bluetooth feature on the switch.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Example

Disabling Bluetooth on the switch. *<XXXX>* is the switch platform and *<NNNNNNNNNN>* is the device identifier.

```
switch(config)# bluetooth disable
switch# show bluetooth
Enabled              : No
Device name          : <XXXX>-<NNNNNNNNNN>

switch(config)# show running-config
...
bluetooth disabled
...
```

bluetooth enable

Syntax

```
bluetooth enable

no bluetooth enable
```

Description

This command enables the Bluetooth feature on the switch. The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

Default: Bluetooth is enabled by default.

The `no` form of this command disables the Bluetooth feature on the switch.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Usage

The default configuration of the Bluetooth feature is `enabled`. The output of the `show running-config` command includes Bluetooth information only if the Bluetooth feature is disabled.

The Bluetooth feature includes both Bluetooth Classic and Bluetooth Low Energy (BLE).

The Bluetooth feature requires the USB feature to be enabled. If the USB feature has been disabled, you must enable the USB feature before you can enable the Bluetooth feature.

Examples

```
switch(config)# bluetooth enable
```

clear events

Syntax

`clear events`

Description

Clears up event logs. Using the `show events` command will only display the logs generated after the `clear events` command.

Command context

Manager (`#`)

Authority

Administrators or local user group members with execution rights for this command.

Examples

Clearing all generated event logs:

```
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2018-10-14:06:57:53.534384|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource utilization poll interval is changed to 27
2018-10-14:06:58:30.805504|lldpd|103|LOG_INFO|MSTR|1|Configured LLDP tx-timer to 36
2018-10-14:07:01:01.577564|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource utilization poll interval is changed to 49

switch# clear events

switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2018-10-14:07:03:05.637544|hpe-sysmond|6301|LOG_INFO|MSTR|1|System resource utilization poll interval is changed to 34
```

clear ip errors

Syntax

`clear ip errors`

Description

Clears all IP error statistics.

Command context

Manager (#)

Authority

Administrators or local user group members with execution rights for this command.

Example

Clearing and showing ip errors:

```
switch# clear ip errors
switch# show ip errors
----------------------------------
Drop reason                Packets
----------------------------------
Malformed packets                0
IP address errors                0
...
```

domain-name

Syntax

`domain-name <NAME>`

`no domain-name [<NAME>]`

Description

Specifies the domain name of the switch.

The `no` form of this command sets the domain name to the default, which is no domain name.

Command context

`config`

Parameters

<NAME>

Specifies the domain name to be assigned to the switch. The first character of the name must be a letter or a number. Length: 1 to 32 characters.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

Setting the domain name to the default value:

```
switch(config)# no domain-name
switch(config)# show domain-name

switch(config)#
```

hostname

Syntax

`hostname <HOSTNAME>`

`no hostname [<HOSTNAME>]`

Description

Sets the host name of the switch.

The `no` form of this command sets the host name to the default value, which is `switch`.

Command context

`config`

Parameters

<HOSTNAME>

Specifies the host name. The first character of the host name must be a letter or a number. Length: 1 to 32 characters. Default: `switch`

Authority

Administrators or local user group members with execution rights for this command.

Examples

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

Setting the host name to the default value:

```
myswitch(config)# no hostname
switch(config)#
```

led locator

Syntax

`led locator {on | off | slow_blink | flashing | fast_blink | half_bright}`

Description

Sets the state of the locator LED.

Command context

Manager (#)

Parameters

on

Turns on the LED.

off

Turns off the LED, which is the default value.

slow_blink

Sets the LED to slow blink on and off.

flashing

Sets the LED to blink on and off repeatedly.

fast_blink

Sets the LED to fast blink on and off.

half_bright

Sets the LED intensity to half bright.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting the state of the locator LED:

```
switch# led locator flashing
```

mtrace

Syntax

```
mtrace <IPV4-SRC-ADDR> <IPV4-GROUP-ADDR> [lhr <IPV4-LHR-ADDR>] [ttl <HOPS>]
    [vrf <VRF-NAME>]
```

Description

Traces the specified IPv4 source and group addresses.

Command context

Manager (#)

Parameters

IPV4-SRC-ADDR

Specifies the source IPv4 address to trace.

IPV4-GROUP-ADDR

Specifies the group IPv4 address to trace.

lhr *<IPV4-LHR-ADDR>*

Specifies the last hop router address from which to start the trace.

ttl *<HOPS>*

Specifies the Time-To-Live duration in hops. Range: 1 to 255 hops. Default: 8 hops.

vrf *<VRF-NAME>*

Specifies the name of the VRF. If a name is not specified the default VRF will be used.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Tracing with source, group, and LHR addresses and TTL:

```
(switch)# mtrace 20.0.0.1 239.1.1.1 lhr 10.1.1.1 ttl 10

Type escape sequence to abort.
Mtrace from 10.0.0.1 for Source 20.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0  10.0.0.1
-1  30.0.0.1 PIM  0 ms
-2  40.0.0.1 PIM  2 ms
-3  50.0.0.1 PIM  100 ms
-4  60.0.0.1 PIM  156 ms
-5  20.0.0.1 PIM  123 ms
```

Tracing with source and group addresses:

```
(switch)# mtrace 200.0.0.1 239.1.1.1

Type escape sequence to abort.
Mtrace from self for Source 200.0.0.1 via Group 239.1.1.1
From destination(?) to source (?)...
Querying ful reverse path...
0  10.0.0.1
-1  30.0.0.1 PIM  0 ms
-2  40.0.0.1 PIM  2 ms
-3  50.0.0.1 PIM  100 ms
-4  60.0.0.1 PIM  156 ms
-5  200.0.0.1 PIM  123 ms
```

show bluetooth

Syntax

```
show bluetooth
```

Description

Shows general status information about the Bluetooth wireless management feature on the switch.

Command context

Operator (>) or Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

This command shows status information about the following:

- The USB Bluetooth adapter
- Clients connected using Bluetooth
- The switch Bluetooth feature.

The output of the `show running-config` command includes Bluetooth information only if the Bluetooth feature is disabled.

The device name given to the switch includes the switch serial number to uniquely identify the switch while pairing with a mobile device.

The management IP address is a private network address created for managing the switch through a Bluetooth connection.

Examples

Example output when Bluetooth is enabled but no Bluetooth adapter is connected. *<XXXX>* is the switch platform and *<NNNNNNNNNN>* is the device identifier.

```
switch# show bluetooth
Enabled              : Yes
Device name          : <XXXX>-<NNNNNNNNNN>
Adapter State        : Absent
```

Example output when Bluetooth is enabled and there is a Bluetooth adapter connected:

```
switch# show bluetooth
Enabled              : Yes
Device name          : <XXXX>-<NNNNNNNNNN>
Adapter State        : Ready
Adapter IP address   : 192.168.99.1
Adapter MAC address  : 480fcf-af153a

Connected Clients
-----------------
Name              MAC Address       IP Address     Connected Since
--------------    --------------    ------------   ------------------------
Mark's iPhone     089734-b12000     192.168.99.10   2018-07-09 08:47:22 PDT
```

Example output when Bluetooth is disabled:

```
switch# show bluetooth
Enabled                : No
Device name            : <XXXX>-<NNNNNNNNNN>
```

show capacities

Syntax

`show capacities <FEATURE> [vsx-peer]`

Description

Shows system capacities and their values for all features or a specific feature.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies a feature. For example, `aaa` or `vrrp`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing all available capacities for BGP:

```
switch# show capacities bgp

System Capacities: Filter BGP
Capacities Name                                                             Value
-----------------------------------------------------------------------------------
Maximum number of AS numbers in as-path attribute                              32
...
```

Showing all available capacities for mirroring:

```
switch# show capacities mirroring

System Capacities: Filter Mirroring
Capacities Name                                                         Value
------------------------------------------------------------------------------
Maximum number of Mirror Sessions configurable in a system                 4
Maximum number of enabled Mirror Sessions in a system                      4
```

Showing all available capacities for MSTP:

```
switch# show capacities mstp

System Capacities: Filter MSTP
Capacities Name                                                            Value
-----------------------------------------------------------------------------------
Maximum number of mstp instances configurable in a system                     64
```

Showing all available capacities for VLAN count:

```
switch# show capacities vlan-count

System Capacities: Filter VLAN Count
Capacities Name                                                            Value
-----------------------------------------------------------------------------------
Maximum number of VLANs supported in the system                             4094
```

show capacities-status

Syntax

```
show capacities-status <FEATURE>
     [vsx-peer]
```

Description

Shows system capacities status and their values for all features or a specific feature.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies the feature, for example `aaa` or `vrrp` for which to display capacities, values, and status. Required.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status

System Capacities Status
Capacities Status Name                                                  Value Maximum
------------------------------------------------------------------------------------------
Number of active gateway mac addresses in a system                         0      16
Number of aspath-lists configured                                          0      64
Number of community-lists configured                                       0      64
...
```

Showing the system capacities status for BGP:

```
switch# show capacities-status bgp

System Capacities Status: Filter BGP
Capacities Status Name                                                      Value Maximum
-----------------------------------------------------------------------------------------
Number of aspath-lists configured                                               0      64
Number of community-lists configured                                            0      64
Number of neighbors configured across all VRFs                                  0      50
Number of peer groups configured across all VRFs                                0      25
Number of prefix-lists configured                                               0      64
Number of route-maps configured                                                 0      64
Number of routes in BGP RIB                                                     0  256000
Number of route reflector clients configured across all VRFs                    0      16
```

show core-dump

Syntax

`show core-dump [all | <SLOT-ID>]`

Description

Shows core dump information about the specified module. When no parameters are specified, shows only the core dumps generated in the current boot of the management module. When the `all` parameter is specified, shows all available core dumps.

Command context

Manager (`#`)

Parameters

`all`

Shows all available core dumps.

`<SLOT-ID>`

Shows the core dumps for the management module or line module in *`<SLOT-ID>`*. *`<SLOT-ID>`* specifies a physical location on the switch. Use the format `member/slot/port` (for example, `1/3/1`) for line modules. Use the format `member/slot` for management modules.

You must specify the slot ID for either the active management module, or the line module.

Authority

Administrators or local user group members with execution rights for this command.

Usage

When no parameters are specified, the `show core-dump` command shows only the core dumps generated in the current boot of the management module. You can use this command to determine when any crashes are occurring in the current boot.

If no core dumps have occurred, the following message is displayed: `No core dumps are present`

To show core dump information for the standby management module, you must use the `standby` command to switch to the standby management module and then execute the `show core-dump` command.

In the output, the meaning of the information is the following:

Daemon Name

Identifies name of the daemon for which there is dump information.

Instance ID

Identifies the specific instance of the daemon shown in the `Daemon Name` column.

Present

Indicates the status of the core dump:

Yes

The core dump has completed and available for copying.

In Progress

Core dump generation is in progress. Do not attempt to copy this core dump.

Timestamp

Indicates the time the daemon crash occurred. The time is the local time using the time zone configured on the switch.

Build ID

Identifies additional information about the software image associated with the daemon.

Examples

Showing core dump information for the current boot of the active management module only:

```
switch# show core-dump
===================================================================================
Daemon Name     | Instance ID | Present     | Timestamp                  | Build ID
===================================================================================
hpe-fand          1399          Yes          2017-08-04 19:05:34          1246d2a
hpe-sysmond       957           Yes          2017-08-04 19:05:29          1246d2a
===================================================================================
Total number of core dumps : 2
===================================================================================
```

Showing all core dumps:

```
switch# show core-dump all
==============================================================================
Management Module core-dumps
==============================================================================
Daemon Name     | Instance ID | Present     | Timestamp                | Build ID
==============================================================================
hpe-sysmond       513           Yes          2017-07-31 13:58:05    e70f101
hpe-tempd         1048          Yes          2017-08-13 13:31:53    e70f101
hpe-tempd         1052          Yes          2017-08-13 13:41:44    e70f101

Line Module core-dumps
==============================================================================
Line Module : 1/1
==============================================================================
dune_agent_0      18958         Yes          2017-08-12 11:50:17    e70f101
dune_agent_0      18842         Yes          2017-08-12 11:50:09    e70f101
==============================================================================
Total number of core dumps : 5
==============================================================================
```

show domain-name

Syntax

```
show domain-name [vsx-peer]
```

Description

Shows the current domain name.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Usage

If there is no domain name configured, the CLI displays a blank line.

Example

Setting and showing the domain name:

```
switch# show domain-name

switch# config
switch(config)# domain-name example.com
switch(config)# show domain-name
example.com
switch(config)#
```

show environment fan

Syntax

```
show environment fan [vsf | vsx-peer]
```

Description

Shows the status information for all fans and fan trays (if present) in the system.

Command context

Manager (#)

Parameters

vsf

Shows output from the VSF member-id on switches that support VSF.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

For fan trays, `Status` is one of the following values:

ready

The fan tray is operating normally.

fault

The fan tray is in a fault event. The status of the fan tray does not indicate the status of fans.

empty

The fan tray is not installed in the system.

For fans:

Speed

Indicates the relative speed of the fan based on the nominal speed range of the fan. Values are:

Slow

The fan is running at less than 25% of its maximum speed.

Normal

The fan is running at 25-49% of its maximum speed.

Medium

The fan is running at 50-74% of its maximum speed.

Fast

The fan is running at 75-99% of its maximum speed.

Max

The fan is running at 100% of its maximum speed.

N/A

The fan is not installed.

Direction

The direction of airflow through the fan. Values are:

front-to-back

Air flows from the front of the system to the back of the system.

N/A

The fan is not installed.

Status

Fan status. Values are:

uninitialized

The fan has not completed initialization.

ok

The fan is operating normally.

fault

The fan is in a fault state.

empty

The fan is not installed.

Examples

Showing output for a system without a fan tray:

```
switch# show environment fan

Fan information
---------------------------------------------------------------
Fan   Serial Number  Speed   Direction      Status        RPM
---------------------------------------------------------------
1     SGXXXXXXXXXX   slow    front-to-back  ok            6000
2     SGXXXXXXXXXX   normal  front-to-back  ok            8000
3     SGXXXXXXXXXX   medium  front-to-back  ok            11000
4     SGXXXXXXXXXX   fast    front-to-back  ok            14000
5     SGXXXXXXXXXX   max     front-to-back  fault         16500
6     N/A            N/A     N/A            empty
...
```

show environment led

Syntax

`show environment led [vsf <MEMBER-ID>| vsx-peer]`

Description

Shows state and status information for all the configurable LEDs in the system.

Command context

Operator (>) or Manager (#)

Parameters

vsf *<MEMBER-ID>*

Shows output from the specified VSF member-id on switches that support VSF.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing state and status for LED for 8400 or 8320 switch series:

```
switch# show environment led

Name            State     Status
-----------------------------------
locator         flashing  ok
```

show environment power-supply

Syntax

`show environment power-supply [vsf | vsx-peer]`

Description

Shows status information about all power supplies in the switch.

Command context

Operator (>) or Manager (#)

Parameters

vsf

Shows output from the VSF member-id on switches that support VSF.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

The following information is provided for each power supply:

Mbr/PSU

Shows the member and slot number of the power supply.

Product Number

Shows the product number of the power supply.

Serial Number

Shows the serial number of the power supply, which uniquely identifies the power supply.

PSU Status

The status of the power supply. Values are:

OK

Power supply is operating normally.

OK*

Power supply is operating normally, but it is the only power supply in the chassis. One power supply is not sufficient to supply full power to the switch. When this value is shown, the output of the command also shows a message at the end of the displayed data.

Absent

No power supply is installed in the specified slot.

Input fault

The power supply has a fault condition on its input.

Output fault

The power supply has a fault condition on its output.

Warning

The power supply is not operating normally.

Wattage Maximum

Shows the maximum amount of wattage that the power supply can provide.

Example

show environment temperature

Syntax

```
show environment temperature [detail] [vsf | vsx-peer]
```

Description

Shows the temperature information from sensors in the switch that affect fan control.

Command context

Operator (>) or Manager (#)

Parameters

detail

Shows detailed information from each temperature sensor.

vsf

Shows output from the VSF member-id on switches that support VSF.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Temperatures are shown in Celsius.

Valid values for status are the following:

normal

Sensor is within nominal temperature range.

min

Lowest temperature from this sensor.

max

Highest temperature from this sensor.

low_critical

Lowest threshold temperature for this sensor.

critical

Highest threshold temperature for this sensor.

fault

Fault event for this sensor.

emergency

Over temperature event for this sensor.

Examples

show events

Syntax

```
show events [ -e <EVENT-ID> |
     -s {alert | crit | debug | emer | err | info | notice | warn} |
     -r | -a | -n <count> |
     -c {lldp | ospf | ... | } |
     -d {lldpd | hpe-fand | ... |}]
```

Description

Shows event logs generated by the switch modules since the last reboot.

Command context

Manager (#)

Parameters

-e *<EVENT-ID>*

Shows the event logs for the specified event ID. Event ID range: 101 through 99999.

-s {alert | crit | debug | emer | err | info | notice | warn}

Shows the event logs for the specified severity. Select the severity from the following list:

- `alert`: Displays event logs with severity alert and above.
- `crit`: Displays event logs with severity critical and above.
- `debug`: Displays event logs with all severities.

- `emer`: Displays event logs with severity emergency only.
- `err`: Displays event logs with severity error and above.
- `info`: Displays event logs with severity info and above.
- `notice`: Displays event logs with severity notice and above.
- `warn`: Displays event logs with severity warning and above.

-r

Shows the most recent event logs first.

-a

Shows all event logs, including those events from previous boots.

-n *<count>*

Displays the specified number of event logs.

-c {lldp | ospf | ... | }

Shows the event logs for the specified event category. Enter `show event -c` for a full listing of supported categories with descriptions.

-d {lldpd | hpe-fand | ... |}

Shows the event logs for the specified process. Enter `show event -d` for a full listing of supported daemons with descriptions.

Authority

Auditors or Administrators or local user group members with execution rights for this command. Auditors can execute this command from the auditor context (auditor>) only.

Examples

Showing event logs:

```
switch# show events
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in Hardware
```

Showing the most recent event logs first:

```
switch# show events -r
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in Hardware
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for bridge_normal interface
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
```

Showing all event logs:

```
switch# show events -a
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
2016-12-01:12:37:31.734541|intfd|4001|INFO|AMM|1|Interface port_admin set to up for bridge_normal interface
2016-12-01:12:37:32.583256|switchd|24002|ERR|AMM|1|Failed to create VLAN 1 in Hardware
```

Showing event logs related to the DHCP relay agent:

```
switch# show events -c dhcp-relay
2016-05-31:06:26:27.363923|hpe-relay|110001|LOG_INFO|DHCP Relay Enabled
2016-05-31:07:08:51.351755|hpe-relay|110002|LOG_INFO|DHCP Relay Disabled
```

Showing event logs related to the DHCPv6 relay agent:

```
switch# show events -c dhcpv6-relay
2016-05-31:06:26:27.363923|hpe-relay|109001|LOG_INFO|DHCPv6 Relay Enabled
2016-05-31:07:08:51.351755|hpe-relay|109002|LOG_INFO|DHCPv6 Relay Disabled
```

Showing event logs related to IRDP:

```
switch# switch# show events -c irdp
2016-05-31:06:26:27.363923|hpe-rdiscd|111001|LOG_INFO|IRDP enabled on interface %s
2016-05-31:07:08:51.351755|hpe-rdiscd|111002|LOG_INFO|IRDP disabled on interface %s
```

Showing event logs related to LACP:

```
switch# show events -c lacp
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
```

Showing event logs as per the specified process:

```
switch# show events -d lacpd
---------------------------------------------------
show event logs
---------------------------------------------------
2016-12-01:12:37:31.733551|lacpd|15007|INFO|AMM|1|LACP system ID set to 70:72:cf:51:50:7c
```

Displaying the specified number of event logs:

```
switch# show events -n 5
---------------------------------------------------
show event logs
---------------------------------------------------
2018-03-21:06:12:15.500603|arpmgrd|6101|LOG_INFO|AMM|-|ARPMGRD daemon has started
2018-03-21:06:12:17.734405|lldpd|109|LOG_INFO|AMM|-|Configured LLDP tx-delay to 2
2018-03-21:06:12:17.740517|lacpd|1307|LOG_INFO|AMM|-|LACP system ID set to 70:72:cf:d4:34:42
2018-03-21:06:12:17.743491|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity 42cc3df7-1113-412f-b5cb-e8227b8c22f2
2018-03-21:06:12:17.904008|vrfmgrd|5401|LOG_INFO|AMM|-|Created a vrf entity 4409133e-2071-4ab8-adfe-f9662c06b889
```

show hostname

Syntax

```
show hostname [vsx-peer]
```

Description

Shows the current host name.

Command context

Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Example

Setting and showing the host name:

```
switch# show hostname
switch
switch# config
switch(config)# hostname myswitch
myswitch(config)# show hostname
myswitch
```

show images

Syntax

```
show images [vsx-peer]
```

Description

Shows information about the software in the primary and secondary images.

Command context

Operator (>) or Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing the primary and secondary images on a 8320 switch:

```
switch# show images
---------------------------------------------------------------------------
ArubaOS-CX Primary Image
---------------------------------------------------------------------------
Version : TL.10.05.0001I
Size    : 405 MB
Date    : 2020-04-23 02:49:04 PDT
SHA-256 : 7efe86a445e87e40f47de156add25720b7277cae1a8db2f9c4ea5f49e74f2a5a

---------------------------------------------------------------------------
ArubaOS-CX Secondary Image
---------------------------------------------------------------------------
Version : TL.10.05.0001I
Size    : 405 MB
Date    : 2020-04-23 02:49:04 PDT
SHA-256 : 7efe86a445e87e40f47de156add25720b7277cae1a8db2f9c4ea5f49e74f2a5a
```

```
Default Image : primary

------------------------------------------------------
Management Module 1/1 (Active)
------------------------------------------------------
Active Image       : primary
Service OS Version : TL.01.05.0002-internal
BIOS Version       : TL-01-0013
```

show ip errors

Syntax

`show ip errors [vsx-peer]`

Description

Shows IP error statistics for packets received by the switch since the switch was last booted.

Command context

Operator (>) or Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

IP error info about received packets is collected from each active line card on the switch and is preserved during failover events. Error counts are cleared when the switch is rebooted.

Drop reasons are the following:

Malformed packet

The packet does not conform to TCP/IP protocol standards such as packet length or internet header length.

A large number of malformed packets can indicate that there are hardware malfunctions such as loose cables, network card malfunctions, or that a DOS (denial of service) attack is occurring.

IP address error

The packet has an error in the destination or source IP address. Examples of IP address errors include the following:

- The source IP address and destination IP address are the same.
- There is no destination IP address.
- The source IP address is a multicast IP address.

- The forwarding header of an IPv6 address is empty.
- There is no source IP address for an IPv6 packet.

Invalid TTLs

The TTL (time to live) value of the packet reached zero. The packet was discarded because it traversed the maximum number of hops permitted by the TTL value.

TTLs are used to prevent packets from being circulated on the network endlessly.

Example

Showing ip error statistics for packets received by the switch:

```
switch# show ip errors
----------------------------------
Drop reason                Packets
----------------------------------
Malformed packets                1
IP address errors               10
...
```

show module

Syntax

`show module [<SLOT-ID>] [vsx-peer]`

Description

Shows information about installed line modules and management modules.

Command context

Operator (>) or Manager (#)

Parameters

<SLOT-ID>

Specifies the member and slot numbers in format `member/slot`. For example, to show the module in member 1, slot 3, enter `1/3`.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Identifies and shows status information about the line modules and management modules that are installed in the switch.

If you use the *<SLOT-ID>* parameter to specify a slot that does not have a line module installed, a message similar to the following example is displayed:

```
Module 1/4 is not physically present.
```

To show the configuration information—if any—associated with that line module slot, use the `show running-configuration` command.

Status is one of the following values:

`Active`

This module is the active management module.

`Standby`

This module is the standby management module.

`Deinitializing`

The module is being deinitialized.

`Diagnostic`

The module is in a state used for troubleshooting.

`Down`

The module is physically present but is powered down.

`Empty`

The module hardware is not installed in the chassis.

`Failed`

The module has experienced an error and failed.

`Failover`

This module is a fabric module or a line module, and it is in the process of connecting to the new active management module during a management module failover event.

`Initializing`

The module is being initialized.

`Present`

The module hardware is installed in the chassis.

`Ready`

The module is available for use.

`Updating`

A firmware update is being applied to the module.

Examples

Showing all installed modules:

```
switch(config)# show module

Management Modules
==================

     Product                                                   Serial
Name Number  Description                                       Number     Status
---- ------- -------------------------------------- ---------- ----------------
1/1  JL581A  8320 Mgmt Mod                                     TW87KCW00X Ready
```

```
Line Modules
============

     Product                                                Serial
Name Number  Description                                    Number     Status
---- ------- ---------------------------------------------- ---------- ----------------
1/1  JL581A  8320                                           TW87KCW00X Ready
```

Showing a slot that does not contain a line module:

```
switch(config)# show module 1/3
Module 1/3 is not physically present
```

show running-config

Syntax

`show running-config [<FEATURE>] [all] [vsx-peer]`

Description

Shows the current nondefault configuration running on the switch. No user information is displayed.

Command context

Manager (#)

Parameters

<FEATURE>

Specifies the name of a feature. For a list of feature names, enter the `show running-config` command, followed by a space, followed by a question mark (?). When the `json` parameter is used, the `vsx-peer` parameter is not applicable.

all

Shows all default values for the current running configuration.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the current running configuration:

```
switch> show running-config
Current configuration:
!
!Version ArubaOS-CX 10.0X.XXXX
!
lldp enable
linecard-module LC1 part-number JL363A
vrf green
!
!
```

```
!
!
!
!
aaa authentication login default local
aaa authorization commands default none
!
!
!
!
router ospf 1 vrf green
    area 0.0.0.0
router pim vrf green
    enable
    rp-address 30.0.0.4
vlan 1
    no shutdown
vlan 20
    no shutdown
vlan 30
    no shutdown
interface 1/1/1
    no shutdown
    no routing
    vlan access 30
interface 1/1/32
    no shutdown
    no routing
    vlan access 20
interface bridge_normal-1
    no shutdown
interface bridge_normal-2
    no shutdown
interface vlan20
    no shutdown
    vrf attach green
    ip address 20.0.0.44/24
    ip ospf 1 area 0.0.0.0
    ip pim-sparse enable

interface vlan30
    no shutdown
    vrf attach green
    ip address 30.0.0.44/24
    ip ospf 1 area 0.0.0.0
    ip pim-sparse enable

    ip pim-sparse hello-interval 100
```

Showing the current running configuration in json format:

```
switch> show running-config json
Running-configuration in JSON:
{
    "Monitoring_Policy_Script": {
        "system_resource_monitor_mm1.1.0": {
            "Monitoring_Policy_Instance": {
                "system_resource_monitor_mm1.1.0/system_resource_monitor_mm1.1.0.default": {
                    "name": "system_resource_monitor_mm1.1.0.default",
                    "origin": "system",
                    "parameters_values": {
                        "long_term_high_threshold": "70",
                        "long_term_normal_threshold": "60",
                        "long_term_time_period": "480",
                        "medium_term_high_threshold": "80",
```

```
                        "medium_term_normal_threshold": "60",
                        "medium_term_time_period": "120",
                        "short_term_high_threshold": "90",
                        "short_term_normal_threshold": "80",
                        "short_term_time_period": "5"
                    }
                }
            },
...
...
...
...
```

Show the current running configuration without default values:

```
switch(config)# show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
!
!
!
!
!
vlan 1
switch(config)# show running-config all
Current configuration:
!
!Version ArubaOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
!
!
!
!
!
vlan 1
switch(config)#
```

Show the current running configuration with default values:

```
switch(config)# snmp-server vrf mgmt
switch(config)# show running-config
Current configuration:
!
!Version ArubaOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
snmp-server vrf mgmt
!
!
!
!
!
```

```
vlan 1
switch(config)#
switch(config)#
switch(config)# show running-config all
Current configuration:
!
!Version ArubaOS-CX Virtual.10.04.0000-6523-gbb15c03~dirty
led locator on
!
!
!
!
snmp-server vrf mgmt
snmp-server agent-port 161
snmp-server community public
!
!
!
!
!
vlan 1
switch(config)#
```

show running-config current-context

Syntax

`show running-config current-context`

Description

Shows the current non-default configuration running on the switch in the current command context.

Command context

`config` or a child of `config`. See Usage.

Authority

Administrators or local user group members with execution rights for this command.

Usage

You can enter this command from the following configuration contexts:

- Any child of the global configuration (`config`) context. If the child context has instances—such as interfaces—you can enter the command in the context of a specific instance.

  Support for this command is provided for one level below the `config` context. For example, entering this command for a child of a child of the `config` context not supported.

  If you enter the command on a child of the `config` context, the current configuration of that context and the children of that context are displayed.

- The global configuration (`config`) context.

  If you enter this command in the global configuration (`config`) context, it shows the running configuration of the entire switch. Use the `show running-configuration` command instead.

Examples

Showing the running configuration for the current interface:

```
switch(config-if)# show running-config current-context
interface 1/1/1
    vsx-sync qos vlans
    no shutdown
    description Example interface
    no routing
vlan access 1
    exit
```

Showing the current running configuration for the management interface:

```
switch(config-if-mgmt)# show running-config current-context
interface mgmt
    no shutdown
    ip static 10.0.0.1/24
    default-gateway 10.0.0.8
    nameserver 10.0.0.1
```

Showing the running configuration for the external storage share named `nasfiles`:

```
switch(config-external-storage-nasfiles)# show running-config current-context
external-storage nasfiles
   address 192.168.0.1
   vrf default
   username nasuser
   password ciphertext AQBapalKj+XMsZumHEwIc9OR6YcOw5Z6Bh9rV+9ZtKDKzvbaBAAAAB1CTrM=
   type scp
   directory /home/nas
   enable
switch(config-external-storage-nasfiles)#
```

Showing the running configuration for a context that does not have instances:

```
switch(config-vsx)# show run current-context
vsx
    inter-switch-link 1/1/1
    role secondary
    vsx-sync sflow time
```

show startup-config

Syntax

`show startup-config [json]`

Description

Shows the contents of the startup configuration.

NOTE: Switches in the `factory-default` configuration do not have a startup configuration to display.

Command context

Manager (#)

Parameters

json

Display output in JSON format.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing the startup-configuration in non-JSON format for an 8320 switch:

```
Leaf2(config)# show startup-config
Startup configuration:
!
!Version ArubaOS-CX TL.xx.xx.xxxx
hostname Leaf2
user admin group administrators password ciphertext
AQBapaGi+KZp4g8gw63UqK+zCtvO5zigFLv2DFBEH+lztqjdYgAAABwrJ+5GayUWArgv9tVFo9AzMY6gmI7x/
KBEkGBJDXjpFson2qM83CXBUI673qWHDQ0pEIZXeuig0XogCVuId4oZiQVZlOe2MfxnqZL+E9hXaMNVowBwbD0
cli-session
    timeout 0
!
!
!
ssh server vrf mgmt
```

Showing the startup-configuration in JSON format:

```
switch# show startup-config json
Startup configuration:
{
    "AAA_Server_Group": {
        "local": {
            "group_name": "local"
        },
        "none": {
            "group_name": "none"
        }
    },
...
```

show system

Syntax

```
show system [vsx-peer]
```

Description

Shows general status information about the system.

Command context

Operator (>) or Manager (#)

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

CPU utilization represents the average utilization across all the CPU cores.

System Contact, System Location, and System Description can be set with the `snmp-server` command.

Examples

Showing system information for the VSX primary and secondary (peer) switch on an 8320:

```
vsx-primary# show system
Hostname           : vsx-primary
System Description : TL.10.xx.xxxxx
System Contact     :
System Location    :

Vendor             : Aruba
Product Name       : JL479A 8320
Chassis Serial Nbr : TW82K7200Q
Base MAC Address   : 98f2b3-68792e
ArubaOS-CX Version : TL.10.xx.xxxxx

Time Zone          : UTC

Up Time            : 19 hours, 51 minutes
CPU Util (%)       : 50
Memory Usage (%)   : 36

vsx-primary# show system vsx-peer
Hostname           : vsx-secondary
System Description : TL.10.xx.xxxxx
System Contact     :
System Location    :

Vendor             : Aruba
Product Name       : JL479A 8320
Chassis Serial Nbr : TW73JQH024
Base MAC Address   : e0071b-cb72e4
ArubaOS-CX Version : TL.10.xx.xxxxx

Time Zone          : UTC

Up Time            : 21 hours, 23 minutes
CPU Util (%)       : 14
Memory Usage (%)   : 36
```

show system resource-utilization

Syntax

```
show system resource-utilization [daemon <DAEMON-NAME> | module <SLOT-ID>] [vsx-peer]
```

Description

Shows information about the usage of system resources such as CPU, memory, and open file descriptors.

Command context

Manager (#)

Parameters

daemon *<DAEMON-NAME>*

Shows the filtered resource utilization data for the process specified by *<DAEMON-NAME>* only.

NOTE: For a list of daemons that log events, enter `show events -d ?` from a switch prompt in the manager (#) context.

module *<SLOT-ID>*

Shows the filtered resource utilization data for the line module specified by *<SLOT-ID>* only.

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Administrators or local user group members with execution rights for this command.

Examples

Showing all system resource utilization data:

```
switch# show system resource-utilization
System Resources:
Processes: 70
CPU usage(%): 20
Memory usage(%): 25
Open FD's: 1024

Process              CPU Usage(%)         Memory Usage(%)    Open FD's
-----------------------------------------------------------------------
pmd                    2                       1                 14
hpe-sysmond            1                       2                 11
hpe-mgmdd              0                       1                 5
...
```

Showing the resource utilization data for the pmd process:

```
switch# show system resource-utilization daemon pmd
Process              CPU Usage         Memory Usage    Open FD's
-----------------------------------------------------------------------
pmd                      2                1               14
```

Showing resource utilization data when system resource utilization polling is disabled:

```
switch# show system resource-utilization
System resource utilization data poll is currently disabled
```

Showing resource utilization data for a line module:

```
switch# show system resource-utilization module 1/1
System Resource utilization for line card module: 1/1
CPU usage(%): 0
Memory usage(%): 35
Open FD's: 512
```

show tech

Syntax

```
show tech [basic | <FEATURE>] [local-file]
```

Description

Shows detailed information about switch features by automatically running the `show` commands associated with the feature. If no parameters are specified, the `show tech` command shows information about all switch features. Technical support personnel use the output from this command for troubleshooting.

Command context

Manager (#)

Parameters

`basic`

Specifies showing a basic set of information.

`<FEATURE>`

Specifies the name of a feature. For a list of feature names, enter the `show tech` command, followed by a space, followed by a question mark (?).

`local-file`

Shows the output of the `show tech` command to a local text file.

Authority

Administrators or local user group members with execution rights for this command.

Usage

To terminate the output of the `show tech` command, enter **Ctrl+C**.

If the command was not terminated with **Ctrl+C**, at the end of the output, the `show tech` command shows the following:

- The time consumed to execute the command.
- The list of failed `show` commands, if any.

To get a copy of the local text file content created with the show tech command that is used with the local-file parameter, use the `copy show-tech local-file` command.

Example

Showing the basic set of system information:

```
switch# show tech basic
====================================================
Show Tech executed on Wed Sep  6 16:50:37 2017
====================================================
====================================================
[Begin] Feature basic
====================================================

*******************************
Command : show core-dump all
```

```
******************************
no core dumps are present

...
============================================================
[End] Feature basic
============================================================

============================================================
1 show tech command failed
============================================================
Failed command:
  1. show boot-history
============================================================
Show tech took 3.000000 seconds for execution
```

Directing the output of the **show tech basic** command to the local text file:

```
switch# show tech basic local-file
Show Tech output stored in local-file. Please use 'copy show-tech local-file'
to copy-out this file.
```

show usb

Syntax

`show usb`

Description

Shows the USB port configuration and mount settings.

Command context

Operator (>) or Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Examples

If USB has not been enabled:

```
switch> show usb
Enabled: No
Mounted: No
```

If USB has been enabled, but no device has been mounted:

```
switch> show usb
Enabled: Yes
Mounted: No
```

If USB has been enabled and a device mounted:

```
switch> show usb
Enabled: Yes
Mounted: Yes
```

show usb file-system

Syntax

`show usb file-system [<PATH>]`

Description

Shows directory listings for a mounted USB device. When entered without the <PATH> parameter the top level directory tree is shown.

Command context

Operator (>) or Manager (#)

Parameters

<PATH>

Specifies the file path to show. A leading "/" in the path is optional.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Usage

Adding a leading "/" as the first character of the *<PATH>* parameter is optional.

Attempting to enter '..' as any part of the *<PATH>* will generate an invalid path argument error. Only fully-qualified path names are supported.

Examples

Showing the top level directory tree:

```
switch# show usb file-system
/mnt/usb:
'System Volume Information'  dir1'

/mnt/usb/System Volume Information':
IndexerVolumeGuid  WPSettings.dat

/mnt/usb/dir1:
dir2  test1

/mnt/usb/dir1/dir2:
test2
```

Showing available path options from the top level:

```
switch# show usb file-system /
total 64
drwxrwxrwx 2 32768 Jan 22 16:27 'System Volume Information'
drwxrwxrwx 3 32768 Mar  5 15:26 dir1
```

Showing the contents of a specific folder:

```
switch# show usb file-system /dir1
total 32
drwxrwxrwx 2 32768 Mar  5 15:26 dir2
-rwxrwxrwx 1     0 Feb  5 18:08 test1

switch# show usb file-system dir1/dir2
total 0
-rwxrwxrwx 1 0 Feb  6 05:35 test2
```

Attempting to enter an invalid character in the path:

```
switch# show usb file-system dir1/../..
Invalid path argument
```

show version

Syntax

`show version [vsx-peer]`

Description

Shows version information about the network operating system software, service operating system software, and BIOS.

Command context

Operator (>) or Manager (#)

Parameters

[vsx-peer]

Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX.

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing version information for an 8320 switch:

```
switch(config)# show version
-----------------------------------------------------------------------------
ArubaOS-CX
(c) Copyright 2017-2020 Hewlett Packard Enterprise Development LP
-----------------------------------------------------------------------------
Version      : TL.xx.xx.xxxx
Build Date   : 2020-08-20 10:56:02 PDT
Build ID     : ArubaOS-CX:xx.xx.xxxx:feb590a400a5:201908201736
Build SHA    : feb590a400a57ed818b01614f92010d74fbc9a4b
Active Image : secondary

Service OS Version : TL.01.03.0008
BIOS Version       : TL-01-0013
```

system resource-utilization poll-interval

Syntax

```
system resource-utilization poll-interval <SECONDS>
```

Description

Configures the polling interval for system resource information collection and recording such as CPU and memory usage.

Command context

```
config
```

Parameters

<SECONDS>

Specifies the poll interval in seconds. Range: 10-3600. Default: 10.

Authority

Administrators or local user group members with execution rights for this command.

Example

Configuring the system resource utilization poll interval:

```
switch(config)# system resource-utilization poll-interval 20
```

top cpu

Syntax

```
top cpu
```

Description

Shows CPU utilization information.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing top CPU information:

```
switch# top cpu
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks:  76 total, 2 running, 74 sleeping,  0 stopped,  0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total,  2487508 free,  897040 used,   661948 buff/cache
KiB Swap:       0 total,        0 free,       0 used,  2859196 avail Mem
```

```
  PID USER      PRI  NI    VIRT    RES    SHR S  %CPU %MEM      TIME+ COMMAND
...
```

top memory

Syntax

`top memory`

Description

Shows memory utilization information.

Command context

Manager (#)

Authority

Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Example

Showing top memory:

```
switch> top memory
top - 09:42:55 up 3 min, 3 users, load average: 3.44, 3.78, 1.70
Tasks:  76 total, 2 running, 74 sleeping,  0 stopped,  0 zombie
%Cpu(s): 31.4 us, 32.7 sy, 0.5 ni, 34.4 id, 04. wa, 0.0 hi, 0.6 si, 0.0 st
KiB Mem : 4046496 total,  2487508 free,  897040 used,   661948 buff/cache
KiB Swap:       0 total,        0 free,       0 used,  2859196 avail Mem

  PID USER      PRI  NI    VIRT    RES    SHR S  %CPU %MEM      TIME+ COMMAND
...
```

usb

Syntax

`usb`

`no usb`

Description

Enables the USB ports on the switch. This setting is persistent across switch reboots and management module failovers. Both active and standby management modules are affected by this setting.

The `no` form of this command disables the USB ports.

Command context

`config`

Authority

Administrators or local user group members with execution rights for this command.

Example

Enabling USB ports:

```
switch(config)# usb
```

Disabling USB ports when a USB drive is mounted:

```
switch(config)# no usb
```

usb mount | unmount

Syntax

`usb {mount | unmount}`

Description

Enables or disables the inserted USB drive.

Command context

Manager (#)

Parameters

mount

Enables the inserted USB drive.

unmount

Disables the inserted USB drive in preparation for removal.

Authority

Administrators or local user group members with execution rights for this command.

Usage

A USB drive must be unmounted before removal.

The supported USB file systems are FAT16 and FAT32.

Examples

Mounting a USB drive in the USB port:

```
switch# usb mount
```

Unmounting a USB drive:

```
switch# usb unmount
```

Accessing Aruba Support

Aruba Support Services	**https://www.arubanetworks.com/support-services/**
Aruba Support Portal	**https://asp.arubanetworks.com/**
North America telephone	1-800-943-4526 (US & Canada Toll-Free Number) +1-408-754-1200 (Primary - Toll Number) +1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working)
International telephone	**https://www.arubanetworks.com/support-services/contact-support/**

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

Airheads social forums and Knowledge Base	**https://community.arubanetworks.com/**
Software licensing	**https://lms.arubanetworks.com/**
End-of-Life information	**https://www.arubanetworks.com/support-services/end-of-life/**
Aruba software and documentation	**https://asp.arubanetworks.com/downloads**

Accessing updates

To download product updates:

Aruba Support Portal

https://asp.arubanetworks.com/downloads

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

My Networking

https://www.hpe.com/networking/support

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

https://support.hpe.com/portal/site/hpsc/aae/home/

IMPORTANT: Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

https://asp.arubanetworks.com/notifications/subscriptions (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

Warranty information

To view warranty information for your product, go to **https://www.arubanetworks.com/support-services/product-warranties/**.

Regulatory information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at **https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts**

Additional regulatory information

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